

# The Three-Prime Running LCM: Kernel Rank and Tail Arithmetic

*Erdős Problem #269*

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## ABSTRACT

For every finite set  $P$  of at least two primes, the sum of the reciprocals of the distinct running least common multiples of the  $P$ -smooth integers is irrational, as Erdős asserted in a 1973 letter that gives no proof. For two primes this follows from a Hecke–Mahler transcendence theorem. A rational value makes the tails of the sum, scaled by the running value, integers of one denominator, so the tails after a prime power  $t$  and after  $tp^n$  are equal for a good return  $n$ . Their words of primes differ by rearranging short runs, and equality forces an integer invariant to agree on every run. A point of the closed subtorus generated by the logarithmic flow that lies on exactly two coordinate walls, found by an Euler characteristic count, produces a run of two primes on which it differs. The proof is ordinary, checked by a second AI agent with no human review, and has no Lean proof. For  $\{2, 3, 5\}$  a separate proof, by five affine maps with disjoint images on dyadic blocks, is checked in Lean. A finite-state criterion of the same form, checked by exact computation, proves irrationality for every set of three primes at most 31 and for 292 of the 330 sets of four primes at most 31; it fails at every depth for  $\{2, 3, 5, 7\}$ , where two different words of blocks have the same affine map. The same comparison of two tails proves that the sub-sum of the catalogue series over the powers of one prime is irrational. The reciprocal three-prime running-LCM kernel has nonsingular minors of every order: row and column rescaling leaves a two-valued matrix, and density supplies the required threshold pattern. For the rescaled matrix, we compute the rank of every finite restriction and the least uniform approximation error over all matrices of finite separated rank. For the repeated  $\{2, 3, 5\}$  sum, we derive an integer-coefficient tail recurrence, determine exactly when a hypothetical rational denominator would clear, and prove a residue criterion equivalent to irrationality. The unproved step is to find such windows for every positive multiplier coprime to 30 and after every prescribed start; the tails grow quadratically, and the finite-state argument has no counterpart there. We also give the full two-prime comparison, following Fan’s earlier Hecke–Mahler reduction, and identify the boundary terms that survive a proposed finite-difference cancellation.

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\**Authorship and AI use.* Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

# 1 The problem, and what is settled

Let  $P$  be a finite set of primes with  $|P| \geq 2$ , and let  $a_1 < a_2 < \dots$  enumerate the positive integers all of whose prime factors lie in  $P$ . Erdős Problem #269 asks whether

$$\sum_{n \geq 1} \frac{1}{[a_1, \dots, a_n]}$$

is irrational, where  $[a_1, \dots, a_n]$  is the least common multiple [1, p. 65][2, p. 106]. The problem number is that of Bloom's catalogue [4].

Write  $\mathcal{R}_P$  for the sum above and  $\mathcal{D}_P$  for the sum in which each distinct running-LCM value contributes its reciprocal once. These differ because the running value can repeat. In a letter of 1 January 1973, Erdős called the irrationality of  $\mathcal{R}_P$  a conjecture and wrote that he could show  $\mathcal{D}_P$  to be irrational, without giving an argument [3, p. 335].

**Theorem 1.1** (the distinct-height sums). *For every finite set  $P$  of primes with  $|P| \geq 2$ , the sum  $\mathcal{D}_P$  is irrational.*

Ordinary proof in Section 4.1. It was found within this project on 26 September 2026 and checked step by step by a second, independent AI agent, whose five small corrections are incorporated; it has had no human review. It has no Lean proof, so Comparator does not apply. Theorem 1.2 is its Lean-checked case.

For two primes the theorem follows from Theorem 1.3 below, whose transcendence input is due to Loxton and van der Poorten and to Bugeaud and Laurent.

**Theorem 1.2** (the distinct-height sum at  $\{2, 3, 5\}$ ). *The sum  $\mathcal{D}_{\{2,3,5\}}$  is irrational.*

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Checked in Lean (Lean proof) for the sum as written in (4); Comparator has not yet been run on it. Section 4 gives the ordinary proof.

Section 4 proves both theorems. The proof of Theorem 1.1 compares the tails after two nearby prime powers: their words of primes differ by rearranging short runs, a rational value would force an integer invariant to agree on every run, and a point on exactly two coordinate walls of a torus produces a run of two primes on which it differs. The Lean-checked proof of Theorem 1.2 reads the sum one dyadic block at a time. The normalised tails then follow five affine maps whose images of a common interval are disjoint, so a rational value would make the powers of 3 fall periodically among the dyadic blocks. The same section turns this into a criterion for every finite set of primes (Theorem 4.6); exact computation verifies it for every set of three primes at most 31 and for 292 of the 330 sets of four primes at most 31 (Corollary 4.7), and it fails at every depth for  $\{2, 3, 5, 7\}$ . Section 11.1 discusses Erdős's assertion for general  $P$ , the singleton and infinite-prime cases, and the literature search behind this record.

For the catalogue sum  $\mathcal{R}_P$  the case of two primes is settled. Theorem 1.3 proves transcendence of both sums for every pair of primes, and Fan posted that argument first [16]; this record leaves the repeated finite cases with  $|P| \geq 3$  unresolved.

**Theorem 1.3** (two-prime transcendence). *Let  $p$  and  $q$  be distinct primes. Then  $\mathcal{R}_{\{p,q\}}$  and  $\mathcal{D}_{\{p,q\}}$  are transcendental.*

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The Lean proof assumes the transcendence theorem of Bugeaud and Laurent.

Section 3 proves this by expressing both values as nonconstant polynomials over  $\mathbb{Q}$  in the same Hecke–Mahler boundary sum. Transcendence of that sum is the theorem of Loxton and van der Poorten [15, Theorem 8, p. 40] in the modern form of Bugeaud and Laurent [14, Theorem 1.1]. The two-prime argument therefore consists of an elementary identity followed by that external value theorem.

Steve Fan’s post of 26 June 2026 on the erdosproblems.com forum gives the running-LCM identity for finite prime sets, the two-prime factorisation, the Hecke–Mahler reduction and its transcendence conclusion [16]. We include the calculation to distinguish repeated from distinct-height sums, without claiming priority for that argument.

**What the third prime changes.** With two primes, row and column rescaling reduces the kernel to a constant matrix. With three, the remaining entry is 1 or  $1/r$ , according to whether two fractional parts add to at least one. Their separate density lets us choose nonsingular minors of every order. No joint density assumption is required.

The arithmetic argument starts elsewhere. We group the supported integers between consecutive powers of two, retaining all multiplicities. The resulting normalised tails satisfy an integer-coefficient recurrence. Were the repeated  $\{2, 3, 5\}$  sum rational, the part of its denominator coprime to 30 would make every sufficiently late normalised tail a positive integer after multiplication. These integers have a quadratic bound in the index. A least positive residue exceeding that bound would contradict rationality. We prove an equivalence: irrationality holds exactly when suitable windows exist for every remaining denominator and after every prescribed index. The existence of such windows with both quantifiers unrestricted remains unproved. Section 9 gives finite tests, and Section 10 explains the remaining arithmetic question.

We use  $\mathbb{N} = \{0, 1, 2, \dots\}$  for exponent indices and braces for fractional parts. Until the prime set is fixed to  $\{2, 3, 5\}$  in Section 6, take  $P = \{p, q, r\}$  with pairwise distinct primes  $p, q, r$ . An integer is  $P$ -smooth when its prime factors lie in  $P$ , equivalently when it has the form  $p^i q^j r^k$  with  $i, j, k \geq 0$ . For  $x \geq 1$  write

$$L(x) = \text{lcm}\{n \leq x : n \text{ smooth}\}, \quad H(x) = p^{\lfloor \log_p x \rfloor} q^{\lfloor \log_q x \rfloor} r^{\lfloor \log_r x \rfloor},$$

for the running least common multiple and the product of the three maximal prime powers. We call the latter the height. Here  $\lfloor \log_b x \rfloor$  is the largest  $e$  with  $b^e \leq x$ . Since  $a_1, \dots, a_n$  are exactly the smooth numbers up to  $a_n$ , we have  $[a_1, \dots, a_n] = L(a_n)$ . We regard the reciprocal height as a function of the exponent triple:

$$K(i, j, k) = \frac{1}{H(p^i q^j r^k)}.$$

For example, 5 and 6 are both supported on  $\{2, 3, 5\}$  and have height 60: they contribute twice to  $\mathcal{R}_P$  but only once to  $\mathcal{D}_P$ . For  $P = \{2, 3, 5\}$  these are precisely the usual 5-smooth integers. The parameter here is the size of the integer, not a growing smoothness bound  $y$  in  $\Psi(x, y)$  [11]; no smooth-number density asymptotic is used below. The appropriate fixed-support context is Tijdeman–Meijer [27, Sections 1–4]. The modern two-prime treatment of Languasco, Luca, Moree and Togbé [28] also makes the lattice-triangle geometry explicit; its gap estimates are not inputs to our shell bound. The jumps occur at positive powers of one of the three primes.

Kovač and Tao [13] treat several other irrationality problems for unit-fraction series by elementary means; their results are not inputs to the present arguments. The formal statement of Problem #269 and its rational normalisation are discussed in Section 11.1.

**Keywords.** irrationality; transcendence; least common multiple; smooth numbers; Cantor series; separated rank; Lean 4. **MSC 2020.** 11J72 (primary); 11A05, 11N25, 68V20 (secondary).

**Relation to the short paper.** The short paper leads with the distinct-height theorem for every finite set of primes, its Lean-checked case  $\{2, 3, 5\}$  and the finite-state corollary for three primes, then gives the determinant construction and proves a quadratic tail bound sufficient for the residue criterion. This companion supplies the finite geometry, full two-prime calculation, the distinct-height criterion with its certificates, and the approximation arguments in Sections 2–5. Sections 6–8 derive the tail coefficients, sharpen the bound, determine the denominator-clearing index and prove the exact growth condition for other window bounds. Section 9 contains the finite tests and the twelve-shell denominator certificate. Section 10 keeps the weighted differences, recodings and value-theorem comparisons separate from the proved residue criterion: none is a premise of that criterion. The tail notation agrees with the short paper and is defined when first used.

*Status.* The problem treated here is open, and this note does not close it. Every Lean mark in the margin points to propositions that the pinned Lean kernel accepts, with no sorry, no added axiom, and no unchecked evaluation. That claim concerns the formal statements only; it says nothing about their mathematical interest, their novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

*Formal proofs.* A result with a kernel-checked Lean proof carries a mark in the margin. *Lean* opens the proof: the declaration itself when one declaration states the whole result, otherwise the list of declarations that together state it. *Comparator* opens the record of an independent check, in which the same statement, written again from Mathlib alone in a separate repository, was compared with our proof by Lean’s Comparator tool, allowing only the three standard axioms. A dagger on the Lean mark means that the Lean proof assumes an input named just below the result. A result without a mark has no Lean proof of its whole statement; what is checked is said below it. The [evidence record](#) gives every declaration, version and check. These checks show that the stated propositions are proved; whether they are the right propositions is a question the reader can settle by comparing them with the text.

## 2 The finite geometry of the running value

The prime-exponent maximum rule for the least common multiple is classical. Applied to all integers up to  $N$ , it gives  $\text{lcm}(1, \dots, N) = \prod_{t \leq N} t^{\lfloor \log_t N \rfloor}$ , where the product is over primes, or equivalently  $\log \text{lcm}(1, \dots, N) = \psi(N)$ ; see Apostol [10] and Montgomery and Vaughan [12]. The same rule applies to the supported prefix below.

The smooth numbers up to  $x$  are indexed by the exponent triples  $(i, j, k)$  with  $i \leq \lfloor \log_p x \rfloor$ ,  $j \leq \lfloor \log_q x \rfloor$ ,  $k \leq \lfloor \log_r x \rfloor$  and  $p^i q^j r^k \leq x$ . The coordinate bounds alone need not describe the prefix: each prime-power factor may be at most  $x$  while their product exceeds  $x$ . For instance, at  $x = 6$  the factors 4, 3 and 5 satisfy their coordinate bounds, but the corresponding smooth number is 60.

**Theorem 2.1** (the running least common multiple). *Let  $p, q, r$  be pairwise distinct primes and  $x \geq 1$ . Then  $L(x) = H(x)$ .*

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*Proof.* Every smooth  $n \leq x$  has exponents bounded by the corresponding integer logarithms, so  $n \mid H(x)$  and hence  $L(x) \mid H(x)$ . Conversely the three pure powers  $p^{\lfloor \log_p x \rfloor}$ ,  $q^{\lfloor \log_q x \rfloor}$  and  $r^{\lfloor \log_r x \rfloor}$  are themselves smooth numbers not exceeding  $x$ , so each divides  $L(x)$ , and distinct primes have coprime powers, so their product divides  $L(x)$  as well.  $\square$

At  $(p, q, r) = (2, 3, 5)$  the first ten values are

| $x$    | 1 | 2 | 3 | 4  | 5  | 6  | 7  | 8   | 9   | 10  |
|--------|---|---|---|----|----|----|----|-----|-----|-----|
| $L(x)$ | 1 | 2 | 6 | 12 | 60 | 60 | 60 | 120 | 360 | 360 |

So  $L(6) = 4 \cdot 3 \cdot 5 = 60$ , which exceeds 6: the running value at a smooth cutoff already contains powers of the other two primes that the cutoff itself does not. The kernel must therefore account for all three maximal powers, not just the factorisation of the cutoff. Also  $H(x) \leq x^3$ , since each factor is at most  $x$ . The exponent is the number of generating primes.

Say that  $x$  and  $y$  lie in the same *logarithmic cell* when  $\lfloor \log_b x \rfloor = \lfloor \log_b y \rfloor$  for each of  $b = p, q, r$ . By Theorem 2.1 the running value depends on  $x$  only through the three integer logarithms, so it is constant on cells and moves only where one logarithm moves.

**Proposition 2.2** (constancy and jump ratios). *If  $x, y \geq 1$  lie in the same logarithmic cell then  $L(x) = L(y)$ , and the same holds for the kernel at two smooth points of one cell. If  $\lfloor \log_p y \rfloor = \lfloor \log_p x \rfloor + 1$  while the other two logarithms agree, then  $L(y) = p L(x)$ , and similarly with  $q$  or  $r$  in place of  $p$ .*

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*Proof.* Both parts are immediate from Theorem 2.1: the height depends on  $x$  only through the three integer logarithms, and advancing one of them multiplies exactly one factor by its base.  $\square$

**Proposition 2.3** (jump count). *Let  $n \geq 0$ . The set of the first  $n$  positive powers of  $p$ , of  $q$  and of  $r$  has exactly  $3n$  elements, and adjoining the common origin 1 gives exactly  $3n + 1$ .*

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*Proof.* For a fixed prime  $b$ , the powers  $b, b^2, \dots, b^n$  are distinct. A common value for two different primes would contradict unique factorisation. Finally 1 is not a positive power of any prime.  $\square$

The later tail estimates use two elementary counting facts. Write  $\mathcal{B}(h_p, h_q, h_r)$  for the exponent triples with  $i \leq h_p$ ,  $j \leq h_q$  and  $k \leq h_r$ , and  $F(H)$  for the points of this box whose height equals  $H$ .

**Proposition 2.4** (grouping equal heights). *For every box  $\mathcal{B}$ ,*

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$$\sum_{(i,j,k) \in \mathcal{B}} K(i,j,k) = \sum_H \#F(H)/H,$$

*the outer sum ranging over the heights attained on  $\mathcal{B}$ .*

*Proof.* Partition  $\mathcal{B}$  into the fibres of the height map. On  $F(H)$  every summand is  $1/H$ , so the fibre contributes  $\#F(H)/H$ .  $\square$

Now fix an interval  $[\lambda, \eta)$  with  $0 \leq \lambda < \eta$  and write  $\mathcal{S}$  for the exponent triples of  $\mathcal{B}(h_p, h_q, h_r)$  whose value  $p^i q^j r^k$  lies in it. In the next lemma, the exponents  $a, a'$  are nonnegative integers, and  $w \geq 0$  is the product of the fixed factors.

**Lemma 2.5** (uniqueness in a short interval). *Let  $b \geq 1$  and  $\eta \leq b\lambda$ . If  $b^a w$  and  $b^{a'} w$  both lie in  $[\lambda, \eta)$  then  $a = a'$ .*

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*Proof.* If  $a < a'$  then  $\eta \leq b\lambda \leq b^{a+1}w \leq b^{a'}w < \eta$ , which is impossible; the case  $a > a'$  is symmetric.  $\square$

The short-interval condition says that multiplying by the omitted base moves a point beyond the interval. It holds for  $[L, 2L)$  when that base is at least two. A wider interval need not have this property: 1 and 2 both lie in  $[1, 3)$  and have the same odd part. Thus the ratio bound, not just finiteness of the interval, permits the injective projection.

**Proposition 2.6** (counting a shell by two coordinates). *If  $\eta \leq r\lambda$  then  $\#\mathcal{S} \leq (h_p + 1)(h_q + 1)$ , and if  $\eta \leq p\lambda$  then  $\#\mathcal{S} \leq (h_q + 1)(h_r + 1)$ . If moreover  $\eta \leq r\lambda$  and  $h_p \leq h_q \leq h_r$  with  $h_p + h_q + h_r = j$ , then  $9\#\mathcal{S} \leq (j + 3)^2$ .*

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*Proof.* Suppose  $\eta \leq r\lambda$ . If two triples of  $\mathcal{S}$  agree in their first two coordinates, Lemma 2.5 with  $b = r$  and  $w = p^i q^j$  forces their third coordinates to agree, so the projection forgetting the third coordinate is injective on  $\mathcal{S}$  and its image lies in a rectangle with  $(h_p + 1)(h_q + 1)$  points. The other case is the same with the first coordinate projected away. Under the sorting hypothesis the two surviving coordinates are the two smallest, so it suffices that  $a \leq b \leq c$  with  $a + b + c = j$  gives  $9(a + 1)(b + 1) \leq (j + 3)^2$ . From  $a \leq b \leq c$  we get  $a + 2b \leq j$ , so it is enough that  $9(a + 1)(b + 1) \leq (a + 2b + 3)^2$ ; writing  $b = a + d$  with  $d \geq 0$ , the difference of the two sides is  $d(3a + 4d + 3) \geq 0$ .  $\square$

### 3 One Hecke–Mahler value controls both two-prime sums

Temporarily let  $P = \{p, q\}$  with  $p < q$ , and write  $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$ , which is the running least common multiple of the  $\{p, q\}$ -smooth numbers up to  $t$  by the argument of Theorem 2.1 with one coordinate omitted. The distinct-height sum retains the initial value 1 and one reciprocal for every later distinct running value, so

$$\mathcal{D}_{\{p,q\}} = 1 + \sum_{t \in \{p, p^2, \dots\} \cup \{q, q^2, \dots\}} \frac{1}{L_{p,q}(t)}, \quad \mathcal{R}_{\{p,q\}} = \sum_{i,j \geq 0} \frac{1}{L_{p,q}(p^i q^j)}.$$

*Proof of Theorem 1.3.* Set

$$\theta = \frac{\log p}{\log q}, \quad x = \frac{1}{p}, \quad y = \frac{1}{q}, \quad m_n = \lfloor n\theta \rfloor, \quad \delta_n = m_{n+1} - m_n.$$

Here  $0 < \theta < 1$ , and  $\theta$  is irrational, since a rational value would give  $p^b = q^a$  for positive integers  $a, b$ . Consequently  $\delta_n \in \{0, 1\}$ . Put

$$A = \sum_{n \geq 0} x^n y^{m_n}, \quad B_* = \sum_{n \geq 0} \delta_n x^n y^{m_{n+1}}.$$

The initial value and the positive powers of  $p$  contribute  $A$ , since  $L_{p,q}(p^n) = p^n q^{m_n}$ . A power of  $q$  lies strictly between  $p^n$  and  $p^{n+1}$  exactly when  $\delta_n = 1$ , it is then  $q^{m_{n+1}}$ , and its post-jump reciprocal is  $x^n y^{m_{n+1}}$ ; so the positive powers of  $q$  contribute  $B_*$  and  $\mathcal{D}_{\{p,q\}} = A + B_*$ . All these series converge absolutely.

Since  $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n} (y - 1)$ , an index shift gives  $A - 1 - xA = x(y - 1)B_*/y$ , so  $B_* = ((p - 1)A - p)/(1 - q)$  and

$$\mathcal{D}_{\{p,q\}} = \frac{(q - p)A + p}{q - 1}. \quad (1)$$

At a smooth point,  $\log_p(p^i q^j) = i + j/\theta$  and  $\log_q(p^i q^j) = j + i\theta$ , so  $L_{p,q}(p^i q^j) = p^{i + \lfloor j/\theta \rfloor} q^{j + m_i}$  and absolute convergence permits the factorisation

$$\mathcal{R}_{\{p,q\}} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor}.$$

For  $j \geq 1$ , the index  $n = \lfloor j/\theta \rfloor$  is precisely the one for which  $p^n < q^j < p^{n+1}$ ; the inequalities are strict because distinct primes have no common positive power. This interval contains at most one power of  $q$ , since  $p < q$ . Thus  $\delta_n = 1$  and  $j = m_n + 1$ . Conversely, every  $n$  with  $\delta_n = 1$  contains that unique power of  $q$ . The second factor is therefore  $1 + B_*$ , giving

$$\mathcal{R}_{\{p,q\}} = \frac{(p + q - 1)A - (p - 1)A^2}{q - 1}. \quad (2)$$

It remains to prove that  $A$  is transcendental. For the Hecke–Mahler series

$$F_\theta(x, y) = \sum_{n \geq 1} \sum_{k=1}^{\lfloor n\theta \rfloor} x^n y^k$$

a finite geometric sum gives  $((1 - y)/y)F_\theta(x, y) = x/(1 - x) - (A - 1)$ , that is

$$A = \frac{1}{1 - x} - \frac{1 - y}{y} F_\theta(x, y). \quad (3)$$

Bugeaud and Laurent's Theorem 1.1 states, in particular, that  $F_\theta(\beta, \alpha)$  is transcendental when  $\theta \in (0, 1)$  is irrational,  $\alpha$  and  $\beta$  are nonzero algebraic numbers,  $|\beta| < 1$  and  $|\beta \alpha^\theta| < 1$

[14, Theorem 1.1]; the  $\rho = 0$  case used here goes back to Loxton and van der Poorten [15, Theorem 8, p. 40]. Take  $(\beta, \alpha) = (x, y)$ : then  $|\beta| = 1/p < 1$  and

$$|xy^\theta| = \frac{1}{p} \left( \frac{1}{q} \right)^{\log p / \log q} = \frac{1}{p^2} < 1.$$

So  $F_\theta(x, y)$  is transcendental, and (3) makes  $A$  transcendental. The coefficient of  $A$  in (1) is  $(q - p)/(q - 1) \neq 0$  and the coefficient of  $A^2$  in (2) is  $-(p - 1)/(q - 1) \neq 0$ , both rational. If either value were algebraic, its identity would exhibit  $A$  as a root of a nonzero polynomial over the algebraic numbers.  $\square$

A product of two transcendental numbers need not be transcendental. What proves the repeated sum transcendental is its nonconstant quadratic expression in the single value  $A$ , not the factorisation by itself. The value theorem adds no unverified hypothesis in this two-prime case: distinct primes give an irrational slope and the displayed reciprocal arguments satisfy its size conditions. Primality is stronger than the calculation needs. For coprime integers  $1 < p < q$ , enumerate the monoid  $\{p^i q^j : i, j \geq 0\}$ , not all integers supported on the prime factors of  $pq$ . Unique exponent pairs, the running-LCM product and the irrationality of  $\log p / \log q$  still hold, so both identities and transcendence conclusions remain valid. For example, this applies to generators 4, 9. By contrast, the monoid generated by 4, 8 has running LCM 8 at the cutoff 8, not  $4^{\lfloor \log_4 8 \rfloor} 8^{\lfloor \log_8 8 \rfloor} = 32$ ; its slope is rational as well. The coprime extension was already noted in the forum discussion (Section 11.1).

A third prime introduces an additional floor term that cannot be separated in this way; Section 5 makes that obstruction precise. For the distinct-height sums, the next section proves irrationality for every finite set of primes by an elementary argument that uses no value theorem.

## 4 The distinct-height sums

This section proves Theorems 1.1 and 1.2, turns the dyadic argument into a criterion for every finite set of primes, and records the exact computations that apply the criterion to sets of three and four primes. The arguments prove irrationality only; they give no transcendence statement of the kind obtained for two primes in Section 3.

Let  $P$  be a finite set of at least two primes, with least element  $p$ , and put  $H_P(t) = \prod_{r \in P} r^{\lfloor \log_r t \rfloor}$ . The proofs of Theorem 2.1 and Propositions 2.2 and 2.3 apply unchanged to any finite set of primes: the running least common multiple of the  $P$ -smooth integers up to  $t$  is  $H_P(t)$ , it changes only at the prime powers  $r^n$  ( $r \in P$ ,  $n \geq 1$ ), where it is multiplied by  $r$ , and distinct prime powers are distinct numbers. List the prime powers as  $t_1 < t_2 < \dots$  and let  $q_k$  be the prime of  $t_k$ . The distinct running values are then 1 and  $H_P(t_k) = q_1 \cdots q_k$ , so

$$\mathcal{D}_P = 1 + \sum_{k \geq 1} \frac{1}{H_P(t_k)} = \sum_{k \geq 0} \frac{1}{q_1 \cdots q_k}. \quad (4)$$

This is the sum  $\Sigma'$  of Erdős's letter [3], which counts each distinct value of  $[a_1, \dots, a_n]$  once. For  $P = \{2, 3, 5\}$  the prime powers 2, 3, 4, 5, 8, 9, 16, 25, 27, 32

give the terms  $1, \frac{1}{2}, \frac{1}{6}, \frac{1}{12}, \frac{1}{60}, \frac{1}{120}, \frac{1}{360}, \frac{1}{720}, \frac{1}{3600}, \frac{1}{10800}, \frac{1}{21600},$  and  $\mathcal{D}_{\{2,3,5\}} = 1.77961698329474905045919064931698327 \dots$ . Put  $Q_k = q_1 \cdots q_k = H_P(t_k)$ .

#### 4.1 Every finite set of primes

The proof compares two tails of the series. A rational value makes every scaled tail an integer, so two tails closer than the scale are equal. The tails after  $t_k$  and after  $t_k p^n$ , for a suitable  $n$ , are that close. Their words of primes differ only by rearranging short runs, and equality of the tails forces an integer invariant  $f$  to agree on each run and its rearrangement. A run of two primes violates this, and a point of a torus on exactly two coordinate walls produces one.

**Words.** For an infinite word  $u = u_1 u_2 \cdots$  over  $P$  put  $V(u) = \sum_{i \geq 1} (u_1 \cdots u_i)^{-1}$ . For a finite word  $\sigma = \sigma_1 \cdots \sigma_n$  put  $\Pi(\sigma) = \sigma_1 \cdots \sigma_n$  and

$$f(\sigma) = \sum_{t=1}^n \sigma_{t+1} \cdots \sigma_n,$$

the term  $t = n$  being 1. Then  $f(\sigma)$  is a positive integer and

$$V(\sigma u) = \frac{f(\sigma) + V(u)}{\Pi(\sigma)}. \quad (5)$$

Put  $x_k = V(q_{k+1} q_{k+2} \cdots) = \sum_{j > k} Q_k / Q_j$ . Every suffix of  $q_1 q_2 \cdots$  contains every prime of  $P$ , in particular one larger than  $\min P$ , so its value lies in  $(0, 1/(\min P - 1)) \subseteq (0, 1)$ .

**Lemma 4.1** (integral tails). *If  $\mathcal{D}_P = N/K$  with integers  $N$  and  $K \geq 1$ , then  $Kx_k$  is an integer for every  $k \geq 1$ .*

*Proof.*  $Q_k \mathcal{D}_P = Q_k + \sum_{i \leq k} Q_k / Q_i + x_k$ , and  $Q_i$  divides  $Q_k$  for  $i \leq k$ . So  $Kx_k = Q_k N - K(Q_k + \sum_{i \leq k} Q_k / Q_i)$ .  $\square$

**Lemma 4.2** (rearranged blocks). *Let  $u = \sigma_1 \sigma_2 \cdots$  and  $u' = \sigma'_1 \sigma'_2 \cdots$  be infinite words, where each  $\sigma_i$  is a nonempty finite word and  $\sigma'_i$  is a rearrangement of it, and suppose that every suffix of  $u$  or  $u'$  starting at a block boundary has value in  $(0, 1)$ .*

(i) *If  $V(u) = V(u')$ , then  $f(\sigma_i) = f(\sigma'_i)$  for every  $i$ .*

(ii) *If  $\sigma_i = \sigma'_i$  for every  $i < i_0$ , then  $|V(u) - V(u')| < 1/\Pi(\sigma_1 \cdots \sigma_{i_0-1})$ .*

*Proof.* Let  $U_i$  and  $U'_i$  be the values of the suffixes starting at block  $i$ . Since  $\Pi(\sigma_i) = \Pi(\sigma'_i)$ , (5) gives  $\Pi(\sigma_i)(U_i - U'_i) = f(\sigma_i) - f(\sigma'_i) + U_{i+1} - U'_{i+1}$ . If  $U_i = U'_i$ , the integer  $f(\sigma_i) - f(\sigma'_i)$  equals  $U'_{i+1} - U_{i+1} \in (-1, 1)$ , so both vanish. Induction from  $U_1 = V(u) = V(u')$  proves (i). For (ii),  $V(u) - V(u') = (U_{i_0} - U'_{i_0})/\Pi(\sigma_1 \cdots \sigma_{i_0-1})$ .  $\square$

**Lemma 4.3** (short rearrangements). *The map  $f$  is injective on the orderings of any set of at most three distinct primes.*

*Proof.* For two primes,  $f(ab) = 1 + b$ . For three,  $f(\sigma_1 \sigma_2 \sigma_3) = 1 + \sigma_3(1 + \sigma_2)$ . Let  $\sigma, \sigma'$  be orderings of  $\{a, b, c\}$  with  $f(\sigma) = f(\sigma')$ . If  $\sigma_3 = \sigma'_3$ , then  $\sigma_2 = \sigma'_2$  and  $\sigma = \sigma'$ . Otherwise say  $\sigma_3 = a$  and  $\sigma'_3 = b$ , so  $a(1 + \sigma_2) = b(1 + \sigma'_2)$  with  $\sigma_2 \in \{b, c\}$  and  $\sigma'_2 \in \{a, c\}$ . The pairs  $(b, a)$  and  $(c, c)$  give  $a = b$ , the pair  $(b, c)$  gives  $b \mid a$ , and  $(c, a)$  gives  $a \mid b$ .  $\square$

For four primes injectivity fails:  $f(5, 7, 3, 2) = f(7, 2, 3, 5) = 51$ . The proof therefore looks for a run of two primes, for which  $f(ab) - f(ba) = b - a \neq 0$ .

**The torus.** Let  $\mathbb{T} = \prod_{q \in P} \mathbb{R} / (\log q)\mathbb{Z}$  and  $\Theta(s) = (s \bmod \log q)_{q \in P}$  for  $s \in \mathbb{R}$ . For  $\theta \in \mathbb{T}$  a  $q$ -crossing is a real  $t$  with  $\theta_q + t \in (\log q)\mathbb{Z}$ , and  $W(\theta)$  is the word of the primes of the positive crossings in increasing order of time. For  $\theta_k = \Theta(\log t_k)$  the positive  $q$ -crossings are the times  $\log(q^j/t_k)$  with  $q^j > t_k$ , so  $W(\theta_k) = q_{k+1}q_{k+2}\dots$ , and crossings of different primes never coincide.

Let  $T_0$  be the closure of  $\Theta(\mathbb{R})$ , a subtorus containing every  $\theta_k$ . For  $t > 0$  and  $\varepsilon > 0$ , the pigeonhole principle applied to the points  $\Theta(jt')$ ,  $j \geq 1$ , with  $t' > t$ , gives  $d > t$  with  $\Theta(d)$  within  $\varepsilon$  of 0, so  $\Theta(-t)$  is within  $\varepsilon$  of  $\Theta(d - t)$ . Hence  $\Theta([0, \infty))$  is dense in  $T_0$ , and so is every forward orbit  $\{\theta + \Theta(t) : t \geq t_0\}$  with  $\theta \in T_0$ .

In the coordinates  $\varphi_q = \theta_q / \log q$  the torus is  $\mathbb{R}^P / \mathbb{Z}^P$ , and by Kronecker's theorem  $T_0$  is the image of the smallest rational subspace  $L \subseteq \mathbb{R}^P$  containing  $\omega = (1/\log q)_{q \in P}$ . The annihilator of  $L$  is a rational subspace spanned by the integer vectors  $c$  with  $\sum_q c_q / \log q = 0$ . If it contained a nonzero vector supported on at most two primes  $a, b$ , it would contain an integer one, and  $\log a / \log b$  would be rational, which is false because no power of  $a$  is a power of  $b$ . So each  $\varphi_q$  is nonzero on  $L$  and any two are linearly independent on  $L$ . In particular the linear form  $\theta_b - \theta_a = (\log b)\varphi_b - (\log a)\varphi_a$  is nonzero on  $L$ , although it vanishes on the flow direction  $\omega$ .

**Lemma 4.4** (two walls). *If  $|P| \geq 4$ , there is  $x \in T_0$  with  $\theta_q(x) = 0$  for exactly two primes  $q \in P$ .*

*Proof.* For  $q \neq q'$  the function  $(u_1, u_2) \mapsto \varphi_q(u_1)\varphi_{q'}(u_2) - \varphi_q(u_2)\varphi_{q'}(u_1)$  is a nonzero polynomial on  $L \times L$ , since  $\varphi_q$  and  $\varphi_{q'}$  are independent on  $L$ . Rational points are dense in  $L \times L$ , so there are rational  $u_1, u_2 \in L$  at which all these determinants are nonzero, and a change of basis of their span multiplies every determinant by the same nonzero constant. Their span is a rational plane; with a  $\mathbb{Z}$ -basis  $e_1, e_2$  of its integer points,  $s \mapsto s_1e_1 + s_2e_2$  identifies  $\mathbb{R}^2/\mathbb{Z}^2$  with a subtorus  $S \subseteq T_0$ . On  $S$  the condition  $\theta_q = 0$  reads  $\langle \gamma_q, s \rangle \in \mathbb{Z}$  with  $\gamma_q = (\varphi_q(e_1), \varphi_q(e_2)) \in \mathbb{Z}^2$ , and these vectors are nonzero and pairwise non-parallel. Each set  $\{\theta_q = 0\} \cap S$  is therefore a family of finitely many disjoint parallel closed geodesics.

Two non-parallel closed geodesics of the flat torus meet. So the union of the families is a graph whose vertices are the points on at least two families, and every geodesic contains a vertex. Let  $k_v$  be the number of families through a vertex  $v$ . A geodesic with  $r$  vertices is cut into  $r$  edges, and geodesics of one family are disjoint, so  $E = \sum_v k_v$ . For  $q \neq q'$  the map  $s \mapsto (\langle \gamma_q, s \rangle, \langle \gamma_{q'}, s \rangle)$  is a covering of  $\mathbb{R}^2/\mathbb{Z}^2$  of degree  $|\det(\gamma_q, \gamma_{q'})|$ , so each component of the complement of these two families is an open parallelogram. Each face of the arrangement lies in such a parallelogram, where the other families are straight chords, so it is an open convex polygon with at least three sides. Euler's formula  $V - E + F = 0$  for the torus and  $2E \geq 3F$  give  $\sum_v (k_v - 3) \leq 0$ . The origin lies on all  $|P| \geq 4$  families, so some vertex has  $k_v = 2$ , and it is the required  $x$ .  $\square$

*Proof of Theorem 1.1.* Suppose  $\mathcal{D}_P = N/K$ . Fix  $p \in P$  and an index  $k$  with  $q_k = p$ . If  $|P| \geq 4$ , fix  $a, b$  and  $x$  as in Lemma 4.4, and put  $\eta = \min(1/10, \min_{c \neq a, b} \text{dist}(\theta_c(x), 0)) >$

0, distances being taken in  $\mathbb{R}/(\log c)\mathbb{Z}$ . Let  $\beta = \min_{q \neq p} \text{dist}(\theta_{k,q}, 0)$ , which is positive because  $t_k$  is a power of  $p$ . Choose  $T > 0$  such that  $2^\nu \geq K$ , where  $\nu$  is the number of crossings of  $\theta_k$  in  $(0, T]$ , and let  $g > 0$  be the least distance between two crossings of  $\theta_k$  in  $[0, T + 1]$ .

For  $n \geq 1$  let  $t_m = t_k p^n$ , so that  $\theta_m = \theta_k + \Theta(n \log p)$ . Write  $\delta_p = 0$  and  $\delta_q = n \log p - m_q \log q$  for  $q \neq p$ , with  $m_q$  the nearest integer to  $n \log p / \log q$ ; then  $\Theta(n \log p)$  has coordinates  $\delta_q \bmod \log q$ . By Dirichlet's simultaneous approximation theorem we may choose  $n$  with

$$\varepsilon = \max_q |\delta_q| < \min\left(\beta, \frac{g}{2}, \frac{\log 2}{2|P|}, \frac{\eta}{8}\right),$$

the last term only when  $|P| \geq 4$ . Then every  $m_q \geq 1$ , and the numbers  $\delta_q$  are pairwise distinct because  $p^n$  and the  $q^{m_q}$  are pairwise distinct.

*Chains.* The  $q$ -crossings of  $\theta_m$  are those of  $\theta_k$  shifted by  $-\delta_q$ . Every  $q$ -crossing  $s$  of  $\theta_k$  with  $q \neq p$  has  $|s| \geq \beta > |\delta_q|$ , so the shift preserves signs and matches the positive  $q$ -crossings of  $\theta_k$  and  $\theta_m$  in order. Cut the positive crossings of  $\theta_k$  wherever two consecutive ones are more than  $2\varepsilon$  apart, and call the pieces *chains*. Crossings in different chains are more than  $2\varepsilon$  apart and their shifts differ by less than  $2\varepsilon$ , so they keep their order. Hence  $W(\theta_m)$  is the concatenation of the chains of  $W(\theta_k)$  in the same order, each rearranged. Two crossings of one prime are at least  $\log 2$  apart, and any  $|P| + 1$  consecutive crossings of one chain lie within  $2\varepsilon|P| < \log 2$ ; they would belong to  $|P| + 1$  distinct primes, which is impossible. So a chain has at most  $|P|$  crossings, and they belong to distinct primes.

*Equal tails.* A chain meeting  $(0, T]$  is a single crossing, since the crossings in  $[0, T + 1]$  are at least  $g > 2\varepsilon$  apart and a crossing beyond  $T + 1$  is more than 1 away. So the first  $\nu$  blocks of the two chain decompositions agree, and Lemma 4.2(ii) gives  $|x_k - x_m| < 2^{-\nu} \leq 1/K$ . By Lemma 4.1,  $Kx_k$  and  $Kx_m$  are integers less than 1 apart, so  $x_k = x_m$ . Lemma 4.2(i) then gives  $f(C) = f(C')$  for every chain  $C$  of  $W(\theta_k)$  and its rearrangement  $C'$  in  $W(\theta_m)$ .

*Case  $|P| \leq 3$ .* Take  $q \neq p$ , and suppose  $\delta_q > 0$ ; the other sign is symmetric. The positive  $p$ -crossings of  $\theta_k$  are the numbers  $j \log p$  with  $j \geq 1$ , and its positive  $q$ -crossings are the numbers  $i \log q - \log t_k$  with  $i$  large. Since  $\log p / \log q$  is irrational, the differences  $j \log p - i \log q + \log t_k$  with  $i, j$  large are dense in  $\mathbb{R}$ . So some  $p$ -crossing  $j \log p$  and  $q$ -crossing  $s$  of  $\theta_k$  satisfy  $s - \delta_q < j \log p < s$ . Their shifts reverse their order, so they lie in one chain, which is genuinely rearranged in  $W(\theta_m)$ . It consists of at most three distinct primes, and Lemma 4.3 gives  $f(C) \neq f(C')$ , a contradiction.

*Case  $|P| \geq 4$ .* Put  $D = \delta_a - \delta_b$ , so  $0 < |D| < 2\varepsilon$ . Since  $\theta_b - \theta_a$  is a nonzero linear form on  $L$ , there is  $y \in L$  with  $|\theta_q(y)| < \eta/8$  for every  $q$  and  $\theta_b(y) - \theta_a(y)$  strictly between 0 and  $D$ . Let  $z$  be the point  $x + y + \Theta(-\eta/2)$  of  $T_0$ , with  $y$  read in the torus. The  $a$ -crossings of  $z$  are the times  $\eta/2 - \theta_a(y) + j \log a$ , so exactly one of them,  $\tau_a$ , lies in  $[-\eta/4, 5\eta/4]$ , and it lies in  $(\eta/4, 3\eta/4)$ ; the same holds for  $b$ , and  $\tau_a - \tau_b$  lies strictly between 0 and  $D$ . Every  $c$ -crossing of  $z$  with  $c \neq a, b$  is at distance at least  $7\eta/8$  from  $\eta/2$ , so it lies outside  $[-\eta/4, 5\eta/4]$ . Let  $U$  be the set of  $\theta \in T_0$  with exactly one  $a$ -crossing and one  $b$ -crossing in  $[-\eta/4, 5\eta/4]$ , both in  $(\eta/4, 3\eta/4)$ , with their difference  $\tau_a - \tau_b$  strictly between 0 and

$D$ , and with no other crossing in  $[-\eta/4, 5\eta/4]$ . Each condition is open, so  $U$  is an open subset of  $T_0$  containing  $z$ , and by density  $\theta_k + \Theta(t^*) \in U$  for some  $t^* > T + 2$ .

Then  $\theta_k$  has an  $a$ -crossing  $t^* + \tau_a$  and a  $b$ -crossing  $t^* + \tau_b$  less than  $|D| < 2\varepsilon$  apart, and every other crossing of  $\theta_k$  is more than  $\eta/2 > 4\varepsilon$  away from both. So these two crossings form a chain  $C$ . Their shifts differ by  $(\tau_a - \tau_b) - D$ , whose sign is opposite to that of  $\tau_a - \tau_b$ , so  $C'$  is  $C$  reversed. But  $f(ab) - f(ba) = b - a \neq 0$ , a contradiction.  $\square$

**The second agent's corrections.** The first write-up of this proof was checked step by step by a second AI agent, who found it correct and closed five small gaps, all incorporated above. The set  $U$  must be defined by open conditions, including the strict inequality on  $\tau_a - \tau_b$ , since a prescribed value of that difference is a closed condition. The order of choices must be explicit: first  $a, b, x$  and  $\eta$ , then  $k, T$  and  $g$ , then  $n$ , which fixes  $\varepsilon$  and  $D$ , then  $y, z$  and  $U$ , and finally  $t^*$ . The faces of the arrangement in Lemma 4.4 need the covering argument to be convex. For  $|P| \leq 3$  the existence of a genuinely rearranged chain needs the density argument, and the  $i$ th  $q$ -crossing of  $\theta_k$  corresponds to the  $i$ th  $q$ -crossing of  $\theta_m$ . And the form  $\theta_b - \theta_a$  vanishes on the flow direction; it is nonzero on  $L$  because  $L$  is larger than the flow line.

**Why the two walls are needed.** A shorter route would take, at a return of the orbit near the origin of  $T_0$ , the chain of crossings there. It fails when  $\dim T_0 = 2$ , which happens exactly when the numbers  $1/\log q$  span a two-dimensional space over  $\mathbb{Q}$ ; nothing known excludes this, and Baker's theorem concerns linear forms in logarithms and says nothing about their reciprocals. In that case every small vector of  $L$ , in the coordinates  $\theta_q$ , has the form  $\alpha(1, \dots, 1) + \beta w$  for one fixed  $w$ , so near the origin all the forms  $\theta_b - \theta_a$  are proportional. The chain at such a return contains every prime of  $P$ , and the only rearrangement it can undergo is the full reversal. Full reversals can collide:

$$f(43, 13, 3, 53) = f(53, 3, 13, 43) = 2280,$$

and among the sets of four primes below 100 there are five such collisions. Lemma 4.4 avoids the question by producing a point on exactly two walls for every  $P$  with  $|P| \geq 4$ ; a point on three walls would also serve, by Lemma 4.3. Swapping one adjacent pair always changes  $f$ , since  $f(\sigma) - f(\sigma') = (\sigma_{i+1} - \sigma_i)\sigma_{i+2} \cdots \sigma_n$  when  $\sigma'$  swaps  $\sigma_i$  and  $\sigma_{i+1}$ ; a rearranged chain of four or more primes can differ from its image by more than one swap. Nor can the argument stop at the first rearranged chain: for four primes that chain can collide, as  $f(5, 7, 3, 2) = f(7, 2, 3, 5) = 51$  shows, which is why Lemma 4.2(i) is applied to every chain.

**Exact illustrations.** The program `two_tail.py` of the [distinct-height programs](#) checks the finite facts above in exact arithmetic; the proof uses none of them. Among the sets of two or three primes below 200,  $f$  has no collision; among the 2380 sets of four primes below 60 it has 30. For  $P = \{2, 3, 5, 7\}$ ,  $t_k = 2^{60}$  and the return  $n = 283526$ , with  $\varepsilon \approx 0.0117$ , the first 4000 crossings correspond exactly, every chain stays in place, and of the 39 rearranged chains 35 are transpositions; none has equal  $f$ -values. For the closer return  $n = 2547450$ , with  $\varepsilon \approx 0.0034$ , the difference  $x_k - x_m$  computed exactly has  $\log_{10} |x_k - x_m| = -501.1234$ , the value predicted by its first rearranged chain, (5, 2) against (2, 5) after 1005 letters. On 400 random arrangements of four to six line families

on  $\mathbb{R}^2/\mathbb{Z}^2$ , some with non-primitive normal vectors,  $\sum_v (k_v - 3) \leq -4$ ; the square with both diagonals attains 0 and still has a vertex on exactly two lines.

## 4.2 Dyadic blocks and affine maps

The blocks are the intervals  $(p^a, p^{a+1}]$  for  $a \geq 0$ . Put  $P_a = H_P(p^a)$  and

$$Y_a = P_a \sum_{t_k > p^a} \frac{1}{H_P(t_k)}.$$

The *type*  $\tau_a$  of block  $a$  is the sequence  $(r_1, \dots, r_v)$  of the primes whose powers lie strictly inside the block, listed in increasing order of those powers; the empty sequence is written  $\emptyset$ .

**Lemma 4.5** (blocks, tails and integrality). *Let  $P$  be a finite set of at least two primes with least element  $p$ .*

- (i) *Block  $a$  ends with the jump at  $p^{a+1}$  and contains at most one power of each prime  $q \in P \setminus \{p\}$ , lying strictly inside it. The power  $q^f$  with  $f \geq 1$  lies in block  $\lfloor f \log_p q \rfloor$ .*
- (ii) *If block  $a$  has type  $\tau = (r_1, \dots, r_v)$ , then  $Y_a = G_\tau(Y_{a+1})$ , where*

$$G_\tau(y) = \frac{\mu_\tau + y}{b_\tau}, \quad b_\tau = pr_1 \cdots r_v, \quad \mu_\tau = 1 + b_\tau \sum_{j=1}^v \frac{1}{r_1 \cdots r_j}.$$

*Each  $G_\tau$  is increasing with slope  $1/b_\tau \leq 1/p$  and maps  $[0, 1/(p-1)]$  into itself, and  $0 < Y_a \leq 1/(p-1)$ .*

- (iii) *If  $\mathcal{D}_P = N/K$  with integers  $N$  and  $K \geq 1$ , then  $KY_a$  is an integer for every  $a \geq 0$ .*

Ordinary proof. Its case  $P = \{2, 3, 5\}$  is checked in Lean as part of the proof of Theorem 1.2.

*Proof.* (i) The endpoint  $p^{a+1}$  is a jump. Consecutive powers of  $q$  have ratio  $q > p$ , so the block contains at most one of them, and no power of  $q$  equals a power of  $p$ . The power  $q^f$  lies in  $(p^a, p^{a+1})$  exactly when  $a < f \log_p q < a + 1$ .

(ii) The jumps of block  $a$  are the interior powers, with heights  $P_a r_1, P_a r_1 r_2, \dots, P_a r_1 \cdots r_v$ , followed by the jump at  $p^{a+1}$ , whose height is  $P_{a+1} = b_\tau P_a$ . Splitting off the block,

$$Y_a = \sum_{j=1}^v \frac{1}{r_1 \cdots r_j} + \frac{1}{b_\tau} + \frac{P_a}{P_{a+1}} Y_{a+1} = \frac{\mu_\tau + Y_{a+1}}{b_\tau}.$$

Each jump multiplies the height by at least  $p$ , so  $0 < Y_a \leq \sum_{k \geq 1} p^{-k} = 1/(p-1)$ . Since every  $r_j > p$ ,  $G_\tau(1/(p-1)) \leq \sum_{j=1}^v p^{-j} + p^{-v}/(p-1) = 1/(p-1)$ , and  $G_\tau(0) > 0$ .

(iii) If  $t \leq p^a$ , each exponent in  $H_P(t)$  is at most the corresponding exponent in  $P_a$ , so  $H_P(t) \mid P_a$ . Hence  $P_a \mathcal{D}_P - Y_a = P_a(1 + \sum_{t_k \leq p^a} H_P(t_k)^{-1})$  is an integer, and so is  $KY_a = P_a N - K(P_a \mathcal{D}_P - Y_a)$ .  $\square$

### 4.3 Proof of Theorem 1.2

Take  $P = \{2, 3, 5\}$ . The possible types, written as words, are  $\emptyset$ , 3, 5, 35 and 53, and Lemma 4.5(ii) gives

| $\tau$     | $\emptyset$ | 3 | 5  | 35 | 53 |
|------------|-------------|---|----|----|----|
| $b_\tau$   | 2           | 6 | 10 | 30 | 30 |
| $\mu_\tau$ | 1           | 3 | 3  | 13 | 9  |

For instance  $\mu_{35} = 1 + 30(\frac{1}{3} + \frac{1}{15}) = 13$ . The first twelve blocks have types  $\emptyset, 3, 5, 3, 53, \emptyset, 35, 3, \emptyset, 53, \emptyset, 35$ : block 4 = (16, 32] contains  $25 < 27$ , and block 6 = (64, 128] contains  $81 < 125$ .

*Proof of Theorem 1.2. Bounds.* By Lemma 4.5(ii),  $Y_a \geq \mu_{\tau_a}/b_{\tau_a} \geq 3/10$ , the minimum being attained by the types 5 and 53, and  $Y_a \leq 1$ . By Lemma 4.5(i), the power  $5^f$  lies in block  $\lfloor f \log_2 5 \rfloor$ , and  $2 < \log_2 5 < 3$ . So the blocks containing a power of 5 begin with block 2 and are two or three apart, and every three consecutive blocks include one. On  $[0, 1]$  the three maps whose block contains a power of 5 are at most  $G_{35}(1) = 7/15$ , and  $G_\emptyset(y) - G_3(y) = y/3 \geq 0$ . If  $a + i$ , with  $0 \leq i \leq 2$ , is the first block from  $a$  onwards that contains a power of 5, then  $Y_{a+i} \leq 7/15$ , and each of the at most two maps leading back from  $Y_{a+i}$  to  $Y_a$  is  $G_\emptyset$  or  $G_3$ , hence at most  $G_\emptyset$ . Therefore

$$\frac{3}{10} \leq Y_a \leq G_\emptyset(G_\emptyset(7/15)) = \frac{13}{15} \quad (a \geq 0).$$

*Separation.* The five maps send  $I = [3/10, 13/15]$  to the intervals

| $\tau$      | 53                                 | 5                                 | 35                                   | 3                                | $\emptyset$                      |
|-------------|------------------------------------|-----------------------------------|--------------------------------------|----------------------------------|----------------------------------|
| $G_\tau(I)$ | $[\frac{31}{100}, \frac{74}{225}]$ | $[\frac{33}{100}, \frac{29}{75}]$ | $[\frac{133}{300}, \frac{104}{225}]$ | $[\frac{11}{20}, \frac{29}{45}]$ | $[\frac{13}{20}, \frac{14}{15}]$ |

in increasing order, and consecutive intervals are separated by  $1/900$ ,  $17/300$ ,  $79/900$  and  $1/180$ . Since  $Y_{a+1} \in I$ , the value  $Y_a$  lies in  $G_{\tau_a}(I)$  and in no other image. So  $Y_a$  determines  $\tau_a$ , then  $Y_{a+1} = b_{\tau_a} Y_a - \mu_{\tau_a}$ , and by induction the whole word  $\tau_a \tau_{a+1} \dots$ .

*Return.* Suppose  $\mathcal{D}_{\{2,3,5\}} = N/K$  with integers  $N$  and  $K \geq 1$ . By Lemma 4.5(iii), every  $KY_a$  is an integer in  $[3K/10, 13K/15]$ . There are finitely many such integers, so  $Y_a = Y_{a'}$  for some  $a < a'$ . Then  $\tau_{a+j} = \tau_{a'+j}$  for every  $j \geq 0$ , and the word  $(\tau_a)$  is eventually periodic, with period  $\ell = a' - a$ .

*Contradiction.* Put  $\alpha = \log 2 / \log 3$ , which is irrational. By Lemma 4.5(i),  $3^e$  lies in block  $a$  exactly when  $a\alpha < e < (a+1)\alpha$ , so block  $a$  contains  $\lfloor (a+1)\alpha \rfloor - \lfloor a\alpha \rfloor$  powers of 3, a number recorded by  $\tau_a$ . Summing over one period, the number  $c = \lfloor (a+\ell)\alpha \rfloor - \lfloor a\alpha \rfloor$  does not depend on  $a$  once  $a$  is large, hence  $\lfloor (a+n\ell)\alpha \rfloor = \lfloor a\alpha \rfloor + nc$  for every  $n \geq 1$ . Dividing by  $n$  and letting  $n \rightarrow \infty$  gives  $\ell\alpha = c$ , that is  $2^\ell = 3^c$ , which is impossible.  $\square$

The Lean proof follows this route. It identifies  $H(t)$  with the running least common multiple, shows that the sum counts each distinct value once, identifies the contribution of each dyadic block with one of the five pairs  $(b_\tau, \mu_\tau)$ , and proves that every three consecutive blocks contain a power of 5 and that the blocks containing a power of 3 are

not eventually periodic. It then applies an [irrationality criterion for block series with these five pairs](#), whose proof contains the bounds, the separation and the return above.

The delicate step is the upper bound. Because  $G_3(y) = G_\emptyset(y/3)$  and  $G_{53}(y) = G_5(y/3)$ , the images of the empty block and of the block with one power of 3 are disjoint exactly when the largest tail is less than three times the smallest, and the same holds for the blocks 53 and 5. The gaps  $1/180$  and  $1/900$  are  $\frac{1}{6}$  and  $\frac{1}{30}$  of  $3 \cdot \frac{3}{10} - \frac{13}{15} = \frac{1}{30}$ . With the trivial bound  $Y_a \leq 1$  in place of  $13/15$  these images overlap; the spacing of the powers of 5 is what makes the difference. Over the first 3000 blocks the tails lie in  $[0.3186, 0.7799]$ , well inside  $I$ .

Grouping the jumps into dyadic blocks matters. One jump at a time, the tails  $x_k = (q_1 \cdots q_k) \sum_{j>k} (q_1 \cdots q_j)^{-1}$  satisfy  $x_k = (1 + x_{k+1})/q_{k+1}$  and  $1/4 \leq x_k \leq 1$ , and the images of  $[1/4, 1]$  under  $x \mapsto (1 + x)/2$  and  $x \mapsto (1 + x)/3$  overlap on  $[5/8, 2/3]$ . A tail before a jump by 2 cannot be told from a tail before a jump by 3 from these bounds alone.

In the form (4),  $\mathcal{D}_P$  is a Cantor series with every digit equal to 1 and every base in  $P$ . The criteria of Erdős–Straus [17, Theorem 2.1] and Hančl–Tijdeman [18, Theorem 3.1], recalled in Section 7, assume that each digit divided by the product of the last two bases in its denominator tends to 0. Here that quotient is at least  $(\max P)^{-2}$ , so neither criterion applies.

#### 4.4 A decoding criterion for every finite set of primes

For  $\{2, 3, 5\}$  a single interval contained every tail. For other sets the bound is allowed to depend on how recently each prime appeared. Put  $Q = P \setminus \{p\}$  and, for  $q \in Q$ , let  $g_q$  be the integer with  $p^{g_q} < q < p^{g_q+1}$ . By Lemma 4.5(i), two consecutive blocks containing a power of  $q$  have indices  $\lfloor f \log_p q \rfloor$  and  $\lfloor (f+1) \log_p q \rfloor$ , which differ by  $g_q$  or  $g_q + 1$ .

These constraints define a finite automaton. Its states are the tuples  $s = (s_q)_{q \in Q}$  with  $1 \leq s_q \leq g_q + 1$ ; the entry  $s_q$  counts the blocks since the last block containing a power of  $q$ . A transition from  $s$  is a pair  $(\tau, s')$  in which  $\tau$  is an ordering of a set  $T \subseteq Q$  with

$$\{q : s_q = g_q + 1\} \subseteq T \subseteq \{q : s_q \geq g_q\},$$

and  $s'_q = 1$  for  $q \in T$ ,  $s'_q = s_q + 1$  for  $q \notin T$ . A path from  $s$  is a sequence of transitions  $(u_1, s_1), (u_2, s_2), \dots$  with  $(u_1, s_1)$  a transition from  $s$  and each  $(u_{i+1}, s_{i+1})$  a transition from  $s_i$ . The value of an infinite path is the limit of  $G_{u_1} \circ \cdots \circ G_{u_n}(y)$  as  $n \rightarrow \infty$ ; by Lemma 4.5(ii) these numbers lie, for  $y \in [0, 1/(p-1)]$ , in nested intervals of length at most  $p^{-n}/(p-1)$ , so the limit exists and does not depend on  $y$ .

Every prime  $q \in Q$  lies in block  $g_q$ . For  $a > \max_q g_q$  let  $s_a$  record, for each  $q$ , the number  $a - \ell$ , where  $\ell < a$  is the index of the last block before  $a$  containing a power of  $q$ . The next such block has index  $\ell + g_q$  or  $\ell + g_q + 1$ , and it is at least  $a$ . Hence  $s_a$  is a state, block  $a$  contains a power of  $q$  when  $(s_a)_q = g_q + 1$  and does not when  $(s_a)_q < g_q$ , and  $(\tau_a, s_{a+1})$  is a transition from  $s_a$ . Since  $Y_a = G_{\tau_a} \circ \cdots \circ G_{\tau_{a+n-1}}(Y_{a+n})$  with  $Y_{a+n} \in [0, 1/(p-1)]$ , the tail  $Y_a$  is the value of the actual path  $(\tau_a, s_{a+1}), (\tau_{a+1}, s_{a+2}), \dots$  from  $s_a$ .

**Theorem 4.6** (a decoding criterion). *Let  $P$  be a finite set of at least two primes, and for every state  $s$  let  $J(s)$  be a closed interval containing the values of all infinite paths from  $s$ . For  $k \geq 1$  and each type  $\tau$ , let  $U_\tau$  be the union of the intervals  $G_{u_1} \circ \cdots \circ G_{u_k}(J(s_k))$  over all states  $s$  and*

all paths  $(u_1, s_1), \dots, (u_k, s_k)$  from  $s$  with  $u_1 = \tau$ . If the sets  $U_\tau$  are pairwise disjoint, then  $\mathcal{D}_P$  is irrational.

Ordinary proof. Lean checks only the case  $P = \{2, 3, 5\}$  with  $k = 1$  and the single interval used in Section 4.3.

*Proof.* Let  $a > \max_q g_q$ . The first  $k$  transitions of the actual path from  $s_a$  form a path of length  $k$  with first type  $\tau_a$ , and  $Y_a = G_{\tau_a} \circ \dots \circ G_{\tau_{a+k-1}}(Y_{a+k})$ , where  $Y_{a+k} \in J(s_{a+k})$  because it is the value of an infinite path from  $s_{a+k}$ . Thus  $Y_a \in U_{\tau_a}$ , and by disjointness  $\tau_a$  is the only type with this property. So  $Y_a$  determines  $\tau_a$ , then  $Y_{a+1} = b_{\tau_a} Y_a - \mu_{\tau_a}$ , and by induction the word  $\tau_a \tau_{a+1} \dots$ .

Suppose  $\mathcal{D}_P = N/K$ . By Lemma 4.5(ii) and (iii), the numbers  $KY_a$  are integers in  $[0, K/(p-1)]$ , so  $Y_a = Y_{a'}$  for some  $\max_q g_q < a < a'$ , and the word  $(\tau_a)$  is eventually periodic. Fix  $q \in Q$  and put  $\beta = \log p / \log q$ , which lies in  $(0, 1)$  and is irrational because no power of  $p$  equals a power of  $q$ . By Lemma 4.5(i), block  $a$  contains  $\lfloor (a+1)\beta \rfloor - \lfloor a\beta \rfloor$  powers of  $q$ , a number recorded by  $\tau_a$ . As in the proof of Theorem 1.2, eventual periodicity with period  $\ell$  forces  $\ell\beta \in \mathbb{Z}$ , a contradiction.  $\square$

Intervals  $J(s)$  with the required property are easy to produce. The constant interval  $[0, 1/(p-1)]$  qualifies by Lemma 4.5(ii). If a family  $J$  qualifies, so does any family  $J'$  with

$$J'(s) \supseteq G_\tau(J(s')) \quad \text{for every transition } (\tau, s') \text{ from } s,$$

because an infinite path from  $s$  is a transition  $(\tau, s')$  followed by an infinite path from  $s'$ . Starting from the constant family and repeatedly taking the smallest interval containing these images, with endpoints rounded outwards, therefore gives valid intervals after any number of rounds. The tightest choice is  $J(s) = [L(s), U(s)]$ , where  $L(s)$  and  $U(s)$  are the least and greatest values of the infinite paths from  $s$ . They satisfy  $L(s) = \min G_\tau(L(s'))$  and  $U(s) = \max G_\tau(U(s'))$  over the transitions from  $s$ , and these equations have unique solutions because each  $G_\tau$  contracts by at least the factor  $1/p$ .

For  $\{2, 3, 5\}$  the constant interval  $I = [3/10, 13/15]$  has the required property: every path takes a transition with  $5 \in T$  within any three steps, which is the only fact the bounds used. With  $k = 1$  the criterion is then the proof given above.

## 4.5 Certificates for three and four primes

**Corollary 4.7** (three and four primes up to 31). *Let  $P$  be a set of primes, each at most 31.*

- (i) *If  $|P| = 3$ , then  $\mathcal{D}_P$  is irrational.*
- (ii) *If  $|P| = 4$  and  $P$  is not one of the 38 sets listed below, then  $\mathcal{D}_P$  is irrational.*

Ordinary proof by Theorem 4.6, applied through an exact finite computation for each set; no Lean proof.

*Proof.* For each set, the program `certificate.py` of the [distinct-height programs](#) builds the automaton, computes intervals  $J(s)$  by 80 rounds of the operation above from  $[0, 1/(p-1)]$ , with endpoints rounded outwards to multiples of  $2^{-96}$ , and tests the disjointness of the sets  $U_\tau$  for  $k = 1, 2, \dots$ . Every comparison is made in exact rational arithmetic, and each  $U_\tau$  is computed as a union of intervals merged where they meet.

The test passes for each of the 165 sets of three primes with some  $k \leq 4$ , and for each set of four primes outside the list with some  $k \leq 5$ . Theorem 4.6 then gives irrationality.  $\square$

For three primes the least passing depth is  $k = 1$  for 141 sets and  $k = 2$  for the 17 sets  $\{2, 3, 11\}$ ,  $\{2, 3, 13\}$ ,  $\{2, 3, 17\}$ ,  $\{2, 3, 19\}$ ,  $\{2, 3, 23\}$ ,  $\{2, 3, 29\}$ ,  $\{2, 3, 31\}$ ,  $\{2, 5, 7\}$ ,  $\{2, 11, 17\}$ ,  $\{2, 13, 19\}$ ,  $\{2, 17, 19\}$ ,  $\{2, 19, 29\}$ ,  $\{2, 19, 31\}$ ,  $\{3, 13, 17\}$ ,  $\{3, 17, 23\}$ ,  $\{3, 23, 31\}$  and  $\{3, 29, 31\}$ . It is  $k = 3$  for  $\{2, 3, 7\}$ ,  $\{2, 11, 13\}$ ,  $\{2, 17, 29\}$  and  $\{2, 19, 23\}$ , and  $k = 4$  for  $\{2, 17, 31\}$ ,  $\{2, 23, 29\}$  and  $\{2, 29, 31\}$ . The smallest gap between sets  $U_\tau$  of different types, among all passing triples, is about  $9.4 \cdot 10^{-6}$ , for  $\{2, 17, 31\}$ ; for  $\{2, 3, 5\}$  it is  $27983/963915 \approx 0.029$  at depth 1, much larger than the gap  $1/900$  of the single interval used above. For four primes the least passing depth is 1, 2, 3, 4, 5 for 184, 34, 28, 42 and 4 sets respectively, and the smallest gap is about  $1.8 \cdot 10^{-8}$ , for  $\{3, 7, 17, 19\}$ . All these gaps are far above the rounding scale  $2^{-96}$ . The 38 four-prime sets with no passing depth  $k \leq 5$  are  $\{2, 3, 5, 7\}$ ,  $\{2, 3, 5, 11\}$ ,  $\{2, 3, 5, 13\}$ ,  $\{2, 3, 5, 17\}$ ,  $\{2, 3, 5, 19\}$ ,  $\{2, 3, 5, 23\}$ ,  $\{2, 3, 7, 11\}$ ,  $\{2, 3, 7, 13\}$ ,  $\{2, 3, 7, 17\}$ ,  $\{2, 3, 7, 23\}$ ,  $\{2, 3, 11, 13\}$ ,  $\{2, 3, 11, 17\}$ ,  $\{2, 3, 11, 19\}$ ,  $\{2, 3, 11, 29\}$ ,  $\{2, 3, 11, 31\}$ ,  $\{2, 3, 13, 19\}$ ,  $\{2, 3, 13, 29\}$ ,  $\{2, 3, 17, 19\}$ ,  $\{2, 3, 17, 29\}$ ,  $\{2, 3, 29, 31\}$ ,  $\{2, 5, 7, 11\}$ ,  $\{2, 5, 17, 29\}$ ,  $\{2, 5, 23, 29\}$ ,  $\{2, 5, 23, 31\}$ ,  $\{2, 11, 13, 19\}$ ,  $\{2, 11, 17, 23\}$ ,  $\{2, 11, 17, 29\}$ ,  $\{2, 11, 23, 31\}$ ,  $\{2, 13, 19, 31\}$ ,  $\{2, 19, 23, 31\}$ ,  $\{3, 5, 13, 17\}$ ,  $\{3, 5, 17, 19\}$ ,  $\{3, 5, 23, 31\}$ ,  $\{3, 7, 11, 13\}$ ,  $\{3, 19, 23, 29\}$ ,  $\{3, 19, 23, 31\}$ ,  $\{3, 19, 29, 31\}$  and  $\{7, 13, 29, 31\}$ . Twenty of them contain both 2 and 3.

A second computation replaces the rounded intervals by the exact extremes  $[L(s), U(s)]$ , found by policy iteration and checked by substitution into their defining equations. It returns the same least depth for all 495 sets. A third check compares the automaton with the first 400 actual blocks of every set: every actual word is a path, and every exact enclosure of an actual tail lies in the interval of its state. This comparison is a consistency check only. The whole triple computation takes about three seconds, and the four-prime computation about seventy seconds, on one laptop; the receipts record every depth and gap.

For  $\{2, 3, 5, 7\}$  the automaton has 18 states and 16 types, and the test fails through depth 7. Its tail intervals range from about 0.188 to 0.785, so the largest exceeds three times the smallest, and the images of the empty block and of the block with one power of 3 overlap. The largest overlap between two first types is always of this pair. For each depth  $k$  the table gives the number of merged intervals in the sets  $U_\tau$  and the length of that largest overlap.

| $k$ | intervals | largest overlap     |
|-----|-----------|---------------------|
| 1   | 16        | $2.4 \cdot 10^{-2}$ |
| 2   | 58        | $1.8 \cdot 10^{-3}$ |
| 3   | 386       | $1.0 \cdot 10^{-4}$ |
| 4   | 1413      | $4.6 \cdot 10^{-5}$ |
| 5   | 8395      | $7.6 \cdot 10^{-7}$ |
| 6   | 38494     | $2.4 \cdot 10^{-7}$ |
| 7   | 193905    | $1.8 \cdot 10^{-8}$ |

At depth 7 two further pairs of types still overlap,  $(5, 7, 3)$  with  $(7)$  by about  $1.5 \cdot 10^{-9}$  and  $(3, 5, 7)$  with  $(3, 7)$  by about  $5.8 \cdot 10^{-10}$ .

The second pair explains why no depth succeeds. The two-block words  $(5, 7, 3), \emptyset$  and  $(7), (3, 5)$  have the same map,

$$G_{(5,7,3)} \circ G_{\emptyset}(y) = G_{(7)} \circ G_{(3,5)}(y) = \frac{103 + y}{420},$$

which is the collision  $f(5, 7, 3, 2) = f(7, 2, 3, 5) = 51$  of Section 4.1 read in blocks: the letters are  $5, 7, 3, 2, 2$  and  $7, 2, 3, 5, 2$ , and  $f = 103$  for both. From the states  $(1, 2, 2)$  and  $(1, 2, 3)$ , written as  $(s_3, s_5, s_7)$ , both words continue by the blocks  $(3), (5, 7)$  and end in the state  $(2, 1, 1)$ , so the four-block words have the same map and the same end state. Hence the sets  $U_{(5,7,3)}$  and  $U_{(7)}$  of Theorem 4.6 meet at every depth, for any valid choice of intervals, and the criterion cannot decide  $\{2, 3, 5, 7\}$  with this automaton. Both four-block words occur in the actual block sequence, 12 and 40 times among the first 20000 blocks. A single tail therefore cannot determine its block. The proof of Theorem 1.1 compares two tails, and there one run of two primes suffices.

#### 4.6 Where the argument stops for the repeated sums

For  $P = \{2, 3, 5\}$  the tails of the two sums obey recurrences with the same bases. With  $P_a = H(2^a)$  as in Section 6,  $b_{\tau_a} = P_{a+1}/P_a = b_a$ , and Lemma 4.5(ii) reads  $Y_{a+1} = b_a Y_a - \mu_{\tau_a}$ , while Theorem 6.2 gives  $X_{a+1} = b_a X_a - m_a$  for the repeated sum. The distinct-height sum subtracts one of five numbers  $\mu_{\tau} \leq 13$ . The repeated sum subtracts the weighted count  $m_a$  of the smooth numbers in the shell, and  $m_a = \Theta((a+1)^2)$  by Lemma 6.3. Three steps of the proof above then have no counterpart. The tails are unbounded, since  $X_a > m_a/30$ , so the integers  $d_a = BX_a$  of Theorem 7.3 are unbounded and need not repeat. The maps  $y \mapsto (m_a + y)/b_a$  depend on  $a$ , so even a repeated value  $d_a = d_{a'}$  would not align the two futures. And there is no finite list of maps whose images could be separated. The residue criterion of Section 8 replaces the return to an earlier state by a comparison of residues with the quadratic bound on  $d_a$ . The example after Proposition 11.2 shows that the radices and quadratic growth alone do not decide the question: numerators  $\widehat{m}_a$  of quadratic size with the same radices give the rational value  $\sum_a \widehat{m}_a / P_{a+1} = 9$ .

The two-tail proof of Theorem 1.1 stops at the same place. Over the prime powers the repeated sum is  $\mathcal{R}_P = \sum_{k \geq 0} c_k / Q_k$ , where  $c_k$  counts the smooth integers in  $[t_k, t_{k+1})$ , with  $t_0 = 1$ . The two tails after  $t_k$  and after  $t_k p^n$  still have words that differ by rearranging chains, and a rational value still makes the scaled tails integers of one denominator. But the scaled tails are unbounded, as the  $X_a$  are, and Lemma 4.2 needs every tail in  $(0, 1)$ : the integer  $f(\sigma_i) - f(\sigma'_i)$  is shown to vanish because it equals a difference of two tails. With unbounded tails that step fails, the digits  $c_k$  move with the rearrangement as well, and equality of two tails no longer forces agreement on each rearranged run. The irrationality of  $\mathcal{R}_P$  for  $|P| \geq 3$  remains open.

**The single-prime sub-sums.** The comparison of two tails does close on part of the catalogue series. For  $p \in P$  put

$$E_p = \sum_{\alpha \geq 0} \frac{1}{L(p^\alpha)} = \sum_{\alpha \geq 0} \frac{1}{H_P(p^\alpha)},$$

the sub-sum of  $\mathcal{R}_P$  over the terms at which the smooth integer is a power of  $p$ . For  $P = \{p, q\}$  with  $p < q$  these are the numbers  $A = E_p$  and  $1 + B_* = E_q$  of the proof of Theorem 1.3.

**Proposition 4.8** (sub-sums and the catalogue sum). *Let  $P$  be a finite set of primes with  $|P| \geq 2$ .*

- (i)  $\mathcal{D}_P = \sum_{q \in P} E_q - (|P| - 1)$ .
- (ii) If  $P = \{p, q\}$ , then  $H_P(p^i q^j) = H_P(p^i) H_P(q^j)$  for all  $i, j \geq 0$ , and  $\mathcal{R}_P = E_p E_q$ .
- (iii) If  $P = \{p, q, r\}$ , then  $H_P(p^i q^j r^k) = H_P(p^i) H_P(q^j) H_P(r^k) p^{\kappa_p(j,k)} q^{\kappa_q(i,k)} r^{\kappa_r(i,j)}$ , where  $\kappa_p(j, k) = \lfloor j \log_p q + k \log_p r \rfloor - \lfloor j \log_p q \rfloor - \lfloor k \log_p r \rfloor \in \{0, 1\}$  and  $\kappa_q, \kappa_r$  are defined in the same way. For a general finite  $P$  the corresponding exponents lie in  $\{0, \dots, |P| - 2\}$ .

Ordinary proof; no Lean proof. The `two_tail.py` program of the [distinct-height programs](#) also checks the identities in exact arithmetic on finite ranges.

*Proof.* (i) Each term  $1/Q_k$  of  $\mathcal{D}_P$  with  $k \geq 1$  is a term of  $E_{q_k}$  and of no other  $E_{q_i}$ , and each  $E_q$  also contains the term 1. (ii) and (iii) The exponent of  $p$  in  $H_P(\prod_{q \in P} q^{e_q})$  is  $e_p + \lfloor \sum_{q \neq p} e_q \log_p q \rfloor$ , and its exponent in  $\prod_{q \in P} H_P(q^{e_q})$  is  $e_p + \sum_{q \neq p} \lfloor e_q \log_p q \rfloor$ . For  $m$  nonnegative reals the floor of the sum exceeds the sum of the floors by an integer in  $\{0, \dots, m - 1\}$ . With two primes  $m = 1$ , so the heights factor, and summing  $H_P(p^i)^{-1} H_P(q^j)^{-1}$  over  $i, j \geq 0$  gives  $E_p E_q$ .  $\square$

With three primes  $\mathcal{R}_P$  is therefore the product of the three single-prime sums twisted by the carries  $\kappa$ , and the plain product  $E_p E_q E_r$  differs from it.

**Theorem 4.9** (the single-prime sub-sums). *Let  $P$  be a finite set of primes with  $|P| \geq 2$  and let  $p \in P$ . Then  $E_p$  is irrational.*

Ordinary proof, given below. It was found within this project on 26 September 2026, in an attempt on  $\mathcal{R}_P$ , and checked step by step by a second, independent AI agent, whose corrections are incorporated; it has had no human review. It has no Lean proof, so Comparator does not apply. For  $|P| = 2$  the statement follows from a Hecke–Mahler transcendence theorem. For  $|P| \geq 3$  we have not searched the literature.

For  $P = \{p, q\}$ ,  $E_p$  is  $A$  or  $1 + B_*$  in the proof of Theorem 1.3, an affine function of the Hecke–Mahler value  $F_\theta(1/p, 1/q)$  with nonzero slope; for  $p < q$ ,  $E_p = p/(p - 1) - (q - 1)F_\theta(1/p, 1/q)$  by (3). It is transcendental by the theorem of Bugeaud and Laurent [14, Theorem 1.1], in the case due to Loxton and van der Poorten [15, Theorem 8, p. 40]. We cite these sources for this case and have not re-verified its earlier attribution history.

*Proof.* For a word  $u$  over  $P$  put  $V_p(u) = \sum_{i: u_i = p} (u_1 \cdots u_i)^{-1}$ , and for a finite word  $\sigma = \sigma_1 \cdots \sigma_n$  put  $f_p(\sigma) = \sum_{t: \sigma_t = p} \sigma_{t+1} \cdots \sigma_n$ . Then  $V_p(\sigma u) = (f_p(\sigma) + V_p(u))/\Pi(\sigma)$ . With

$$y_k = V_p(q_{k+1}q_{k+2}\cdots),$$

$$E_p = 1 + \sum_{k: q_k=p} \frac{1}{Q_k}, \quad Q_k E_p = Q_k + \sum_{j \leq k, q_j=p} \frac{Q_k}{Q_j} + y_k.$$

In a suffix  $u$  of  $q_1q_2\cdots$  the  $j$ th letter  $p$  sits at a prefix product at least  $p^j$ , strictly larger once a letter other than  $p$  has occurred, so  $0 < V_p(u) < 1/(p-1) \leq 1$ . Hence, if  $E_p = N/K$ , every  $Ky_k$  is an integer, and Lemma 4.2 holds with  $V_p$  and  $f_p$  in place of  $V$  and  $f$ , with the same proof. For a word of distinct primes in which  $p$  occurs once,  $f_p$  is the product of the letters after  $p$ ; by unique factorisation, two orderings of one set of distinct primes containing  $p$  have the same  $f_p$  exactly when  $p$  has the same followers in both.

Since  $\theta_p - \theta_q$  is a nonzero linear form on  $L$  for every  $q \neq p$  (Section 4.1), there is  $v \in L$  with  $|\theta_q(v)| \leq 1$  for all  $q$  and  $\theta_q(v) \neq \theta_p(v)$  for all  $q \neq p$ . Write  $v_q = \theta_q(v)$  and  $\mu = \min_{q \neq p} |v_p - v_q| > 0$ .

Suppose  $E_p = N/K$ . Fix  $k$  with  $q_k = p$ , define  $\beta, T, \nu$  and  $g$  as in the proof of Theorem 1.1, put  $\eta = 1/10$ , and choose  $n$  by Dirichlet's theorem with  $\varepsilon = \max_q |\delta_q| < \min(\beta, g/2, \log 2/(2|P|), \eta/4, \eta\mu/4)$ . Each  $\delta_q$  with  $q \neq p$  is nonzero. The steps *Chains* and *Equal tails* of that proof apply word for word, with  $y$  in place of  $x$ :  $W(\theta_m)$  is the concatenation of the chains of  $W(\theta_k)$ , each rearranged,  $y_k = y_m$ , and  $f_p(C) = f_p(C')$  for every chain  $C$  and its rearrangement  $C'$ .

Let  $b_0 = \min_{q \neq p} |\delta_q|/|v_p - v_q|$ , so  $0 < b_0 \leq \varepsilon/\mu < \eta/4$ . Let  $B = \{q \neq p : \delta_q < 0\}$ ,  $A_+ = \{q \neq p : v_q < v_p\}$  and  $A_- = \{q \neq p : v_q > v_p\}$ . The sets  $A_+$  and  $A_-$  partition the nonempty set  $P \setminus \{p\}$ , so they differ, and one of them,  $A_s$ , differs from  $B$ . Put  $b = sb_0/2$  and let  $z$  be the point  $bv + \Theta(-\eta/2)$  of  $T_0$ . Its  $q$ -crossings are the times  $\eta/2 - bv_q + j \log q$ ; put  $\tau_q = \eta/2 - bv_q \in (3\eta/8, 5\eta/8)$ , and note that the other  $q$ -crossings lie outside  $(-\eta/4, 5\eta/4)$ . Moreover  $\tau_q - \tau_p = b(v_p - v_q)$  is nonzero, has the sign of  $s(v_p - v_q)$ , and has absolute value at most  $|\delta_q|/2$ . Let  $U$  be the set of  $\theta \in T_0$  with no crossing in  $[-\eta/4, \eta/4] \cup [3\eta/4, 5\eta/4]$ , a  $q$ -crossing  $\tau_q(\theta)$  in  $(\eta/4, 3\eta/4)$  for every  $q \in P$ , and, for every  $q \neq p$ ,  $\tau_q(\theta) - \tau_p(\theta)$  of the sign of  $b(v_p - v_q)$  and of absolute value less than  $|\delta_q|$ . Each condition is open, so  $U$  is an open subset of  $T_0$  containing  $z$ , and  $\theta_k + \Theta(t^*) \in U$  for some  $t^* > T + 2$ .

The crossings of  $\theta_k$  near  $t^*$  are then one crossing of each prime, each within  $|\delta_q| < \varepsilon$  of the  $p$ -crossing, and every other crossing is more than  $\eta/2 > 2\varepsilon$  away. So they form one chain  $C$ , in which the followers of  $p$  are the primes of  $A_s$ . In  $C'$  the  $q$ -crossing moves by  $-\delta_q$  and the  $p$ -crossing stays, so  $q$  follows  $p$  exactly when  $\tau_q - \tau_p > \delta_q$ , that is, exactly when  $\delta_q < 0$ . The followers of  $p$  in  $C'$  form  $B \neq A_s$ , so  $f_p(C) \neq f_p(C')$ , a contradiction.  $\square$

The two-wall lemma is not needed here:  $f_p$  detects every rearrangement that changes the followers of  $p$ , and a passage near the origin offers such a change for one sign of  $b$ . For  $|P| = 1$ ,  $E_p = p/(p-1)$  is rational, and the proof breaks exactly there, since  $P \setminus \{p\}$  is empty. The second agent's corrections were small: the two follower sets at the origin are complementary but one of them can be empty, the open set  $U$  must carry the order and size conditions on  $\tau_q - \tau_p$ , and a larger chain threshold proposed in the first write-up is unnecessary. For  $P = \{2, 3, 5\}$ ,  $p = 2$ ,  $t_k = 2^{20}$  and  $n = 65$ , the first rearranged chain is (2,3) against (3,2) after 131 letters, with  $f_2$  equal to 3 and 1, and the difference of the two tails, computed exactly to 300 letters, has  $\log_{10} |y_k - y_m| = -58.098$ .

For  $\mathcal{R}_p$  itself the comparison fails at the first step. Between the scales  $t_k$  and  $t_k p^n$  the tail of  $\mathcal{R}_p$  gains the early terms of every ray  $p^\alpha w$ , with  $w$  a  $(P \setminus \{p\})$ -smooth integer, that enters between the two scales, so the two scaled tails are far apart.  $E_p$  is the case of the single ray  $w = 1$ , where nothing enters.

## 5 Why the third prime prevents finite separation

With two generators the reciprocal-height kernel is one product of a row function and a column function. With three primes, no finite sum of products separating one exponent from the other two can equal the kernel. The first proposition is algebraic and even allows real generators; the later rank theorem uses distinct primes.

**Proposition 5.1** (two generators separate). *For real  $p, q > 1$ , with  $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$  as above, and all integers  $i, j \geq 0$ , the two-prime kernel  $K_2(i, j) = 1/L_{p,q}(p^i q^j)$  is the outer product*

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$$K_2(i, j) = (p^i q^{\lfloor \log_q p^i \rfloor})^{-1} (p^{\lfloor \log_p q^j \rfloor} q^j)^{-1},$$

so every two-by-two minor of  $K_2$  vanishes.

*Proof.* Since  $i$  and  $j$  are integers,  $\lfloor \log_p(p^i q^j) \rfloor = i + \lfloor \log_p q^j \rfloor$  and  $\lfloor \log_q(p^i q^j) \rfloor = j + \lfloor \log_q p^i \rfloor$ . Hence  $L_{p,q}(p^i q^j)$  is the product of  $p^i q^{\lfloor \log_q p^i \rfloor}$ , which depends on  $i$  alone, and  $p^{\lfloor \log_p q^j \rfloor} q^j$ , which depends on  $j$  alone. A matrix whose entries are a product of a row function and a column function has vanishing two-by-two minors. For distinct primes  $p, q$  the product  $L_{p,q}$  is the running least common multiple by the argument of Theorem 2.1.  $\square$

At three generators the smallest rectangle already fails to factor. A factorisation  $f(i)g(j)h(k)$  would force  $K(0,0,0)K(1,1,0) = K(1,0,0)K(0,1,0)$ .

**Proposition 5.2** (non-separability at  $\{2,3,5\}$ ). *With  $(p, q, r) = (2, 3, 5)$ ,*

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$$\det \begin{pmatrix} K(0,0,0) & K(0,1,0) \\ K(1,0,0) & K(1,1,0) \end{pmatrix} = \det \begin{pmatrix} 1 & 1/6 \\ 1/2 & 1/60 \end{pmatrix} = -\frac{1}{15} \neq 0.$$

*Proof.* The four values are computed from  $H(1) = 1$ ,  $H(2) = 2$ ,  $H(3) = 2 \cdot 3 = 6$  and  $H(6) = 4 \cdot 3 \cdot 5 = 60$ , so the determinant is  $1/60 - 1/12 = -1/15$ .  $\square$

**Theorem 5.3** (no finite separation of the kernel). *Let  $p, q, r$  be primes with  $p \neq q$ ,  $p \neq r$  and  $q \neq r$ . For every  $n \geq 1$  there are injective maps  $I, J : \{0, \dots, n-1\} \rightarrow \mathbb{N}$  such that, for every  $k \geq 0$ ,*

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$$\det(K(I(a), J(b), k))_{0 \leq a, b < n} \neq 0.$$

Consequently, for no finite  $d$  do there exist rational-valued functions  $f_\ell : \mathbb{N} \rightarrow \mathbb{Q}$  and  $G_\ell : \mathbb{N}^2 \rightarrow \mathbb{Q}$ ,  $0 \leq \ell < d$ , satisfying  $K(i, j, k) = \sum_{\ell < d} f_\ell(i) G_\ell(j, k)$  for all  $i, j, k$ .

*Proof.* Put  $\alpha = \log_r p$ ,  $\beta = \log_r q$ ,  $x_i = \{i\alpha\}$  and  $y_j = \{j\beta\}$ . Write  $v_p(N)$  for the exponent of the prime  $p$  in the positive integer  $N$ . The three height exponents are

$$\begin{aligned} v_p(H(p^i q^j r^k)) &= i + \lfloor j \log_p q + k \log_p r \rfloor, \\ v_q(H(p^i q^j r^k)) &= j + \lfloor i \log_q p + k \log_q r \rfloor, \\ v_r(H(p^i q^j r^k)) &= k + \lfloor i\alpha \rfloor + \lfloor j\beta \rfloor + \lfloor x_i + y_j \rfloor. \end{aligned}$$

Every term except the last floor depends on  $i$  and  $k$  alone or on  $j$  and  $k$  alone, so with positive rational  $R_i(k)$  and  $C_j(k)$ ,

$$K(i, j, k) = R_i(k) C_j(k) t^{\lfloor x_i + y_j \rfloor}, \quad t = r^{-1}. \quad (6)$$

The remaining matrix is independent of  $k$ , and  $x_i + y_j \in [0, 2)$ , so its entries are 1 and  $t$ .

Each of  $\alpha$  and  $\beta$  is irrational, since a rational value would give an equality of positive powers of distinct primes, so each fractional-part orbit is dense in  $[0, 1]$  and each is injective. The row and column indices are chosen independently, so no density theorem for a single orbit of pairs is required. First choose indices with  $0 < x_{I(0)} < \dots < x_{I(n-1)} < 1$ . We want the columns to cross their thresholds at successive selected rows, so that consecutive row differences will leave a triangular matrix. Write  $s_b = 1 - y_{J(b)}$ , and use density of the second orbit to choose

$$s_0 \in (0, x_{I(0)}), \quad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \leq b < n).$$

The intervals are disjoint, so the  $J(b)$  are distinct. Their strict endpoints ensure that no selected entry lies on a threshold, and  $x_{I(a)} + y_{J(b)} \geq 1$  holds exactly when  $b \leq a$ . Dividing row  $a$  by  $R_{I(a)}(k)$  and column  $b$  by  $C_{J(b)}(k)$  therefore leaves

$$C_n(t) = \begin{pmatrix} t & 1 & 1 & \dots & 1 \\ t & t & 1 & \dots & 1 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ t & t & \dots & t & 1 \\ t & t & \dots & t & t \end{pmatrix}, \quad \det C_n(t) = t(t-1)^{n-1} \neq 0,$$

the determinant following by subtracting from each row its predecessor, working upwards from the last. Before the division, the determinant equals

$$\det C_n(t) \prod_{a < n} R_{I(a)}(k) \prod_{b < n} C_{J(b)}(k),$$

which is nonzero for every  $k$ ; this also explains why the same indices work in every layer.

For the last assertion, suppose that a representation with  $d$  summands exists and fix  $k$ . On the selected rows  $I(0), \dots, I(d)$  and columns  $J(0), \dots, J(d)$  the resulting matrix factors as

$$(f_\ell(I(a)))_{0 \leq a \leq d, \ell < d} (G_\ell(J(b), k))_{\ell < d, 0 \leq b \leq d}.$$

It has rank at most  $d$  and hence zero determinant, contradicting the nonzero minor of order  $d+1$ .  $\square$

Index selection is essential. In the  $k = 0$  layer of the  $\{2, 3, 5\}$  kernel, the leading  $4 \times 4$  minor is singular because  $K(3, j, 0) = K(0, j, 0)/120$  for  $0 \leq j \leq 3$ . This is not a proportionality of the full rows: at the next column,

$$K(3, 4, 0) - \frac{K(0, 4, 0)}{120} = -\frac{1}{19440000}.$$

A singular leading minor therefore does not settle the rank.

In each fixed layer  $k$ , the same threshold description determines every finite sampled rank, rather than only producing one nonsingular minor.

**Proposition 5.4** (rank of threshold columns). *Let  $m \geq 1$  and let  $c$  lie in a field with  $c \neq 0, 1$ . For  $0 \leq h \leq m$ , let  $v_h$  be the length- $m$  column whose first  $h$  entries are 1 and whose remaining entries are  $c$ . If the distinct columns of a matrix are the  $v_h$  with  $h$  in a nonempty set  $E \subseteq \{0, \dots, m\}$ , then its rank is*

$$|E| - \mathbf{1}_{\{0, m\} \subseteq E}.$$

*Proof.* Write  $E = \{h_1 < \dots < h_s\}$ . The  $s - 1$  consecutive differences are

$$v_{h_{a+1}} - v_{h_a} = (1 - c)\mathbf{1}_{\{h_a, \dots, h_{a+1}-1\}},$$

so they are linearly independent because their nonempty supports are disjoint. If  $h_1 > 0$  or  $h_s < m$ , those supports miss a coordinate on which  $v_{h_1}$  is nonzero, and adjoining  $v_{h_1}$  gives rank  $s$ . If  $h_1 = 0$  and  $h_s = m$ , the differences sum to  $(1 - c)\mathbf{1}$  while  $v_0 = c\mathbf{1}$ , so  $v_0$  is already in their span and the rank is  $s - 1$ .  $\square$

The restrictions  $c \neq 0, 1$  exclude the zero column at  $c = 0$  and the collapse of all columns at  $c = 1$ . They hold automatically at  $c = 1/r$  over  $\mathbb{Q}$ . To apply the proposition, fix  $k$  and sort the chosen row phases  $x_i$ . Each normalised column is a threshold column  $v_h$ , where  $h$  is the number of sampled phases strictly below  $1 - y_j$ . Repeated threshold positions give proportional columns before column normalisation. The row and column factors in (6) are nonzero, so the formula gives the exact rank of every nonempty rectangular sample within this layer, independently of  $k$ . Repeated rows or columns do not change rank.

This also gives an exact algorithm. Order the sampled rows by the rational numbers  $p^i / r^{\lfloor \log_r p^i \rfloor}$ . For column  $j$ , count those strictly below  $r^{\lfloor \log_r q^j \rfloor + 1} / q^j$ , then apply the end-point correction to the set of resulting counts. Indeed, raising  $x_i < 1 - y_j$  to base  $r$  gives exactly this rational comparison, and  $x_i + y_j = 1$  belongs to the  $c$  side of the threshold. The integer logarithms are found by comparing powers, so the whole calculation uses integer arithmetic rather than numerical logarithms.

The fixed-layer restriction is essential. With  $(p, q, r) = (2, 3, 5)$ , rows  $i = 0, 1$  and columns indexed by  $(j, k) = (0, 0), (0, 1)$  give

$$\begin{pmatrix} K(0, 0, 0) & K(0, 0, 1) \\ K(1, 0, 0) & K(1, 0, 1) \end{pmatrix} = \begin{pmatrix} 1 & 1/60 \\ 1/2 & 1/360 \end{pmatrix}, \quad \det = -\frac{1}{180}.$$

Both columns have the same phase  $y_0 = 0$ , yet their rank is 2, not 1. When  $k$  varies with the column, the row factor also varies with that column and cannot be removed by one common diagonal rescaling. The threshold formula is not a rank formula for such mixed-layer samples.

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**Scope of the rank theorem.** Fan's comment of 26 June 2026 [16] notes that the two-prime argument does not seem to generalise immediately to  $|P| \geq 3$ . The theorem above gives a precise obstruction: no finite exact separation of the stated form exists. The same finite-dimensional proof works for real or complex factors, without continuity or boundedness assumptions. This does not exclude approximation or imply irrationality or transcendence of the sum. The threshold construction needs  $\log_r p$  and  $\log_r q$  irrational, but not  $p \neq q$ . The latter condition is needed for the three-prime running-LCM interpretation, not for the two independent dense orbits.

### 5.1 The exact uniform approximation error

Exact infinite rank does not by itself give a lower bound on approximation error. Here such a bound follows because any two distinct columns of the normalised matrix have a fixed positive distance. The matrix is

$$C(i, j) = t^{\lfloor x_i + y_j \rfloor}, \quad x_i = \{i \log_r p\}, \quad y_j = \{j \log_r q\}, \quad t = r^{-1}, \quad (7)$$

which is the factor left in (6) after the row and column factors are divided out. Say that a real matrix  $A$  on  $\mathbb{N} \times \mathbb{N}$  has *finite separated rank* when all of its columns lie in one finite-dimensional space of real sequences, equivalently when  $A(i, j) = \sum_{\ell < d} f_\ell(i) g_\ell(j)$  for some finite  $d$  and some sequences  $f_\ell, g_\ell$ , with no continuity or boundedness assumed. This is ordinary finite column rank; a basis of the column space supplies a separated expression.

**Theorem 5.5** (distance from matrices of finite separated rank). *Let  $p, q, r$  be pairwise distinct primes and let  $C$  be as in (7). Then*

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$$\inf_A \sup_{i, j \geq 0} |C(i, j) - A(i, j)| = \frac{1-t}{2} = \frac{r-1}{2r},$$

*the infimum being over all matrices  $A$  of finite separated rank, and it is attained by the constant matrix of value  $(1+t)/2$ .*

*Proof.* Every entry of  $C$  lies in  $\{1, t\}$ , and  $C(i, j) = t$  exactly when  $x_i + y_j \geq 1$ . Fix  $j \neq k$ . The numbers  $y_j$  are pairwise distinct, since  $\log_r q$  is irrational, so we may assume  $y_j < y_k$ , and then  $0 \leq 1 - y_k < 1 - y_j \leq 1$ . Density of the orbit  $(x_i)$  in  $(0, 1)$  supplies an index  $i$  with  $1 - y_k < x_i < 1 - y_j$ , and at that row the two columns carry the entries  $t$  and  $1$ . Hence any two distinct columns of  $C$  are at sup-distance exactly  $1 - t$ .

Let  $A$  have finite separated rank and put  $\varepsilon = \sup_{i, j} |C(i, j) - A(i, j)|$ . Suppose  $\varepsilon < (1 - t)/2$ . Each column  $A_j$  then satisfies  $\|A_j\|_\infty \leq 1 + \varepsilon$ . Let  $V$  be the span of these columns. It is finite-dimensional by hypothesis and consists of bounded sequences, so the supremum norm is defined on  $V$ . No boundedness of the individual separated factors is needed. By the triangle inequality, distinct columns satisfy

$$\|A_j - A_k\|_\infty \geq \|C_j - C_k\|_\infty - 2\varepsilon = 1 - t - 2\varepsilon > 0.$$

The columns would be an infinite family in the bounded ball of radius  $1 + \varepsilon$  in  $V$ , separated by the fixed positive distance  $1 - t - 2\varepsilon$ . This contradicts total boundedness of

bounded subsets of a finite-dimensional normed space. Hence  $\varepsilon \geq (1 - t)/2$  for every  $A$  of finite separated rank.

For sharpness take  $A(i, j) = (1 + t)/2$ , which has separated rank one; every entry of  $C$  is at distance exactly  $(1 - t)/2$  from it.  $\square$

The lower bound concerns the entire normalised matrix, not each finite restriction. For example, at  $t = 1/5$  the matrices

$$T = \begin{pmatrix} 1/5 & 1 \\ 1/5 & 1/5 \end{pmatrix}, \quad A = \begin{pmatrix} 3/10 & 9/10 \\ 1/10 & 3/10 \end{pmatrix}$$

satisfy  $\det A = 0$  and  $\|T - A\|_{\max} = 1/10 < 2/5$ . Here  $\|\cdot\|_{\max}$  is the largest absolute entry. This finite example has no bearing on the infinite family of separated columns used in the proof.

For the original kernel, the decaying row and column factors instead allow approximation in the summation norm. Let  $K^{(N)}(i, j, k) = K(i, j, k)$  for  $i < N$  and  $K^{(N)}(i, j, k) = 0$  otherwise. This is a sum of at most  $N$  terms separated between  $i$  and  $(j, k)$ . Since  $H(x) > x^3/(pqr)$  for  $x \geq 1$ , geometric summation gives

$$\sum_{i,j,k \geq 0} |K(i, j, k) - K^{(N)}(i, j, k)| \leq \frac{pqr p^{-3N}}{(1 - p^{-3})(1 - q^{-3})(1 - r^{-3})}. \quad (8)$$

The same bound controls the supremum of the entrywise errors, so these finite separated-rank approximants converge both in  $\ell^1$  and uniformly to the original kernel. The positive uniform lower bound in Theorem 5.5 belongs to the rescaled matrix. Diagonal rescaling preserves exact rank, but the rescaling factors here are unbounded and do not preserve uniform error estimates. Neither the rank obstruction nor these approximation bounds decide whether the scalar sum is rational.

## 6 The recurrence for tails between powers of two

For the rest of the record set  $P = \{2, 3, 5\}$  and  $S = \mathcal{R}_P$ . Write  $P_a = H(2^a)$  and  $h_a = P_a/2$ . Here  $P_a$  is a boundary height, not a set of primes;  $h_0 = 1/2$ , while  $h_a$  is a positive integer for  $a \geq 1$ . At a dyadic endpoint the power of 2 is exact, while the maximal powers of 3 and 5 exceed  $2^a/3$  and  $2^a/5$ . Hence

$$\frac{8^a}{15} < P_a \leq 8^a \quad (a \geq 0).$$

These bounds will control both the tail scale and the growth of a recurrence error. Group the terms between consecutive powers of two and define

$$s_a = \sum_{\substack{i,j,k \geq 0 \\ 2^a \leq 2^i 3^j 5^k < 2^{a+1}}} \frac{1}{H(2^i 3^j 5^k)}, \quad T_a = \sum_{j \geq a} s_j, \quad X_a = h_a T_a. \quad (9)$$

The factor  $1/2$  in  $h_a$  comes from the strict cutoff. For  $a \geq 1$ , every height  $H(x)$  with  $x < 2^a$  divides  $P_a/2$ : the exponent of 2 is at most  $a - 1$ , and the other exponents are at most their boundary values. We will use this to clear the finite prefix in Lemma 7.2.

The jumps after  $2^a$  and up to  $2^{a+1}$  consist of any powers of 3 or 5 strictly inside that interval, followed by the factor 2 at its right endpoint. There is at most one power of each odd prime: successive powers have ratio greater than two. Let  $I_a$  list the pairs  $(p, e)$  with  $p \in \{3, 5\}$  and  $2^a < p^e < 2^{a+1}$ , ordered by the value  $p^e$ .

**Proposition 6.1** (four possible bases). *For every  $a$ ,*

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$$b_a = \frac{P_{a+1}}{P_a} = 2 \prod_{(p,e) \in I_a} p \in \{2, 6, 10, 30\}, \quad \text{so} \quad 2 \leq b_a \leq 30. \quad (10)$$

*Proof.* A power of 3 may occur and a power of 5 may occur, each at most once. The resulting factor is  $2, 2 \cdot 3, 2 \cdot 5$  or  $2 \cdot 3 \cdot 5$ .  $\square$

The coefficient subtracted at step  $a$  is the shell mass with its denominators cleared:

$$m_a = h_{a+1} s_a = \sum_{\substack{x \text{ smooth} \\ 2^a \leq x < 2^{a+1}}} \frac{P_{a+1}}{2^{\mathbf{H}(x)}}.$$

Each summand is an integer, by the same strict-cutoff argument. For example,  $[2, 4)$  contains just 2 and 3, so  $m_1 = 6(1/2 + 1/6) = 4$  and  $X_2 = 6X_1 - 4$ . The first four pairs  $(b_a, m_a)$ , starting at  $a = 1$ , are  $(6, 4)$ ,  $(10, 7)$ ,  $(6, 7)$  and  $(30, 65)$ . In particular  $m_3 > b_3$ : these coefficients are not positional digits.

The shell  $[16, 32)$  shows why the order of the jumps matters. Its internal jumps are 25 and 27, in that order. The four smooth numbers 16, 18, 20, 24 precede both jumps and have weight 15; 25 precedes only the jump at 27 and has weight 3; and 27, 30 have weight 1. Thus

$$m_4 = 4 \cdot 15 + 1 \cdot 3 + 2 \cdot 1 = 7 + (5 - 1) \cdot 4 \cdot 3 + (3 - 1) \cdot 5 = 65.$$

The second expression starts with the seven points and adds the corrections before 25 and 27. It is this form that extends to every shell.

To compute  $m_a$  in general, count the smooth numbers before each prime-power jump. Let  $\mathcal{N}(t)$  be the number of positive  $\{2, 3, 5\}$ -smooth integers strictly below  $t$ . First give every point in  $[2^a, 2^{a+1})$  weight one. At an internal jump  $p^e$ , the points before that jump need an additional weight  $p - 1$ , multiplied by the prime factors at all later internal jumps. Thus, for  $a \geq 1$ ,

$$\begin{aligned} m_a &= \mathcal{N}(2^{a+1}) - \mathcal{N}(2^a) \\ &\quad + \sum_{(p,e) \in I_a} (p - 1)(\mathcal{N}(p^e) - \mathcal{N}(2^a)) \prod_{\substack{(q,f) \in I_a \\ p^e < q^f}} q, \\ m_0 &= 1. \end{aligned} \quad (11)$$

The first difference counts the whole shell; each later difference counts its points strictly before the indicated jump. The proof below justifies these weights, establishes convergence and derives the tail identities.

**Theorem 6.2** (the tail recurrence). *The shell masses are summable and  $S = \sum_{a \geq 0} s_a$ . For every  $a \geq 0$ ,*

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$$m_a = h_{a+1}s_a \in \mathbb{N}_{>0}, \quad X_{a+1} = b_a X_a - m_a, \quad X_a = \sum_{j \geq a} \frac{m_j}{b_a b_{a+1} \cdots b_j}. \quad (12)$$

*Proof.* Every exponent of 3 or 5 in the  $a$ th shell is at most  $a$ , and for each such pair Lemma 2.5 leaves at most one exponent of 2 placing the point in  $[2^a, 2^{a+1})$ . Each height is at least  $2^a$  and the shell contains  $2^a$ , so  $0 < s_a \leq (a+1)2^{2-a}$ . The majorant is summable, which justifies every tail splitting below, and unique prime factorisation identifies  $\sum_a s_a$  with the original repeated series.

Write the internal jumps as  $t_\ell = p_\ell^{e_\ell}$ ,  $1 \leq \ell \leq v$ , and set  $t_0 = 2^a$ ,  $t_{v+1} = 2^{a+1}$  and  $n_\ell = \mathcal{N}(t_\ell)$ . On  $[t_\ell, t_{\ell+1})$  the height is  $P_a \prod_{u \leq \ell} p_u$ , and the terminal jump has factor two, so

$$h_{a+1}s_a = \sum_{\ell=0}^v (n_{\ell+1} - n_\ell) \prod_{u > \ell} p_u = n_{v+1} - n_0 + \sum_{\ell=1}^v (p_\ell - 1) \left( \prod_{u > \ell} p_u \right) (n_\ell - n_0).$$

The second equality is finite summation by parts and is exactly (11). The count itself is finite: for  $p = 2, 3, 5$ , with  $q, r$  the other primes, counting by the exponent of  $p$  gives

$$\mathcal{N}(p^e) = \sum_{u=1}^e \#\{(i, j) \in \mathbb{N}^2 : q^i r^j < p^u\}.$$

At  $a = 0$  the shell is the single point 1, giving  $m_0 = 1$ . Positivity follows from  $s_a > 0$ . Splitting  $T_a = s_a + T_{a+1}$  and multiplying by  $h_a$  gives the recurrence, and  $b_a \cdots b_j = P_{j+1}/P_a$  with  $m_j = h_{j+1}s_j$  gives  $m_j/(b_a \cdots b_j) = h_a s_j$ , whose summation is the last identity.  $\square$

For estimates, it is useful to count by the odd part  $3^j 5^k$  instead. Each such part below  $2^{a+1}$  has exactly one power-of-two multiple in  $[2^a, 2^{a+1})$ . This turns the same numerator into a two-dimensional count, with weights determined by the remaining odd-prime jumps.

**Lemma 6.3** (the shell numerator as a weighted lattice count). *Put  $\lambda_3 = \log_2 3$ ,  $\lambda_5 = \log_2 5$  and  $\theta_p = 1/\lambda_p$  for  $p = 3, 5$ . For  $j, k \geq 0$  write  $w_{j,k} = j\lambda_3 + k\lambda_5$  and  $t_{j,k} = \{w_{j,k}\}$ . The shell numerator is*

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$$m_a = \sum_{\substack{j,k \geq 0 \\ w_{j,k} < a+1}} 3^{\lfloor (a+1)\theta_3 \rfloor - \lfloor (a+t_{j,k})\theta_3 \rfloor} 5^{\lfloor (a+1)\theta_5 \rfloor - \lfloor (a+t_{j,k})\theta_5 \rfloor}.$$

Every summand is in  $\{1, 3, 5, 15\}$ . In particular, for  $a \geq 0$ ,

$$(\lfloor a/(2\lambda_3) \rfloor + 1)(\lfloor a/(2\lambda_5) \rfloor + 1) \leq m_a \leq 15(a+1)^2.$$

Thus  $m_a = \Theta((a+1)^2)$  and the numerator sequence is unbounded. These are integer numerators, not positional digits restricted to  $\{0, \dots, b_a - 1\}$ .

*Proof.* For each pair with  $w_{j,k} < a + 1$ , exactly one exponent  $i = a - \lfloor w_{j,k} \rfloor \geq 0$  puts  $2^i 3^j 5^k = 2^{a+\lfloor w_{j,k} \rfloor}$  in  $[2^a, 2^{a+1})$ . This pairs each point of the triangle with exactly one smooth integer in the shell. Substituting in  $P_{a+1}/(2H(2^i 3^j 5^k))$  cancels the power of 2 and gives the displayed weight. Each remaining floor increment is zero or one since  $0 < \theta_p < 1$ . Finally  $\lambda_3, \lambda_5 > 1$  implies  $j, k \leq a$  for every summation pair, proving the upper bound. For the lower bound, restrict to  $0 \leq j \leq \lfloor a/(2\lambda_3) \rfloor$  and  $0 \leq k \leq \lfloor a/(2\lambda_5) \rfloor$ . Then  $w_{j,k} \leq a$  and each weight is at least one.  $\square$

The formula shows two sources of variation in  $m_a$ : points enter the triangle as  $a$  increases, and their weights depend on two floors. Quadratic growth does not make this count a quadratic polynomial.

The recurrence also restricts how persistently a nonintegral tail can approach the integers. The following alternative uses only the integer coefficients and the bounds  $2 \leq b_a \leq 30$ .

**Proposition 6.4** (integer tails or repeated separation from the integers). *For every integer  $B \geq 1$ , either  $BX_a \in \mathbb{Z}$  for some  $a \geq 0$  and every later  $a$ , or for every  $a_0$  there is  $a \geq a_0$  with  $|BX_a - z| \geq 1/31$  for every  $z \in \mathbb{Z}$ .*

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*Proof.* If the second alternative fails, there are  $A$  and integers  $z_a$  such that  $e_a = BX_a - z_a$  satisfies  $|e_a| < 1/31$  for all  $a \geq A$ . The recurrence gives  $z_{a+1} - b_a z_a + Bm_a = b_a e_a - e_{a+1}$ . Since  $Bm_a$  is integral, the left side is an integer; the right side has absolute value less than  $(30 + 1)/31 = 1$ . Both sides therefore vanish, so  $e_{a+1} = b_a e_a$ . Hence  $|e_{A+k}| \geq 2^k |e_A|$  for every  $k$ , whereas  $|e_{A+k}| < 1/31$ . Thus  $e_A = 0$ , and the integer recurrence propagates integrality from  $BX_A$  to every later  $BX_a$ .  $\square$

The proposition does not determine which alternative holds for the tails of  $S$ . For comparison, Erdős–Taylor [20, Theorem 1, p. 600] prove a countability statement for increasing integer sequences with bounded successive ratios; Fan [21, Lemma 3.1, arXiv v1] allows an unbounded, not necessarily increasing sequence, still with a uniform upper bound on successive ratios. For our divisibility chain the same error argument identifies the entire exceptional set, not just its cardinality. For each fixed integer  $B \geq 1$ ,

$$\{\zeta \in \mathbb{R} : \text{dist}(Bh_a \zeta, \mathbb{Z}) \rightarrow 0\} = \frac{1}{B} \mathbb{Z}[1/30], \quad \mathbb{Z}[1/30] = \{m/30^t : m \in \mathbb{Z}, t \in \mathbb{N}\}.$$

Indeed, apply the proof above to  $Bh_a \zeta$ , whose successive terms are related by multiplication by  $b_a$ . Convergence of the distances forces  $Bh_A \zeta \in \mathbb{Z}$  for some  $A \geq 1$ . Since  $h_A$  has no prime factors outside 2, 3, 5, this gives membership in the right-hand side. Conversely, every fixed product of powers of 2, 3, 5 eventually divides  $h_a$ , so every member of the right-hand side gives integral values for all large  $a$ .

Separation from the integers for one multiplier does not prove irrationality. For example,  $\zeta = 1/7$  satisfies  $\text{dist}(h_a \zeta, \mathbb{Z}) \geq 1/7$  for every  $a \geq 1$ , because  $7 \nmid h_a$ ; with  $B = 7$  the values  $Bh_a \zeta$  are all integers. For  $S$ , the denominator clearing in Theorem 7.3 below shows which multipliers matter: the integral alternative for the actual tails must be excluded for every  $B \geq 1$  coprime to 30, not merely for  $B = 1$ .

## 7 Bounding the tails and clearing a rational denominator

We first bound the tail without assuming rationality. The sum of the three height exponents will index the height cells. Put

$$n_a = a + \lfloor \log_3(2^a) \rfloor + \lfloor \log_5(2^a) \rfloor, \quad Q(n) = \frac{n^2 + 8n + 18}{9},$$

so  $n_a$  is the sum of the three height exponents at  $2^a$ .

**Theorem 7.1** (a quadratic upper bound). *For every  $a \geq 0$ ,  $0 < X_a \leq Q(n_a)$ .*

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*Proof.* Partition the smooth integers  $x \geq 2^a$  by their height vector

$$(A, B, C) = (\lfloor \log_2 x \rfloor, \lfloor \log_3 x \rfloor, \lfloor \log_5 x \rfloor).$$

The cell of a vector is the interval  $[\lambda, \eta)$ , where

$$\lambda = \max(2^A, 3^B, 5^C), \quad \eta = \min(2^{A+1}, 3^{B+1}, 5^{C+1}).$$

Thus  $\eta \leq 2^{A+1} \leq 2\lambda$ . By Lemma 2.5, fixing the exponents of 3 and 5 leaves at most one exponent of 2. Since  $A \geq B \geq C$ , the cell contains at most

$$(B+1)(C+1) \leq \frac{(A+B+C+3)^2}{9}$$

smooth points: writing  $u = B+1 \geq v = C+1$  and using  $A+1 \geq u$ , the difference  $(2u+v)^2 - 9uv = (u-v)(4u-v)$  is nonnegative. Unique factorisation identifies these exponent triples with distinct smooth integers.

As the cutoff increases all three height exponents are nondecreasing, so two nonempty cells with the same exponent sum are the same cell, and there is at most one nonempty cell with each exponent sum. Every cell above  $2^a$  has exponent sum at least  $n_a$ , and a cell with sum  $n_a + k$  has height at least  $P_a 2^k$ , since each of the  $k$  extra prime factors is at least 2. Nonnegative summation over exponent sums, allowing empty cells, gives

$$X_a \leq \frac{1}{18} \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{2^k} = \frac{n_a^2 + 8n_a + 18}{9},$$

the evaluation using the geometric moments  $\sum 2^{-k} = 2$ ,  $\sum k 2^{-k} = 2$  and  $\sum k^2 2^{-k} = 6$ . Positivity follows from the shell at  $2^a$ .  $\square$

Increasing the exponent sum by one costs a factor of at least two in the denominator, whereas the number of points in a cell grows at most quadratically. Summing those geometric contributions gives the bound. The argument uses unique factorisation and the projection count, not an asymptotic estimate for smooth numbers or a Hecke–Mahler theorem.

The quadratic order is also necessary. The positive next tail and the recurrence give

$$X_a = \frac{m_a + X_{a+1}}{b_a} > \frac{m_a}{b_a} \geq \frac{m_a}{30}.$$

The lower bound of Lemma 6.3, together with  $n_a \leq 3a$ , therefore gives  $X_a = \Theta((a+1)^2)$ . In particular, the normalised tails are not uniformly bounded. This order estimate does not assert an asymptotic constant or optimality of  $Q$ .

The bound is used at the endpoint of a window, where the natural index is the positive prime-power jump count strictly below the cutoff. Put

$$j_a = \#\{p^e < 2^a : p \in \{2, 3, 5\}, e \geq 1\} = n_a - 1 \quad (a \geq 1), \quad (13)$$

the equality holding for  $a \geq 1$  because the powers of 2 below  $2^a$  number  $a - 1$  while the powers of 3 and of 5 below  $2^a$  number  $\lfloor \log_3 2^a \rfloor$  and  $\lfloor \log_5 2^a \rfloor$ ; at  $a = 0$  the count is  $j_0 = 0$  and  $n_0 - 1 = -1$ . Substituting  $n_a = j_a + 1$  into  $Q$  gives the integer bound used throughout the rest of the note:

$$K(B, a) = \lfloor BQ(n_a) \rfloor, \quad K(B, a) = \left\lfloor \frac{B(j_a^2 + 10j_a + 27)}{9} \right\rfloor \quad (a \geq 1). \quad (14)$$

The cutoff in (13) is  $2^a$  and it is strict. The symbol  $K(B, a)$  bounds an *integral* quantity  $BX_a$ : from  $BX_a \leq BQ(n_a)$  one may take the floor only after integrality has been established. We do not assert  $BX_a \leq K(B, a)$  for arbitrary real tails.

**Lemma 7.2** (finite denominator clearing). *For all integers  $0 \leq u \leq b$  the window mass  $h_b \sum_{a=u}^{b-1} s_a$  is a natural number. If  $S = N/D$  with  $N \in \mathbb{Z}$  and  $D \in \mathbb{N}_{>0}$ , then  $DX_a \in \mathbb{Z}$  for every  $a \geq 1$ , and there are indices  $1 \leq i < j \leq D + 1$  for which  $X_i - X_j \in \mathbb{Z}$ .*

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*Proof.* An empty window has mass zero. Otherwise  $b \geq 1$ , and every integer  $x < 2^b$  has 2-height exponent at most  $b - 1$  while its other height exponents are at most those at  $2^b$ , so  $H(x) \mid P_b/2 = h_b$  and every term of the finite window clears at  $h_b$ . Since  $h_a \sum_{u < a} s_u$  is an integer,

$$DX_a = h_a N - D h_a \sum_{u < a} s_u \in \mathbb{Z}.$$

Among  $D + 1$  of these integers two share a residue modulo  $D$ , and the corresponding states differ by an integer.  $\square$

The strict upper endpoint is what permits division by two. A cutoff including  $2^b$  would not clear its term at the normaliser  $h_b = P_b/2$ .

**Theorem 7.3** (rationality gives positive integer tails). *Suppose  $S = N/D$  with  $N \in \mathbb{Z}$ ,  $D \in \mathbb{N}_{>0}$ , and write*

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$$D = 2^u 3^v 5^w B, \quad u, v, w \in \mathbb{N}, \quad B \in \mathbb{N}_{>0}, \quad \gcd(B, 30) = 1, \quad a_D = u + 1 + 2v + 3w.$$

*Then for every  $a \geq a_D$  the number  $d_a = BX_a$  is a positive integer and*

$$d_{a+1} = b_a d_a - B m_a, \quad 1 \leq d_a \leq K(B, a) \leq 90B(a+1)^2.$$

*Proof.* Let  $M = 2^u 3^v 5^w$ . For  $a \geq a_D$  we have  $2^a \geq 2^{u+1}$ ,  $2^a \geq 3^v$  and  $2^a \geq 5^w$ , using  $3 < 2^2$  and  $5 < 2^3$ ; hence  $M \mid h_a$ . By Lemma 7.2,  $X_a$  differs from  $h_a N/D$  by an integer. Since  $M \mid h_a$  and  $D = MB$ , multiplying by  $B$  shows that  $BX_a$  is an integer. Positivity and the recurrence come from (12), and the upper bound is Theorem 7.1 with the floor taken, since  $d_a$  is an integer at most  $BQ(n_a)$ . Finally  $n_a \leq 3a$ , so  $Q(n_a) \leq a^2 + \frac{8}{3}a + 2 \leq 90(a+1)^2$ .  $\square$

Every positive denominator admits the stated factorisation: remove all powers of 2, 3 and 5, leaving  $B$  coprime to 30. Thus the theorem does not impose an extra restriction on a hypothetical rational  $S$ . The removed factor  $2^u 3^v 5^w$  controls how far out the integer tails begin; their bound depends on  $B$ . The displayed  $a_D$  is sufficient, but need not be the first such index. The next proposition gives the first index when the fraction is reduced. Write  $\text{den}(x)$  for the positive denominator of a rational number  $x$  in lowest terms.

**Proposition 7.4** (exact denominators and minimal clearing). *Suppose  $S = N/(MB)$  is in lowest terms, with  $M = 2^u 3^v 5^w$  and  $\gcd(B, 30) = 1$ . For every  $a \geq 1$ ,*

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$$\text{den}(X_a) = \frac{MB}{\gcd(M, h_a)}, \quad \text{den}(BX_a) = \frac{M}{\gcd(M, h_a)}.$$

Hence the first integral reduced tail occurs at

$$a_* = \min\{a \geq 1 : 2^a \geq \max(2^{u+1}, 3^v, 5^w)\} \leq a_D,$$

and  $BX_a$  is integral exactly for  $a \geq a_*$ . This onset is computable by integer powers; it is not an estimate obtained by rounding logarithms.

*Proof.* By Lemma 7.2,  $X_a$  differs from  $h_a N/(MB)$  by an integer. Since  $N$  is coprime to  $MB$  and  $h_a$  is supported on  $\{2, 3, 5\}$ , reduction gives both denominators. Now  $M \mid h_a$  means  $a - 1 \geq u$ ,  $\lfloor \log_3 2^a \rfloor \geq v$  and  $\lfloor \log_5 2^a \rfloor \geq w$ , precisely the three integer inequalities. All three persist when  $a$  increases, giving the first and every later integral reduced tail. The earlier bounds  $3 < 4$  and  $5 < 8$  give  $a_* \leq a_D$ .  $\square$

For example, a hypothetical reduced denominator  $2^3 3^2 5 \cdot 7$  would require  $2^a \geq \max(16, 9, 5)$ , so  $a_* = 4$  rather than the sufficient  $a_D = 11$ . Lowest terms matter for this exact answer: unreduced factors could cancel against the numerator sooner.

The recurrence preserves integrality forward. Its homogeneous equation also determines how fast two distinct solutions separate.

**Proposition 7.5** (propagation of integrality and uniqueness of a small solution). *For every  $a$ ,  $X_a = (m_a + X_{a+1})/b_a > 0$ , and if  $X_a \in \mathbb{Z}$  then  $X_n \in \mathbb{Z}$  for every  $n \geq a$ . Moreover, fix  $A$ , a positive width function  $w$  with  $w(A+k)/8^k \rightarrow 0$ , and a real sequence  $(y_n)_{n \geq A}$  satisfying  $y_{n+1} = b_n y_n - m_n$ . If  $y_n$  and  $X_n$  both lie in  $(m_n/b_n, m_n/b_n + w(n)]$  for every  $n \geq A$ , then  $y_A = X_A$ .*

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*Proof.* The identity is the recurrence solved for  $X_a$ , and positivity holds because every shell contains its dyadic left endpoint. Integer coefficients preserve integrality at every later step. For the last assertion,  $y_{A+k} - X_{A+k} = (P_{A+k}/P_A)(y_A - X_A)$ . The dyadic height bounds from Section 6 give  $P_{A+k}/P_A > 8^k/15$ , whereas the common interval bounds the absolute difference by  $w(A+k)$ . Thus  $|y_A - X_A| < 15w(A+k)/8^k \rightarrow 0$ .  $\square$

The width condition allows every positive polynomial width and widths  $\rho^n$  with  $1 < \rho < 8$ , but not  $8^n$  itself. Both solutions must lie in the stated intervals; the recurrence alone does not supply that condition. More generally, fix an integer  $B \geq 1$ . Every real solution of  $y_{a+1} = b_a y_a - B m_a$  starting at an index  $A$  satisfies

$$y_a = BX_a + \frac{P_a}{P_A}(y_A - BX_A) \quad (a \geq A).$$

Subtracting the recurrence for  $BX_a$  proves the identity. Since  $X_a = O((a+1)^2)$  and  $P_a = \Theta(8^a)$ , the unique solution with  $y_a = o(8^a)$  is  $y_a = BX_a$ ; every other solution has  $|y_a| = \Theta(8^a)$ . This classifies growth; integrality of  $BX_a$  remains a separate question. The same formula explains a numerical precaution: a starting error  $\varepsilon$  is multiplied by  $P_a/P_A$  under forward iteration. A decimal approximation propagated forwards is therefore not a certificate of the tail floors. The digit calculation in Section 10 uses rational intervals with an explicit infinite-tail bound instead.

### 7.1 A smaller bound from the order of the prime-power jumps

The proof of Theorem 7.1 bounded each new prime factor below by 2. But too many powers of 2 cannot occur without an intervening power of 3. Using this restriction gives a smaller geometric majorant.

**Proposition 7.6** (a smaller quadratic bound). *For every  $a \geq 0$ ,*

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$$0 < X_a \leq \tilde{Q}(n_a) < Q(n_a), \quad \tilde{Q}(n) = \frac{1210n^2 + 9130n + 18847}{11979}.$$

*Proof.* For a point  $x \geq 2^a$ , let  $d_p$  be the increase in its  $p$ -height exponent from the boundary  $2^a$ , and put  $k = d_2 + d_3 + d_5$ . The definition of  $d_3$  gives  $x < 3^{\lfloor \log_3 2^a \rfloor + d_3 + 1} \leq 2^a 3^{d_3 + 1} < 2^{a+2d_3+2}$ , hence  $d_2 \leq 2d_3 + 1$ . It follows that  $k \leq 3(d_3 + d_5) + 1$ , so at least  $j = \lfloor (k+1)/3 \rfloor$  of the  $k$  new prime factors are odd. Each of these contributes at least 3, and each remaining factor at least 2. Hence

$$\frac{H(x)}{P_a} \geq 2^{k-j3j} =: d(k).$$

There is at most one height cell with each exponent sum, with at most  $(n_a + k + 3)^2/9$  supported points by the earlier projection bound. Consequently

$$X_a \leq \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{18d(k)}.$$

Now  $d(3m) = 12^m$ ,  $d(3m+1) = 2 \cdot 12^m$  and  $d(3m+2) = 6 \cdot 12^m$ . Grouping in threes and summing the quadratic geometric series gives  $\tilde{Q}(n_a)$ . Finally  $11979(Q(n) - \tilde{Q}(n)) = 121n^2 + 1518n + 5111 > 0$  for  $n \geq 0$ .  $\square$

The estimate uses the original shell multiplicities and the order of the prime-power jumps. It is not a bound for an arbitrary recurrence with the same bases. For an integral  $BX_a$ , the bound can be rounded down to  $\lfloor B\tilde{Q}(n_a) \rfloor$ . This improves the bound available in a rationality contradiction. Theorem 8.2 and the recorded finite tests use  $K$ , not this smaller bound. No optimality claim is made for either quadratic bound.

Section 8 explains how the smaller bound can also replace  $K$  in the residue argument.

**Relation to Cantor-series criteria.** Since  $h_0 = 1/2$ , the value  $S/2 = X_0$  is the Cantor series  $\sum_{a \geq 0} m_a/P_{a+1}$ , with  $P_0 = 1$  and  $P_{a+1} = b_a P_a$ . Its normalised tails  $X_a$  are the usual objects in the rationality criteria for Cantor series. Under a small-numerator hypothesis,

Erdős and Straus characterise rationality by the existence of a positive integer  $B$  and integers  $c_a$  satisfying, eventually,

$$Bm_a = b_a c_a - c_{a+1}, \quad |c_{a+1}| < b_a/2$$

[17, Theorem 2.1 and (2.4), pp. 85–86]. The second condition is part of the criterion, not a consequence of the recurrence alone. Their proof already chooses nearest integers. Hančl and Tijdeman make that choice part of the criterion: they prescribe  $c_a$  as a nearest integer to  $Bm_a/b_a$  and require the recurrence [18, §§2–3 and Theorem 3.1, pp. 372–375]. Koutsoukou-Argraki and Li formalised the Erdős–Straus criteria in Isabelle/HOL [19].

Both cited criteria require  $m_a/(b_{a-1}b_a) \rightarrow 0$  in our notation. Here the weighted-triangle count gives  $m_a = \Theta((a+1)^2)$ , and

$$\frac{m_a}{900} \leq \frac{m_a}{b_{a-1}b_a} \leq \frac{m_a}{4} \quad (a \geq 1).$$

The ratio therefore has quadratic order and tends to infinity. There is also a direct lower count that does not use the real-logarithm triangle estimate. For  $a \geq 1$ , each pair  $0 \leq j, k \leq \lfloor a/5 \rfloor$  has  $3^j 5^k < 2^a$ , since  $15 < 2^5$ , and hence a unique power-of-two multiple in  $[2^a, 2^{a+1})$ . Each such point contributes at least one to  $m_a$ . Thus  $m_a \geq (\lfloor a/5 \rfloor + 1)^2$ , an elementary lower bound sufficient to see directly that the small-numerator condition fails.

The actual scaled tails also satisfy

$$BX_a - \frac{Bm_a}{b_a} = \frac{BX_{a+1}}{b_a} = \Theta((a+1)^2)$$

for each fixed  $B \geq 1$ , by the recurrence, the quadratic tail bounds and  $2 \leq b_a \leq 30$ . Thus any eventual integer tails supplied by a hypothetical rational value could not be the prescribed nearest integers. In fact, those prescribed integers fail the recurrence at every sufficiently large index, regardless of how ties are resolved. If  $c_a$  is a nearest integer to  $Bm_a/b_a$ , then

$$|b_a c_a - Bm_a| \leq 15, \quad c_{a+1} \geq \frac{Bm_{a+1}}{30} - \frac{1}{2} \rightarrow \infty.$$

Consequently  $c_{a+1} = b_a c_a - Bm_a$  is impossible for all sufficiently large  $a$ . This is a failure of the proposed carry construction, not an irrationality contradiction: the criterion's small-numerator hypothesis also fails.

The denominator clearing itself is standard; the calculations specific to this series retain its multiplicities, use the strict endpoint to divide the normaliser by two, and bound the resulting positive tails.

For polynomial Cantor data, Hančl–Tijdeman [24, Theorems 2.2 and 3.1] give a polynomial cancellation criterion and a division mechanism. Their nonconstant polynomial radix is not our bounded  $(b_a)$ . Their separate Theorem 4.2 uses a finite-product rearrangement. We do not use an unrestricted infinite reindexing from that argument: for the integer decompositions considered here, the terminal terms must be checked separately. In the first-order case  $m_a = b_a c_a - c_{a+1}$ , finite summation gives

$$\sum_{a=0}^{N-1} \frac{m_a}{P_{a+1}} = c_0 - \frac{c_N}{P_N}.$$

Every integer  $c_0$  generates an integer solution recursively. Hence  $c_N/P_N \rightarrow c_0 - S/2$ ; the existence of an integer solution gives no rationality information.

For these actual numerators the terminal term cannot vanish for *any* integer  $c_0$ . Indeed,  $P_3 = 120$ ,  $n_3 = 5$  and  $(m_0, m_1, m_2) = (1, 4, 7)$ , so the tail bound gives

$$0 < \frac{S}{2} = \frac{1}{2} + \frac{4}{12} + \frac{7}{120} + \frac{X_3}{120} \leq \frac{107}{120} + \frac{83}{1080} = \frac{523}{540} < 1.$$

Thus  $c_0 - S/2 \neq 0$  for every integer  $c_0$ . Since  $8^N/15 < P_N \leq 8^N$ , every such unscaled integer solution satisfies  $|c_N| = \Theta(8^N)$ . This is a concrete reason that quadratic growth of  $m_a$  does not imply polynomial growth of recursively defined carries. The starting index is essential: this does not exclude the actual tails becoming integral at a later index. After replacing  $m_a$  by  $Dm_a$  for an integer  $D \geq 1$ , however, a vanishing terminal term would require  $DS/2 = c_0$ . Excluding that possibility for every  $D$  would require irrationality itself; the unscaled calculation does not do so.

In the cited rearrangement the transformed integer numerator must be  $o(b_a)$ , which for bounded radices means eventual zero. Only after the terminal terms have been controlled does this cancellation determine the infinite sum. A polynomial bound on each individual decomposition coefficient would suffice for a fixed number of shifts: the corresponding series are absolutely convergent, since their denominators are products of at least 2 at each step. A polynomial bound on the combined numerators is not a substitute, as the exponentially growing carries above demonstrate. Nor does quadratic growth force a carry to be a polynomial in the floor coordinates.

## 8 A residue criterion and the bounds it allows

For example, the first two steps give

$$X_2 = 6X_1 - 4, \quad X_3 = 10X_2 - 7 = 60X_1 - 47.$$

If  $BX_1$  is an integer, then  $BX_3 \equiv -47B \pmod{60}$ . Thus a finite calculation determines the endpoint residue without determining the real tail itself. We next compare such residues with the tail bound.

A window starts at  $\ell \geq 0$  and consists of  $h \geq 0$  steps. Its product of bases and accumulated numerator are integers, even when the tails are not. Set

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \quad W_{\ell,h+1} = b_{\ell+h}W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h}F_{\ell,h} + m_{\ell+h}.$$

Induction gives

$$X_{\ell+h} = W_{\ell,h}X_\ell - F_{\ell,h}, \quad d_{\ell+h} = W_{\ell,h}d_\ell - BF_{\ell,h} \tag{15}$$

for any sequence satisfying  $d_{n+1} = b_nd_n - Bm_n$ . Since  $b_a = P_{a+1}/P_a$ , the product telescopes to  $W_{\ell,h} = P_{\ell+h}/P_\ell$ . The previously proved bounds  $8^a/15 < P_a \leq 8^a$  therefore give  $W_{\ell,h} > 8^h/15$ . For a positive integer  $C$  and an integer  $N$ , use  $\text{lpr}_C(N) = 1 + ((N-1) \bmod C) \in \{1, \dots, C\}$ . In particular  $\text{lpr}_C(0) = C$ . Positivity of this representative is what allows comparison with a positive integral carry.

**Proposition 8.1** (least positive residues). *Let  $C > 0$  and let  $c$  be an integer with  $0 < c$  and  $|c| \leq K$ . If  $c \equiv N \pmod{C}$  and  $K < \text{lpr}_C(N)$ , then the hypotheses are contradictory.*

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*Proof.* Every positive integer congruent to  $N$  is at least  $\text{lpr}_C(N)$ . Thus  $\text{lpr}_C(N) \leq c = |c| \leq K$ , contrary to the strict inequality.  $\square$

For a bound  $G : \mathbb{N}_{>0} \times \mathbb{N} \rightarrow \mathbb{N}$  let  $E(G)$  be the statement

$$\begin{aligned} &\text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1, \\ &\text{there are } \ell \geq a_0 \text{ and } h \geq 1 \text{ with } \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) > G(B, \ell + h). \end{aligned} \quad (16)$$

The window may depend on both  $B$  and  $a_0$ . All quantities in the inequality are finite integers. The quantifiers over denominators and prescribed onsets are nevertheless unbounded; a search over a finite rectangle of  $(B, \ell)$  does not verify them. Every  $b_a > 0$ , so  $W_{\ell,h} > 0$  is automatic.

The two hypotheses on  $G$  in the next theorem serve different purposes. Domination of  $K$  lets a residue above  $G$  exclude a positive integer tail. The limit  $G(B, a)/8^a \rightarrow 0$  lets every sufficiently long window from a fixed nonintegral start overtake  $G$ . It is a sufficient growth condition, not a necessary one: if only existence of a window is required, a subsequence of small endpoint bounds can suffice, as we prove below. The stated theorem already covers every polynomial upper bound that dominates  $K$ , and also  $\max\{K(B, a), \lceil B\rho^a \rceil\}$  for  $1 < \rho < 8$ . The zero bound fails to control the possible integer tail.

**Theorem 8.2** (a residue criterion for every dominating bound of size  $o(8^a)$ ). *Let  $G : \mathbb{N}_{>0} \times \mathbb{N} \rightarrow \mathbb{N}$  satisfy  $K(B, a) \leq G(B, a)$  for all  $B$  and  $a$ , and  $G(B, a)/8^a \rightarrow 0$  as  $a \rightarrow \infty$  for each fixed  $B$ . Then*

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$$E(G) \iff S \notin \mathbb{Q}.$$

Both  $K$  of (14) and  $K_0(B, a) = 90B(a + 1)^2$  satisfy these hypotheses, so  $E(K)$ ,  $E(K_0)$  and irrationality of  $S$  are mutually equivalent. By contrast,  $E(0)$  holds automatically, since every least positive residue is at least 1; its truth alone therefore provides no contradiction to an integral tail.

*Proof.* Suppose  $E(G)$  and suppose  $S = N/D$  were rational. Theorem 7.3 supplies  $B \geq 1$  coprime to 30, an onset  $a_D$ , and positive integers  $d_a = BX_a \leq K(B, a) \leq G(B, a)$  for  $a \geq a_D$  satisfying the cleared recurrence. Apply (16) with  $a_0 = a_D$  to obtain a window  $(\ell, h)$  with  $\ell \geq a_D$ . By (15),  $d_{\ell+h} \equiv -BF_{\ell,h}$  modulo  $W_{\ell,h}$ , while  $0 < d_{\ell+h} \leq G(B, \ell + h) < \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h})$ . Proposition 8.1 is the contradiction, so  $S$  is irrational.

Conversely suppose  $S \notin \mathbb{Q}$ , and fix  $B \geq 1$  coprime to 30 and  $a_0 \geq 1$ . Set  $\ell = a_0$  and  $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0, 1)$ ; the prefix identity makes  $BX_\ell$  irrational. For all sufficiently large  $h$ ,  $0 < \delta + BX_{\ell+h}/W_{\ell,h} < 1$ , because  $X_{\ell+h} = O((\ell + h + 1)^2)$  and  $W_{\ell,h} > 8^h/15$  as shown above. The integer  $\lceil BX_\ell \rceil W_{\ell,h} - BF_{\ell,h}$  is therefore exactly

$$\text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \delta W_{\ell,h} + BX_{\ell+h}.$$

Its first term eventually exceeds  $G(B, \ell + h)$ , since  $G(B, a) = o(8^a)$ . Thus every sufficiently long window from this fixed start escapes, which is stronger than the required existence.

For the two stated bounds,  $n_a \leq 3a$  gives  $K(B, a) \leq BQ(3a) \leq 90B(a+1)^2 = K_0(B, a)$ . Both are  $O((a+1)^2)$  for fixed  $B$ , so both satisfy the required limit. Finally,  $\text{lpr}_C(N) \geq 1 > 0$  always, independently of whether any tail is integral; a residue greater than zero cannot exclude a positive integer.  $\square$

The domination assumption is sufficient, not asserted to be necessary for equivalence. In particular, Proposition 7.6 already supplies the smaller valid bound  $\lfloor B\widehat{Q}(n_a) \rfloor$ . Replacing  $K$  by that bound in the forward implication leaves the argument unchanged; the converse uses only  $G(B, a) = o(8^a)$ . The zero-bound example explains why a size comparison with the possible integer tail is needed in the contradiction, not why this particular  $K$  is indispensable.

The reverse implication gives more than existence: at any fixed start with  $BX_\ell$  non-integral, the least positive residue eventually occupies a fixed positive fraction of the whole modulus. The next proposition states this without assuming that  $S$  is irrational.

**Proposition 8.3** (the fixed-start residue limit). *For fixed integers  $B, \ell \geq 1$ , write*

$$R_h = \text{lpr}_{W_{\ell,h}}(-BF_{\ell,h}).$$

*For all sufficiently large  $h$ ,*

$$R_h = (\lceil BX_\ell \rceil - BX_\ell)W_{\ell,h} + BX_{\ell+h},$$

*and consequently*

$$\lim_{h \rightarrow \infty} \frac{R_h}{W_{\ell,h}} = \lceil BX_\ell \rceil - BX_\ell.$$

*In particular, if  $BX_\ell$  is integral, then  $R_h = BX_{\ell+h}$  for all sufficiently large  $h$ .*

*Proof.* The least positive residue is the modulus times the gap to the next strictly larger integer. Division with remainder gives

$$\frac{R_h}{W_{\ell,h}} = \left\lfloor \frac{BF_{\ell,h}}{W_{\ell,h}} \right\rfloor + 1 - \frac{BF_{\ell,h}}{W_{\ell,h}}.$$

By the window identity,  $BF_{\ell,h}/W_{\ell,h} = BX_\ell - BX_{\ell+h}/W_{\ell,h}$ . These finite sums approach  $BX_\ell$  strictly from below, since  $BX_{\ell+h} > 0$  and  $BX_{\ell+h}/W_{\ell,h} \rightarrow 0$ . Their floors therefore eventually equal  $\lceil BX_\ell \rceil - 1$ , including when  $BX_\ell$  is an integer. Substitution proves both claims. The strict approach from below matters at an integer limit; the residue convention alone would assign a zero congruence class the value  $W_{\ell,h}$ , not zero.  $\square$

Thus a nonintegral  $BX_\ell$ , even for a rational  $S$ , gives arbitrarily long successful windows against any bound  $G(B, a) = o(8^a)$  for that fixed  $B$ . The rationality contradiction instead uses the denominator's specific multiplier and a start beyond its clearing onset. Success for one pair  $(B, \ell)$  does not replace either quantifier.

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**The exact growth restriction on a dominating bound.** A bound need not be small at every late endpoint. For example, set

$$G(B, a) = \begin{cases} K(B, a), & a \text{ even}, \\ K(B, a) + 8^a, & a \text{ odd}. \end{cases}$$

At an odd endpoint  $a = \ell + h$ , no window with  $\ell \geq 1$  can pass: its residue is at most  $W_{\ell, h} = P_a/P_\ell \leq 8^a/2 < G(B, a)$ . At even endpoints the test is unchanged. If  $S$  is irrational, the proof above therefore supplies successful windows ending at every sufficiently large even index, from each fixed start. Thus  $G(B, a)/8^a \rightarrow 0$  is not necessary for the equivalence.

More precisely, for every integer-valued  $G \geq K$ ,

$$E(G) \iff \begin{cases} S \notin \mathbb{Q}, \\ \liminf_{a \rightarrow \infty} \frac{G(B, a)}{8^a} = 0 \text{ for every } B \geq 1 \text{ with } \gcd(B, 30) = 1. \end{cases}$$

Suppose first that  $E(G)$  holds. The earlier rationality contradiction uses only  $G \geq K$ . For a successful window starting at  $\ell \geq a_0$ , put  $a = \ell + h$ . Since a least positive residue is at most its modulus,

$$0 \leq \frac{G(B, a)}{8^a} < \frac{P_a}{P_\ell 8^a} \leq \frac{1}{P_\ell} \leq \frac{1}{P_{a_0}}.$$

Such endpoints satisfy  $a > a_0$ , and  $P_{a_0} \rightarrow \infty$ ; hence the lower limit is zero. Conversely, suppose  $S$  is irrational and these lower limits vanish. Fix  $B$  and  $\ell \geq 1$ . By the fixed-start residue limit, with  $\delta = \lceil BX_\ell \rceil - BX_\ell > 0$ ,

$$\frac{\text{lpr}_{W_{\ell, h}}(-BF_{\ell, h})}{P_{\ell+h}} \rightarrow \frac{\delta}{P_\ell} > 0.$$

The bounds  $8^a/15 < P_a \leq 8^a$  imply  $\liminf_a G(B, a)/P_a = 0$ . Along a sufficiently late subsequence of endpoints the residue therefore exceeds  $G$ . This gives arbitrarily long successful windows from each fixed start, but need not give success at every sufficiently large length, as the parity example shows.

In particular,  $G(B, a) = K(B, a) + 8^a$  allows no window at all. More generally, an eventual lower bound  $G(B, a) \geq c8^a$ , for one eligible  $B$  and some  $c > 0$ , excludes every sufficiently late start: choose  $P_\ell \geq 1/c$  and compare the bound with  $W_{\ell, h} \leq 8^{\ell+h}/P_\ell$ . These are restrictions on the test bound, not additional assumptions about the series. The original quadratic bounds satisfy the lower-limit condition, so the remaining problem is still the existence of the windows. Equivalence alone says nothing about the relative difficulty of the formulations; a useful arithmetic proof must exploit further information about the numerators  $m_a$ .

A fixed window can have a lower bound greater than 1. Write  $W$  and  $F$  for its product and accumulated numerator. Every prime factor of  $W$  lies in  $\{2, 3, 5\}$ , so an eligible multiplier  $B$  is a unit modulo  $W$ . Congruence then gives

$$\gcd(\text{lpr}_W(-BF), W) = \gcd(F, W).$$

For  $(W, F) = (6, 4)$ , the possible residues are 2 and 4, attained at  $B = 1$  and 11; thus 2 is a lower bound for this window, and the residues are not units. For  $(W, F) = (60, 47)$ , however,  $\text{lpr}_{60}(-37 \cdot 47) = 1$ , with 37 coprime to 30. Thus no lower bound greater than 1 works for all windows and eligible multipliers, although a particular window may have one.

### 8.1 How fast the window base grows

Using only  $W_{\ell, h} \geq 2^h$  in the converse proof gives the sufficient condition  $G(B, a) = o(2^a)$  for each fixed  $B$ . The exact height formula instead gives growth comparable to  $8^h$ , uniformly in the start, and hence admits the larger class  $G(B, a) = o(8^a)$ . For an irrational  $S$ , the exact restriction on a dominating bound is the preceding lower-limit condition, not either little- $o$  bound.

**Proposition 8.4** (growth of the window product). *Put  $\theta_3 = \log_3 2$  and  $\theta_5 = \log_5 2$ . For all  $\ell \geq 0$  and  $h \geq 1$ ,*

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$$W_{\ell, h} = 2^h 3^{\lfloor (\ell+h)\theta_3 \rfloor - \lfloor \ell\theta_3 \rfloor} 5^{\lfloor (\ell+h)\theta_5 \rfloor - \lfloor \ell\theta_5 \rfloor}, \quad \frac{8^h}{15} < W_{\ell, h} < 15 \cdot 8^h.$$

*Proof.* Telescoping (10) gives  $W_{\ell, h} = P_{\ell+h}/P_\ell$ , and the displayed formula is that quotient written out. Each floor difference differs from  $h\theta_p$  by less than one, and  $3^{\theta_3} = 5^{\theta_5} = 2$ , so the 3-factor lies strictly between  $2^h/3$  and  $3 \cdot 2^h$  and the 5-factor strictly between  $2^h/5$  and  $5 \cdot 2^h$ . Multiplying the three ranges gives the bounds.  $\square$

**Corollary 8.5** (a fixed maximum length cannot cover arbitrarily late starts). *Fix  $B \geq 1$  coprime to 30 and  $H \geq 1$ . Only finitely many starts  $\ell$  admit an escaping window of length at most  $H$  against the bound  $K$ .*

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*Proof.* A least positive residue never exceeds its modulus, so escape at  $(\ell, h)$  requires  $K(B, \ell + h) < W_{\ell, h} < 15 \cdot 8^h \leq 15 \cdot 8^H$ . On the other hand  $j_a \geq a - 1$ , since the powers  $2, \dots, 2^{a-1}$  already lie below  $2^a$ , so  $K(B, \ell + h) \geq \lfloor B((\ell - 1)^2 + 10(\ell - 1) + 27)/9 \rfloor$ , which tends to infinity with  $\ell$ .  $\square$

Corollary 8.5 is the exact reason a finite scan cannot approach the cofinal quantifier by widening its denominator range alone. For  $\ell \geq 1$ , the inequality  $K(B, \ell + h) > B\ell^2/9$  shows that an escaping window must satisfy

$$3h > \log_2 K(B, \ell + h) - \log_2 15 > \log_2 B + 2\log_2 \ell - \log_2 135.$$

This is only a necessary lower bound, not an asymptotic formula or an upper bound for the first successful length. An explicit eventual escape threshold can be obtained from a positive lower bound for  $\delta = \lceil BX_\ell \rceil - BX_\ell$  and for  $1 - \delta$ , together with the displayed tail and window estimates. No such uniform information about the tails of  $S$  is proved here.

## 9 Finite computations and their limits

The three displayed window tuples and the twelve-shell denominator bound below have directly replayable finite certificates. The larger scan and the two much larger denominator exclusions are also replayable: their programs, machine-readable witnesses and saved outputs are published in the [large-computation directory](#). They remain finite computations outside the theorem inputs, and none supplies the unbounded quantifiers.

### 9.1 Checking individual windows

The three rows below record individual instances of the residue inequality (16). The integer-only [dyadic-window checker](#) computes them, and exact enumeration reproduces them. For each required shell, enumerate the pairs with  $3^j 5^k < 2^{a+1}$ , choose the unique exponent  $i$  that puts  $2^i 3^j 5^k$  in  $[2^a, 2^{a+1})$ , and add  $P_{a+1} / (2 H(2^i 3^j 5^k))$ . All powers and comparisons are integral. The recursions for  $W$  and  $F$  then give the displayed window data. The columns give the denominator  $B$ , the start  $\ell$ , the length  $h$ , the endpoint jump index  $j_{\ell+h}$ , the product  $W$  of the bases, the accumulated numerator  $F$ , the residue  $R = \text{lpr}_W(-BF)$  and the bound  $K$  of (14).

| $B$ | $\ell$ | $h$ | $j_{\ell+h}$ | $W$   | $F$  | $R$ | $K$ |
|-----|--------|-----|--------------|-------|------|-----|-----|
| 1   | 1      | 2   | 4            | 60    | 47   | 13  | 9   |
| 7   | 1      | 3   | 6            | 360   | 289  | 137 | 95  |
| 16  | 1      | 4   | 9            | 10800 | 8735 | 640 | 352 |

The first row reads as follows. The window starts at  $\ell = 1$  and has length 2, so  $W = b_1 b_2 = 6 \cdot 10 = 60$ ; the accumulated numerator is  $F = 47$ ; and  $\text{lpr}_{60}(-47) = 13$ , since  $-47 + 60 = 13$ , which exceeds  $K(1, 3) = \lfloor (16 + 40 + 27)/9 \rfloor = 9$ . The third row lies outside the domain of (16), since  $\gcd(16, 30) = 2$ , and is displayed to illustrate the window arithmetic at greater depth.

The larger scan covers  $B \leq 5000$  coprime to 30 and  $100 \leq \ell \leq 3000$ : all 3,869,934 pairs escape, with first escape length at most 18 in a search to length 24 ([saved run](#)). Its histogram in Section 11.4 is a finite observation that the window-growth bound does not predict. The accompanying integer-only check reconstructs the shells and tests all 2496 pairs with  $1 \leq B \leq 97$ ,  $\gcd(B, 30) = 1$  and  $1 \leq \ell \leq 96$ . Every pair escapes; the first successful lengths range from 1 to 10, with a search limit of 24. This check uses the enumeration and window recursions above, without floating-point logarithms. Neither computation proves escape for unbounded  $B$  or arbitrarily late starts; Corollary 8.5 rules out covering the latter quantifier with any fixed maximum length.

### 9.2 Finite denominator bounds

**A denominator bound from twelve shells.** The actual tail bound gives a small certificate whose integer data fit on the page. Since  $h_1 = 1$ , we have  $X_1 = S - 1$ . Finite summation through shell 11, with  $P_{12} = 27993600000$  and  $Q(n_{12}) = Q(24) = 262/3$ , gives

$$X_1 = 2 \sum_{a=1}^{11} \frac{m_a}{P_{a+1}} + \frac{2X_{12}}{P_{12}},$$

$$\frac{11327881097}{13996800000} < X_1 \leq \frac{33983643553}{41990400000}.$$

The enclosure is strictly inside the interval  $(11498/14207, 6409/7919)$ , as integer cross-multiplication verifies. The endpoint determinant is  $14207 \cdot 6409 - 11498 \cdot 7919 = 1$ . For any rational  $u/v$  strictly between them, with  $v > 0$ , both integers  $14207u - 11498v$  and  $6409v - 7919u$  are positive. Consequently

$$v = 7919(14207u - 11498v) + 14207(6409v - 7919u) \geq 22126.$$

Thus, if  $S$  is rational, its reduced denominator is at least 22126. This is the elementary denominator bound between two fractions with cross-determinant one, often expressed using Farey neighbours. It is also the best lower bound on a possible denominator obtainable from this enclosure alone: the mediant

$$\frac{11498 + 6409}{14207 + 7919} = \frac{17907}{22126}$$

is in lowest terms and lies strictly inside the certified enclosure. This does not identify  $X_1$  with the mediant or assert that  $S$  has that denominator. It says that this interval is still compatible with a rational of denominator 22126.

The certificate uses the actual shell coefficients and proved tail bound. A narrower certified enclosure excludes further denominators; the continued-fraction exclusion below uses one built from 15,500 shells.

**Window-128 lattice exclusion.** A finite lattice computation, outside the theorem inputs, excludes the following denominators. Using windows of length 128 and 64 starting indices, the first 10005, no rational value of the  $\{2, 3, 5\}$  running-LCM series has reduced denominator  $MB$  with  $M$  a divisor supported on  $\{2, 3, 5\}$  of  $2^{10005}3^{6312}5^{4308}$ ,  $\gcd(B, 30) = 1$  and  $1 < B \leq B_{\max}$ , where  $B_{\max}$  is the 106-digit integer

$$B_{\max} = 1134599670999687767349520845707093359257353022286558739363600235 \\ 016103207564063373270305324172145281971729,$$

so that  $\log_2 B_{\max} = 348.9846 \dots$

The exponent triple  $(10005, 6312, 4308)$  is that of  $H(2^{10005})$ . The normalisation therefore uses the full height at its first start, whereas Lemma 7.2 uses the half height  $h_{10005}$ . This family of smooth denominators is the larger one.

The [certificate](#) and its exact [checker](#) are published. Its starts  $a_1 < \dots < a_{64}$  are the first  $a \geq 10005$  with  $W_{a,128} = W_{10005,128} = W = 2^{128}3^{81}5^{56}$ , so  $\log_2 W = 386.40993 \dots$ ; put  $m_0 = W/2$ ,  $F_j = F_{a_j,128}$  and  $w_j = (a_j + 128)^2 + 6(a_j + 128) + 11$ . The certificate lists an integer matrix  $C = (c_{ij})$  with  $\sum_j c_{ij}F_j \equiv 0 \pmod{m_0}$  and  $\sum_j |c_{ij}|w_j \leq 184,560,461,211$  for every row  $i$ , an integer matrix  $E$  with  $CE = EC = m_0I$ , and  $B_{\max} = \lfloor (W - 1)/184,560,461,211 \rfloor$ . If  $B \leq B_{\max}$  and integers  $f_j = Ws_j - 2BF_j$  satisfy  $0 < f_j \leq Bw_j$ , each  $\sum_j c_{ij}f_j$  is a multiple of  $W$  of absolute value less than  $W$ , hence zero, and invertibility of  $C$  forces every  $f_j$  to vanish, a contradiction. The weights bound the tails: since  $n_b \leq 2.07b$ , Theorem 7.1 gives  $2X_b \leq 2Q(n_b) \leq b^2 + 6b + 11$  for every  $b \geq 0$ . The reduction from a rational value  $N/(MB)$  to the 64 window inequalities, with  $s_j = 2BX_{a_j}$  and  $f_j = 2BX_{a_j+128}$ , uses the window identity at full height and is not machine-checked.

**Continued-fraction exclusion.** A finite computation, outside the theorem inputs, shows that neither  $X_1$  nor  $S$  is a fraction with reduced denominator at most  $2^{22482}$ , about  $10^{6768}$ . Let  $W$  and  $F$  be the window base and accumulated numerator of shells 1 to 15,500. The tail bound used for the twelve-shell certificate, at  $Q(n_{15501}) = Q(31956) = 340480534/3$ , gives  $F/W < X_1 \leq (F + Q)/W$ . Every shell is built with integers, and shells 1 to 30 are also checked against a direct enumeration of the smooth numbers. The two endpoints share their first 13,540 partial quotients, and the numbers with a given prefix of partial quotients form an interval [23, Lemma 10.1.2, p. 106]. The first 13,109 digits,  $a_0 = 0$  included, define a cylinder whose endpoints have cross-determinant 1 and strictly contain the enclosure, so every fraction in the enclosure has denominator at least the sum of the two endpoint denominators, a 22,483-bit integer. The [enclosure](#), the [common digits](#), the [regeneration program](#) and the exact [checker](#) are published. A separate continued-fraction probe ([saved run](#)) certifies 13,109 partial quotients after  $a_0$  and records a largest denominator of 22,483 bits, a largest partial quotient of 129,114, a mean partial quotient of 23.4133, observed Gauss–Kuzmin frequencies 0.4208, 0.1665, 0.0917, 0.0575, 0.0391 against the predicted 0.4150, 0.1699, 0.0931, 0.0589, 0.0406, and a Lévy constant of 1.18869 against  $\pi^2/(12 \log 2) = 1.18657$ . The predicted values are the almost-everywhere frequencies of Gauss’s law and Lévy’s almost-everywhere constant [22, pp. 288–289 and footnote 5]. These statistics describe a finite prefix and do not bear on whether  $X_1$  is a Liouville number, algebraic, or rational with a larger denominator.

The two exclusions concern different finite families: one uses a lattice at a fixed starting index and fixed smooth part, the other a continued-fraction enclosure at  $a = 1$ . Both have published witnesses and exact checkers. Each excludes only its stated denominator range, and neither settles an instance of the problem.

## 10 The remaining arithmetic questions

### 10.1 An exact weighted-shift identity for the repeated series

We return to the repeated sum  $S$ ; the distinct-height sum  $\mathcal{D}_{\{2,3,5\}}$  plays no part in this subsection. Set  $\alpha = S/2$  and keep the boundary heights  $P_a$  defined above. A shifted tail represents the same value after a finite rational correction. The following identity combines several such shifts with fixed integer coefficients. Its weights are necessary: ordinary shifts of the numerators alone would not account for the changing denominators. In this subsection and the next,  $r$  denotes a positive integer shift, not a prime generator.

**Lemma 10.1** (weighted shifts preserve the actual value). *Fix integers  $c_0, \dots, c_\sigma$ , not all zero, independently of the positive integer shift  $r$ . Put*

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$$\gamma_{a,t} = \frac{P_t P_{a+1}}{P_{a+t+1}}, \quad D_{r,a} = 15 \sum_{j=0}^{\sigma} c_j \gamma_{a,jr} m_{a+jr}.$$

Then  $\gamma_{a,t} \in \{1, 1/3, 1/5, 1/15\}$  and  $D_{r,a} \in \mathbb{Z}$ . Furthermore, with

$$A_r = 15 \sum_{j=0}^{\sigma} c_j P_{jr}, \quad Z_r = 15 \sum_{j=0}^{\sigma} c_j P_{jr} \sum_{k < jr} \frac{m_k}{P_{k+1}} \in \mathbb{Z},$$

we have the absolutely convergent identity

$$A_r \alpha - Z_r = \sum_{a \geq 0} \frac{D_{r,a}}{P_{a+1}}. \quad (17)$$

One may take  $C = 225 \sum_{j=0}^{\sigma} |c_j| \max(1, j)^2$  in  $|D_{r,a}| \leq C(a+r+1)^2$ . If  $J$  is the largest index with  $c_J \neq 0$ , then  $A_r \neq 0$  whenever  $\sum_{j < J} |c_j| 2^{-(J-j)r} < |c_J|$ ; an empty sum is zero.

*Proof.* The exponent of 2 in  $\gamma_{a,t}$  is  $t + (a+1) - (a+t+1) = 0$ . For  $p = 3, 5$ , the exponent is  $\lfloor t\theta_p \rfloor + \lfloor (a+1)\theta_p \rfloor - \lfloor (a+t+1)\theta_p \rfloor$ , which is 0 or  $-1$  by floor addition. Thus the only possible denominator factors are one 3 and one 5; the prefactor 15 in  $D_{r,a}$  clears both. For  $k < jr$ ,  $P_{k+1}$  divides  $P_{jr}$ , so  $Z_r$  is an integer. For each  $t$ , absolute convergence gives

$$\sum_{a \geq 0} \frac{\gamma_{a,t} m_{a+t}}{P_{a+1}} = P_t \sum_{k \geq t} \frac{m_k}{P_{k+1}} = P_t \left( \alpha - \sum_{k < t} \frac{m_k}{P_{k+1}} \right).$$

The finite linear combination proves the identity. Lemma 6.3,  $0 < \gamma \leq 1$  and  $a+jr+1 \leq \max(1, j)(a+r+1)$  give the stated constant. For  $j < J$ ,  $P_{jr}/P_{Jr} \leq 2^{-(J-j)r}$ . Dividing  $A_r$  by  $15P_{Jr}$  and bounding the remaining terms proves the nonvanishing condition, which holds for all sufficiently large  $r$ .  $\square$

For any fixed nonzero coefficient vector, including  $(1, -3, 3, -1)$ , the nonvanishing condition is automatic for sufficiently large  $r$ : each term on its left tends to zero. No arithmetic assumption on  $S$  is needed. The identity still represents  $S/2$ , but gives no sparsity or cancellation of  $D_{r,a}$ .

## 10.2 Why the growing boundary need not cancel

The numerators count weighted lattice points in growing triangles. Suppose first that a point's weight is unchanged by the four shifts. In a third difference with coefficients  $(1, -3, 3, -1)$ , a point present in all four triangles has total coefficient zero. A point entering only the last three, last two or last triangle has coefficient  $-1, 2$  or  $-1$ , respectively. The common interior therefore cancels, but the new boundary strips need not. We now separate these contributions from changes in the weights caused by floor crossings.

Recall that  $w_{j,k} = j \log_2 3 + k \log_2 5$ ,  $t_{j,k} = \{w_{j,k}\}$  and  $\theta_p = 1/\log_2 p$  for  $p = 3, 5$ . To distinguish the shared points from the new ones, let  $\mathcal{T}_a = \{(j, k) \in \mathbb{N}^2 : w_{j,k} < a+1\}$  and, for  $0 \leq t < 1$ , set

$$\omega_a(t) = \prod_{p \in \{3,5\}} p^{\lfloor (a+1)\theta_p \rfloor - \lfloor (a+t)\theta_p \rfloor}.$$

This weight is defined even for pairs outside  $\mathcal{T}_a$ . For  $v \geq 0$ , put

$$\begin{aligned} \kappa_p(a, t, vr) &= \lfloor (a+t+vr)\theta_p \rfloor - \lfloor (a+t)\theta_p \rfloor - \lfloor vr\theta_p \rfloor, \\ \chi_v(a, t) &= 3^{-\kappa_3(a,t,vr)} 5^{-\kappa_5(a,t,vr)}. \end{aligned}$$

Each  $\kappa_p$  is 0 or 1. A value  $\kappa_p = 1$  records a floor-addition carry. This is different from a new lattice point entering  $\mathcal{T}_a$  as  $a$  increases.

**Proposition 10.2** (an exact interior-and-strip decomposition). *For a fixed operator  $c_0, \dots, c_\sigma$  and  $r \geq 1$ , let  $E_0 = \mathcal{J}_a$  and  $E_s = \mathcal{J}_{a+sr} \setminus \mathcal{J}_{a+(s-1)r}$  for  $1 \leq s \leq \sigma$ . Then*

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$$\frac{D_{r,a}}{15} = \sum_{s=0}^{\sigma} \sum_{(j,k) \in E_s} \omega_a(t_{j,k}) \sum_{v=s}^{\sigma} c_v \chi_v(a, t_{j,k}).$$

For the cubic operator  $(1, -3, 3, -1)$ , if all these crossing bits vanish, then

$$\frac{D_{r,a}}{15} = -M_0 + 2M_1 - M_2, \quad M_s = \sum_{(j,k) \in \mathcal{J}_{a+(s+1)r} \setminus \mathcal{J}_{a+sr}} \omega_a(t_{j,k}).$$

Thus absence of floor crossings cancels the common interior, but not necessarily the three boundary strips.

*Proof.* For each  $p = 3, 5$ , the exponent of  $p$  on either side of

$$\gamma_{a,vr} \omega_{a+vr}(t) = \omega_a(t) \chi_v(a, t)$$

is

$$\lfloor (a+1)\theta_p \rfloor + \lfloor vr\theta_p \rfloor - \lfloor (a+t+vr)\theta_p \rfloor.$$

On the left the upper-boundary exponents cancel; on the right the  $\lfloor (a+t)\theta_p \rfloor$  terms cancel. This proves the pointwise identity. Insert the triangle formula for each  $m_{a+vr}$  and group by the first triangle containing a pair. A pair in  $E_s$  occurs exactly in the terms  $v \geq s$ , proving the first formula. Under the no-crossing hypothesis,  $\chi_v = 1$ . The full cubic sum is zero, while its three successive suffix sums are  $-1, 2, -1$ , proving the second formula.  $\square$

**Example 10.3** (cubic cancellation can fail with no floor crossings). Take  $a = 0, r = 35$  and  $(c_0, c_1, c_2, c_3) = (1, -3, 3, -1)$ . Exact integer comparisons give

| $u$                                                           | 0 | 35  | 70  | 105  |
|---------------------------------------------------------------|---|-----|-----|------|
| $\lfloor \log_3 2^u \rfloor = \lfloor \log_3 2^{u+1} \rfloor$ | 0 | 22  | 44  | 66   |
| $\lfloor \log_5 2^u \rfloor = \lfloor \log_5 2^{u+1} \rfloor$ | 0 | 15  | 30  | 45   |
| $m_u$                                                         | 1 | 195 | 723 | 1582 |

The first two rows imply  $b_u = 2$  and  $\lfloor (u+t)\theta_p \rfloor = \lfloor u\theta_p \rfloor$  for every  $0 \leq t < 1, p = 3, 5$ . Hence every relevant  $\kappa_p(0, t, u)$  vanishes,  $\gamma_{0,u} = 1$ , and all triangle weights at these four indices are 1. The last row is therefore the count of pairs satisfying  $3^j 5^k < 2^{u+1}$ , not a floating-point estimate. The strips have sizes 194, 528, 859, so

$$D_{35,0} = 15(-194 + 2 \cdot 528 - 859) = 45 \neq 0.$$

Exact enumeration reproduces all four counts and checks the endpoint power inequalities using integers. This example disproves the claim that avoiding floor crossings alone forces the third difference to vanish. It does not exclude a different fixed operator, different shifts, a boundary correction or a sparse-defect statement restricted to a suitable infinite subsequence.

**What the Hecke–Mahler comparison does and does not supply.** In the published Luca–Ouaknine–Worrell article [25, Definition 5 and Theorem 6], the same fixed integer coefficients are applied to shifts of one integer sequence. The resulting nonzero terms must have expanding gaps and uniform polynomial variation; the original sequence must also have polynomial growth. Theorem 6 then gives a fixed-base value criterion. Theorem 8 verifies that condition for  $f(\lfloor m\vartheta + \rho \rfloor)$  when  $f \in \mathbb{Z}[x]$  is nonconstant,  $\vartheta, \rho \in (0, 1)$  and  $\vartheta$  is irrational. These interval restrictions belong to that normalised combinatorial statement. Their main value theorem, Theorem 1, allows every real  $\rho$  and every irrational real  $\vartheta$ , with any algebraic base  $\beta$  satisfying  $|\beta| > 1$ . Claim 10 uses a finite difference of order  $\deg f + 1$ , which vanishes when no floor crossing occurs. This is a polynomial evaluated at a floor, not merely an integer sequence of polynomial growth. Our  $m_a$  is instead a weighted lattice count. Proposition 10.2 separates floor crossings from its moving boundary, and Example 10.3 shows that the latter can survive a cubic difference even with no crossings.

There is a second distinction. In our displayed shift identity the coefficient of  $m_{a+jr}$  is  $15c_j\gamma_{a,jr}$ , which can depend on  $a$ . For example,  $P_1 = 2$ ,  $P_2 = 12$  and  $P_3 = 120$  give  $\gamma_{0,1} = 1/3$  but  $\gamma_{1,1} = 1/5$ . These factors are required by the denominator chain. A support estimate for  $D_{r,a}$  is therefore not by itself a verification of Definition 5 for the sequence  $(m_a)$ . No fixed-coefficient representation to which that theorem applies has been established here.

For a variable-denominator analogue, a useful hypothesis would be a fixed choice of coefficients and shifts  $r_n \rightarrow \infty$  for which  $\Delta_n = \{a \geq 0 : D_{r_n,a} \neq 0\}$  is infinite and distinct members are at least  $\eta r_n$  apart, for a fixed  $\eta > 0$ . Even this would leave the uniform polynomial-variation requirement, for example

$$|D_{r_n,a'}| \leq C((a' - a)^d + |D_{r_n,a}|) \quad (a < a', a, a' \in \Delta_n),$$

with  $C, d$  independent of  $n, a$  and  $a'$ . This is a condition on the gaps between nonzero terms, not merely on their density: even a set of density zero can contain adjacent pairs. It also requires infinitely many nonzero terms for each selected shift. A small but nonzero term at every index would fail it, and an identically zero difference would also fail the infinitude requirement. The cited paper constructs suitable shifts for the polynomial-floor sequences just described; no such result is proved for our weighted counts. The bound  $O((a + r + 1)^2)$  controls absolute size: it depends on the location  $a$  itself. The variation condition instead bounds a later nonzero term by the gap from an earlier one and the size of that earlier term, with constants uniform in the shift. The absolute-size bound alone gives no such comparison. The denominator chain itself is not a fixed-base power sequence. The next proposition shows what is lost in two direct fixed-base recodings. A value criterion for the original chain would need a separate proof, including non-cancellation and the height estimates in any Subspace-Theorem argument. Here height means Diophantine height, which controls the numerators and denominators of the approximating algebraic quantities, not the running-LCM height  $H$ . The finite set of prime divisors alone does not provide those estimates.

**Why a finite alphabet of bases is not enough.** Kebis, Luca, Ouaknine, Scoones and Worrell [26, Definition 3, Theorem 6 and Claim 7] work with fixed-base series whose

coefficients lie in a finite algebraic alphabet. Their echoing condition requires long near-repetitions, separated intervals containing the mismatches, and nonzero weighted mismatch sums on at least two of those intervals. Claim 7 supplies the non-cancellation step in the proof of the value theorem. Merely having finitely many letters does not give these properties. The four-letter radix sequence is not the numerator sequence: the actual  $m_a$  is unbounded. A recoding would have to preserve the scalar value, identify its coefficient alphabet and verify the echoing conditions. No fixed-base recoding with verified echoing properties is proved here.

Two natural bases expose the tradeoff. The least integer base whose powers clear every  $P_n$  is 30, while base 8 matches the growth of  $P_n = H(2^n)$ . The former keeps integrality; the latter keeps polynomial size.

**Proposition 10.4** (what direct fixed-base recoding preserves). *The following identities converge absolutely:*

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$$\alpha = \sum_{a \geq 0} \frac{e_a}{30^{a+1}} = \sum_{a \geq 0} \frac{v_a}{8^{a+1}}, \quad e_a = \frac{m_a 30^{a+1}}{P_{a+1}}, \quad v_a = \frac{m_a 8^{a+1}}{P_{a+1}}.$$

Here  $e_a \in \mathbb{Z}_{>0}$  and  $e_a \geq (15/4)^{a+1}$ , whereas  $v_a \in \mathbb{Z}[1/15]$  and  $0 < v_a < 225(a+1)^2$ . For an integer  $q \geq 2$ , the termwise divisibility  $P_n \mid q^n$  for every  $n \geq 1$  holds exactly when  $30 \mid q$ ; that direct recoding then has coefficients at least  $(q/8)^{a+1}$ .

*Proof.* Each exponent in  $P_n$  is at most  $n$ , so  $P_n \mid 30^n$ . The estimate  $8^n/15 < P_n \leq 8^n$  was proved in Section 6. These facts,  $m_a \geq 1$  and  $m_a \leq 15(a+1)^2$ , give all the coefficient bounds and identities. Necessity of  $30 \mid q$  follows because each of 2, 3, 5 divides some  $P_n$ ; sufficiency follows from  $P_n \mid 30^n$ . The general lower bound follows again from  $P_n \leq 8^n$ .  $\square$

The base-30 coefficients are integral but grow exponentially, so the cited polynomial-growth criterion does not apply. The base-8 coefficients have polynomial size but are not all integral: already  $v_1 = 4 \cdot 8^2/12 = 64/3$ . In fact, their reduced denominators grow exponentially even after cancellation. Since  $P_{a+1}/2^{a+1}$  is odd, the expression

$$v_a = \frac{m_a 4^{a+1}}{P_{a+1}/2^{a+1}}$$

can cancel an odd factor only through  $m_a$ . Consequently,

$$\text{den}(v_a) \geq \frac{P_{a+1}}{2^{a+1}m_a} > \frac{4^{a+1}}{225(a+1)^2}.$$

The last inequality uses  $P_{a+1} > 8^{a+1}/15$  and  $m_a \leq 15(a+1)^2$ . Thus the bound on absolute value does not control Diophantine height even for these particular coefficients; fixed denominator-prime support is not enough.

More generally, no fixed integer base gives this termwise recoding both eventually integral coefficients and polynomial growth. If  $30 \mid q$ , the proposition gives exponential

growth. Otherwise choose  $p \in \{2, 3, 5\}$  not dividing  $q$ . Cancellation by  $m_a$  removes at most a factor  $m_a$ , so

$$\text{den}\left(\frac{m_a q^{a+1}}{P_{a+1}}\right) \geq \frac{p^{\lfloor (a+1) \log_p 2 \rfloor}}{m_a} > \frac{2^{a+1}}{15p(a+1)^2}.$$

This denominator tends to infinity. The argument concerns only the displayed termwise rescaling, not regrouping or carrying.

Carrying can in fact make the coefficients into digits, while leaving the radices variable. To see exactly what changes, set

$$\delta_a = m_a + \lfloor X_{a+1} \rfloor - b_a \lfloor X_a \rfloor = \lfloor b_a \{X_a\} \rfloor.$$

The equality follows from the tail recurrence and  $m_a \in \mathbb{Z}$ . Thus  $0 \leq \delta_a < b_a$  and  $\{X_a\} = (\delta_a + \{X_{a+1}\})/b_a$ . Iterating gives the finite identity

$$\{X_0\} = \sum_{a=0}^{N-1} \frac{\delta_a}{P_{a+1}} + \frac{\{X_N\}}{P_N}.$$

The last term is nonnegative, is less than  $1/P_N \leq 2^{-N}$ , and tends to zero. This is the greedy mixed-radix expansion of the fractional part of  $X_0 = S/2$ , with digits in  $\{0, \dots, 29\}$ . For the actual series its first eight digits are

$$(\delta_0, \dots, \delta_7) = (1, 4, 8, 3, 10, 1, 8, 5).$$

To certify them, use the twelve-shell enclosure from Section 9, either propagated by the recurrence or evaluated directly as

$$P_a \sum_{j=a}^{11} \frac{m_j}{P_{j+1}} < X_a \leq P_a \sum_{j=a}^{11} \frac{m_j}{P_{j+1}} + \frac{P_a}{P_{12}} Q(n_{12}) \quad (0 \leq a < 12).$$

Both descriptions give the same rational endpoints. At indices 0 through 8 the whole interval lies strictly between consecutive integers, which certifies the floor; a small interval width alone would not suffice if the interval crossed an integer. For example,  $\lfloor X_4 \rfloor = 2$  and  $\lfloor X_5 \rfloor = 5$ , giving  $\delta_4 = 65 + 5 - 30 \cdot 2 = 10$ , not the uncarried numerator  $m_4 = 65$ . This finite calculation makes no claim about the later digit complexity.

A rational number need not have eventually periodic digits when the radices vary. For example, take radix 5 at the indices 1, 2, 4, 8, ... and radix 3 elsewhere. The constant tail fraction  $1/2$  gives digit 2 at those indices and 1 elsewhere. These digits are not eventually periodic, but their mixed-radix expansion sums to  $1/2$ . This example uses different radices from our four possible bases; it shows why fixed-base eventual periodicity cannot be transferred to a variable-radix expansion. Our digits also depend on the actual tail fractions, not just on the four-letter radix sequence. The construction supplies neither a fixed-base expansion with verified echoing properties nor a bound on its digit complexity.

Adamczewski–Bugeaud’s complexity theorem [29, Theorem 1] concerns the actual digits of an integer-base expansion: for an algebraic irrational their length- $n$  block complexity divided by  $n$  tends to infinity. Neither  $(b_a)$  nor the uncarried  $(e_a)$  is such a digit expansion of  $\alpha$ .

**Problem 10.5** (the repeated three-prime target). Prove irrationality of  $S = \mathcal{R}_{\{2,3,5\}}$ . Equivalent forms are  $E(K)$  and

$$BX_a \notin \mathbb{Z} \quad \text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a \geq 1. \quad (18)$$

The residue formulation uses finite integer computations, but asks for a success for every eligible denominator and beyond every prescribed start. The tail formulation requires that no positive integer multiplier coprime to 30 make any  $X_a$ ,  $a \geq 1$ , integral. These are reformulations of the same irrationality assertion, not extra assumptions on the series, and neither follows from the rank obstruction. The next proposition proves the tail reformulation, complementing Theorem 8.2.

**Proposition 10.6** (irrationality is equivalent to nonintegrality of every reduced tail). *Statement (18), quantified over every  $B \geq 1$  coprime to 30 and every  $a \geq 1$ , is equivalent to irrationality of  $S$ .*

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*Proof.* If  $BX_a \in \mathbb{Z}$  for some such  $B$  and  $a$ , then  $X_a \in \mathbb{Q}$ , and since  $S = \sum_{j < a} s_j + X_a/h_a$  with  $h_a$  a positive rational and the prefix a finite sum of rationals,  $S \in \mathbb{Q}$ . Conversely if  $S \in \mathbb{Q}$ , then Theorem 7.3 produces  $B$  coprime to 30 with  $BX_a \in \mathbb{Z}$  for every  $a \geq a_D$ .  $\square$

Integrality at one index propagates forwards, but nonintegrality at one early index does not propagate forever: a rational denominator can clear later. Proposition 7.4 describes that onset exactly. The dichotomy in Proposition 6.4 also permits integral tails, so it does not settle this question.

### 10.3 What a function-theoretic proof would require

This subsection returns to the distinct-height sum of Section 4, in which each height is counted once, at the jump where it first appears:

$$\mathcal{D}_{\{2,3,5\}} = 1 + \sum_{t \in \{2^n, 3^n, 5^n : n \geq 1\}} \frac{1}{2^{\lfloor \log_2 t \rfloor} 3^{\lfloor \log_3 t \rfloor} 5^{\lfloor \log_5 t \rfloor}}.$$

The bases  $b_a$  in (10) come from the same prime-power jumps. This sum counts each height once; the repeated sum counts it once for every smooth integer at that height. Theorem 1.2 proves  $\mathcal{D}_{\{2,3,5\}}$  irrational by an elementary argument that gives no information about transcendence. For two primes, Section 3 deduces the transcendence of  $\mathcal{D}_{\{p,q\}}$  from one Hecke–Mahler value. A comparable statement at three primes would need a representation of the following kind.

**Problem 10.7** (an exact function representing the distinct-height sum). Transcendence of  $\mathcal{D}_{\{2,3,5\}}$  is not proved here. For a functional approach to it, first specify the function, its coefficient field and convergence domain, then prove an exact identity for  $\mathcal{D}_{\{2,3,5\}}$ , and only then apply a stated value theorem. A conditional theorem must display the extra nondegeneracy assumption; a no-representation theorem must specify the class it excludes.

For  $(p, q, r) = (2, 3, 5)$ , the two slopes in the rank proof are  $\log_5 2 = \theta_5$  and  $\log_5 3 = \theta_5/\theta_3$ , not  $\theta_3$  and  $\theta_5$ . Each is irrational: a rational value would equate a positive power of 5 with a positive power of 2 or 3. The proof chooses the row and column indices independently. It therefore needs no rational independence of  $1, \theta_3, \theta_5$  and no density assertion for the singly indexed orbit  $(\{n\theta_3\}, \{n\theta_5\})$ . Any functional approach that needs such a stronger hypothesis must establish it separately.

A finite-dimensional encoding must preserve equality of the functions it is meant to represent. If two encodings agree but their functions do not, the encoding cannot justify a conclusion about those functions. The Lean sources express this condition as a factorisation through a finite-dimensional space and prove that the resulting space of functions is then finite-dimensional ([identity for the carry](#), [map to functions](#), [finite-dimensional conclusion](#)). They also bound the carry residue and its digit in their stated intervals ([residue bound](#), [digit bound](#)). These conditional linear-algebra statements neither construct a function representing  $\mathcal{D}_{\{2,3,5\}}$  nor show that a corresponding function space must be infinite-dimensional.

**The cited value theorems and their hypotheses.** Pellarin’s rank-one theorem concerns quadratic irrational slopes and algebraic evaluation points in the convergence domain [30, Theorem 1.1 and Remark 1.2, pp. 330–332]. The prime-logarithm slopes used here are not quadratic: each  $\log_r p$  is irrational by unique factorisation, and the Gelfond–Schneider theorem then makes it transcendental, since  $r^{\log_r p} = p$  is algebraic.<sup>1</sup> This rules out direct substitution of those slopes into Pellarin’s quadratic-slope theorem. It does not rule out a different representation of the scalar sum, and it does not affect the two-prime argument above, which uses a theorem for general irrational slopes.

In Pellarin’s reduced-slope normalisation, the rank-one condition says that the evaluation points arise from a common point by the quadratic order’s monomial action, up to multiplication by torsion points. For a single evaluation point it is automatic: use that point itself and the identity action. For several points it is a condition to check, not a consequence of having two lattice coordinates. Under it, algebraic independence of the values is equivalent to  $\mathbb{Q}$ -linear independence of the specified auxiliary formal Laurent series; the rank-one condition alone does not imply independence. This module rank is not the rank of our reciprocal kernel. No exact representation suitable for this theorem has been constructed here, but infinite kernel rank alone does not preclude one.

For one variable, Adamczewski and Faverjon’s proofs of Nishioka’s theorem and the lifting theorem are regular-point results [31, Theorems 1 and 2]. Regularity means that the system matrix stays defined and invertible along the evaluation orbit. Their multivariate lifting theorem [32, Theorem 3.3] additionally assumes an admissible transformation–point pair. Its growth, decay and nonvanishing conditions are specified in Definition 5.1; none follows merely from writing a double sum. Lifting transfers a relation among values to one among functions; a transcendence application must exclude the latter. A Mahler system alone cannot do this, as the constant function 1 illustrates.

<sup>1</sup>For this logarithmic form of the Gelfond–Schneider theorem, see Keith Conrad, *Transcendence of  $e$* , p. 1, footnote 1, <https://kconrad.math.uconn.edu/blurbs/analysis/transcendence-e.pdf> (accessed 18 September 2026).

Regularity of a preselected system is not, however, a universal prerequisite for Mahler's method. The same paper's Theorem 1.1 treats univariate Mahler functions at algebraic points in the punctured unit disc where their values are defined, without that regularity hypothesis. The hypotheses must therefore be matched to the particular theorem being invoked. No exact Mahler representation or suitable evaluation data for this distinct-height sum are constructed here. The locators for this paper refer to its 68-page author manuscript, not the older 52-page arXiv version.

#### 10.4 What the auxiliary carry results assume

The finite set of possible bases extends to any finite list of primes. For ordered primes  $p_1 < \dots < p_s$  an interval  $(p_1^a, p_1^{a+1})$  contains at most one power of each other prime, because consecutive  $p_i$ -powers have ratio  $p_i > p_1$ , so its block radix belongs to the  $2^{s-1}$ -letter alphabet  $\{p_1 \prod_{i=2}^s p_i^{\varepsilon_i} : \varepsilon_i \in \{0, 1\}\}$ . This observation does not give the frequencies or spacing of the bases. An irrationality proof might need estimates for those frequencies, an asymptotic formula with an error term for the weighted shell counts, or information about the kernel after a specified family of shifts.

Some auxiliary results explain why bounded integer approximations alone are insufficient. Here are their hypotheses in explicit form.

Let  $j(n) \in \{2, 3, 5\}$  label a sequence of jumps, and suppose each label occurs. A perturbation  $\varepsilon_n$  in an additive abelian group has sum zero on every interval  $[a, b)$  whose endpoints have the same label exactly when

$$\varepsilon_n = u(j(n+1)) - u(j(n))$$

for three potential values  $u(2), u(3), u(5)$ . One direction follows by telescoping. Conversely, the zero-sum assumption makes  $\sum_{k < n} \varepsilon_k$  the same at any two indices with the same label. Assign that common value to  $u(j(n))$ ; every label occurs, so all three values are defined. Consecutive partial sums then give the displayed identity. In the usual terminology, the perturbation is a coboundary of the function  $u$  on the labels: it is the difference of successive potential values. This is a telescoping condition, not a bound on the size of the perturbation. It does not by itself force the perturbation to vanish: unequal potential values give nonzero differences when the label changes. If the perturbation also vanishes on a genuine  $2 \rightarrow 3$  transition and on a genuine  $2 \rightarrow 5$  transition, all three potential values agree, so every perturbation is zero.

For an integer carry satisfying  $c_{n+1} = b_n c_n - D m_n$ , compare it with an integer sequence  $z_n$ . Define the error  $e_n = D z_n - c_n$  and the perturbation  $\varepsilon_n = b_n z_n - z_{n+1} - m_n$ . Then

$$e_{n+1} = b_n e_n - D \varepsilon_n.$$

Under the preceding zero-sum and transition hypotheses,  $\varepsilon_n = 0$  and hence  $e_N = (\prod_{n < N} b_n) e_0$ . If every  $b_n \geq 2$  and the integer  $e_0$  is nonzero, then  $|e_N| \geq 2^N$ . This contradicts even one bound  $|e_N| < 2^N$ , and therefore also excludes a uniform bound for all  $N$ . The nonzero initial error is essential: the identically zero error causes no contradiction.

A calculation on four states gives a related obstruction. If  $0 < T_i < 1$ ,  $z_i \in \mathbb{Z}$  and  $|z_i - T_i| < 1$ , then  $z_i$  is 0 or 1. For bases 2, 3, 2, 5, the equalities  $2z_0 - z_1 - 1 = 0$  and

$2z_2 - z_3 - 1 = 0$  force all four integers to be 1. The sum of the first two perturbations is then

$$(2z_0 - z_1 - 1) + (3z_1 - z_2 - 1) = 1,$$

not zero. Thus unit accuracy, the two exact equalities and a zero sum on that block cannot all hold. Unit accuracy alone does not give either exact equality.

No integer approximation to the tails of  $S$  is constructed here that satisfies all these conditions. Rationality supplies integer carries of at most quadratic growth, but not the additional zero-sum and transition identities. The carries obtained from  $S$  satisfy a weighted block identity instead. Applying the auxiliary results would require that identity to imply their hypotheses, or would require a different construction of the integer approximations.

There is also a conditional criterion for a sum supported on the powers of 2. Its elementary ingredients can be stated without introducing another formal vocabulary. From  $y' = by - 1$  with  $b > 0$  and  $0 < y' < 1$  one gets  $1/b < y < 2/b$ . For integers  $a > 0$  and  $p \geq 3$ , if  $u > 1/a$  and  $v < 2/(pa)$ , then  $u - v > 1/(3a)$ . Finally, a real number  $x$  is irrational if, for every positive integer  $d$ , there are integers  $n, k$  with  $0 < |nx - k| < 1/d$ .

To use these observations for a series, the conditional result requires exact identities  $y_M = H_M x - z_M$ , with  $H_M, z_M \in \mathbb{Z}$ , and, for every positive  $d$ , two indices with different  $H_M$ , different  $y_M$ , and  $|y_M - y_{M'}| < 1/d$ . Their difference gives the required nonzero linear form. Equal states would give zero and would not suffice; a merely bounded gap would not suffice either. Indeed, for a rational  $x$  of denominator  $d$ , every nonzero difference of this form has size at least  $1/d$ . No concrete series in this record is shown to satisfy all these clearing and arbitrarily small nonzero-gap conditions.

## 10.5 Where the problem stands

The distinct-height sum is irrational for every finite set of at least two primes (Theorem 1.1), by an ordinary proof checked by a second AI agent; its case  $\{2, 3, 5\}$  is checked in Lean (Theorem 1.2). The finite-state criterion of Theorem 4.6 gives an independent proof for the sets in Corollary 4.7 and cannot decide  $\{2, 3, 5, 7\}$  at any depth. For the catalogue sum, the sub-sums  $E_p$  over the powers of one prime are irrational (Theorem 4.9). For the repeated  $\{2, 3, 5\}$  sum, the coefficients, recurrence, quadratic bounds and denominator clearing are derived here from the original multiplicities. What remains is to exclude eventual integral reduced tails, equivalently to prove residue escape for every admissible denominator and beyond every prescribed starting index. The fixed-start residue formula explains these quantifiers, and the strip decomposition shows why absence of floor crossings alone does not give the required cancellation.

The rank and uniform-norm results concern the kernel, not the arithmetic of its sum. Finite tests, countability, radix recoding and denominator support do not close the remaining argument. The settled two-prime case uses Fan's earlier reduction to the cited Hecke–Mahler value theorem.

## Statements and declarations

**Proof sources.** The two-prime deduction rests on the cited external value theorem, which the Lean proof of Theorem 1.3 assumes; Fan posted that deduction first, on the

erdosproblems.com forum [16]. Theorem 1.1 and Lemmas 4.1–4.4 have the ordinary proofs of Section 4.1 and no Lean proofs; a formal proof would need Kronecker’s theorem on the closure of a line in a torus and the Euler characteristic count of Lemma 4.4. The proof of Theorem 1.1 was found by an AI agent on 26 September 2026 and checked by a second AI agent working independently, who re-derived every step and closed the five gaps listed there; it has had no human review. Theorem 4.9 and Proposition 4.8 have the ordinary proofs of Section 4.6 and no Lean proofs; the theorem was checked in the same way. Theorem 1.2 is checked in Lean, as the remark below it states; Comparator has not yet compared it. Lemma 4.5, Theorem 4.6 and Corollary 4.7 have the ordinary proofs given in Section 4, the corollary together with an exact finite computation, and no Lean proofs. Every other theorem, proposition, lemma and corollary has a Lean proof of its whole statement, marked in the margin; Section 12 states what the Lean proof of the two-prime theorem checks.

The weighted-triangle and weighted-shift identities, exact denominator formula, scaled dichotomy, uniqueness argument, residue limit, strip decomposition and direct recoding proposition also have the written proofs given here. The independent integer computations check finite instances, not infinite quantifiers. The twelve-shell rational enclosure, denominator bound and eight mixed-radix digits have the exact finite proofs given above. The two large numerical exclusions have published witnesses and exact checkers, and the larger scan has a published saved run. None of these records proves escape beyond every prescribed starting index.

**Artefact and data availability.** The links in the source inventory and to the dyadic-window checker open a [fixed revision](#) of the repository. The witnesses and exact checkers for the two large numerical exclusions and the saved larger scan are in the [large-computation directory](#). The [distinct-height programs](#) recompute the constants of Theorem 1.2 and the certificates of Corollary 4.7 in exact rational arithmetic, with two independent computations of the tail intervals, and store their outputs; a fast test checks them against the stored receipts. The same directory checks the finite illustrations of Section 4.1 and the block identity for  $\{2, 3, 5, 7\}$ .

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Loxton and van der Poorten cited by them. The problem numbering and its status as of 28 July 2026 are taken from the Erdős Problems catalogue maintained by Thomas Bloom [4].

## 11 Further examples, conditional lemmas and source references

This section collects the historical details, the integer recurrence lemmas in their general form, worked examples and an inventory of the formal sources.

### 11.1 The historical and catalogue record

For  $P = \{p\}$ , the enumeration is  $a_n = p^{n-1}$ , so  $[a_1, \dots, a_n] = p^{n-1}$  and both sums equal  $p/(p-1)$ . In the primary 1988 source Erdős states irrationality for infinite  $P$  as a simple exercise and presents persistence for a finite number of primes greater than one as a probable extension [2, p. 106]. Bloom’s catalogue, as accessed on 28 July 2026, records an open problem [4]; that status is not inferred from the conjectural wording.

In the letter written on 1 January 1973 and published in 1974, Erdős calls the irrationality of  $\mathcal{R}_P$  a conjecture that he cannot prove, and says that he can show the sum over the distinct values of  $[a_1, \dots, a_k]$  to be irrational [3, p. 335]. That sum is  $\mathcal{D}_P$ . The letter introduces its primes as  $p_1, \dots, p_s$ , later writing  $p_r$ , without restricting their number, so the singleton calculation forces the qualification  $|P| \geq 2$ . It prints no argument. The catalogue has cited the letter for Problem #269 since 28 December 2025, after the letter was quoted in the site’s discussion of missing problems [5], and its page records Erdős’s statement about the distinct values. The catalogue page says that Erdős proved the statement; the letter asserts it and prints no argument. Theorem 1.1 proves it for every finite  $P$  with  $|P| \geq 2$ . For  $|P| = 2$  it was already a consequence of the Hecke–Mahler transcendence theorem of Loxton and van der Poorten [15, Theorem 8, p. 40] and Bugeaud and Laurent [14, Theorem 1.1], through the affine identity of Section 3, by the route Fan used for  $\mathcal{R}_{\{p,q\}}$ . We do not know what argument Erdős had in mind.

We have not found a proof of the assertion in print. Erdős and Graham state only the question for  $\mathcal{R}_P$ : the sum is irrational for infinite  $P$ , and they ask what happens for finite  $P$  with more than one element [1, p. 65]; the distinct-height sum does not appear there. Erdős’s survey of 1975 on the irrationality of series [8] does not mention the running least common multiple of the  $P$ -smooth integers, his survey of 1980 on combinatorial number theory [6] does not discuss irrationality, and his survey of 1988 states only the question for  $\mathcal{R}_P$  [2, p. 106]. Every Hecke–Mahler transcendence theorem we found, including those of Loxton and van der Poorten, Bugeaud and Laurent, and Luca–Ouaknine–Worrell [25], has a single slope, and so reaches  $\mathcal{D}_P$  only for  $|P| = 2$ . The rationality criterion of Diananda and Oppenheim [7], which condenses a Cantor series into blocks of consecutive terms, has the form of the dyadic argument of Section 4.3; it restates rationality and does not decide  $\mathcal{D}_P$ . The discussion thread of Problem #269 contains no argument for  $\mathcal{D}_P$  with  $|P| \geq 3$ , and the *Formal Conjectures* statement [9] concerns  $\mathcal{R}_P$  only. The Cantor-series criteria of Erdős–Straus [17, Theorem 2.1] and Hančl–Tijdeman [18, Theorem 3.1] do not apply to  $\mathcal{D}_P$ , as explained in Section 4.3. The search did not reach the sections on irrational series in Guy’s *Unsolved Problems in Number Theory*, or MathSciNet reviews. The unresolved target of this record

is the repeated series  $\mathcal{R}_P$ ; a transcendence statement for  $\mathcal{D}_P$  with  $|P| \geq 3$  is a separate question (Problem 10.7).

On 26 June 2026 Steve Fan posted the two-prime factorisation, the Hecke–Mahler reduction and the transcendence conclusion in the discussion thread of the problem’s page [16]; the comment itself notes that the argument does not seem to generalise immediately to  $|P| \geq 3$ , and a reply there observes that it applies to arbitrary coprime pairs. The first public version of this note is dated 22 July 2026, after Fan’s post. We retain the calculation as exposition and make no priority claim for it.

The statement of the problem has been formalised as a conjecture with an unfilled proof in the *Formal Conjectures* collection [9]. In the cited source, the rational, irrational and infinite-prime assertions end in sorry. Its Nat-indexed series includes the empty-prefix least-common-multiple term, so its value differs from the conventional one by a rational constant; transporting a theorem across that boundary needs an explicit series-identification lemma, which is not supplied here.

## 11.2 The abstract carry lemmas in their general form

The earlier argument used carries obtained from the actual  $\{2, 3, 5\}$  sum. The following statements instead concern arbitrary integer recurrences. Divisibility and escaping windows are explicit hypotheses. For general radices they need separate justification; for the actual radices, the divisibility criterion below shows that every integral carry acquires the smooth factor eventually. This does not supply escaping windows. The Cantor-series comparison is in Section 7.

Let  $D$  and  $B$  be positive integers with  $D = D_{\text{sm}}B$ , where  $D_{\text{sm}} = 2^u 3^v 5^w$ ,  $u, v, w \in \mathbb{N}$  and  $\gcd(B, 30) = 1$ . Let  $(c_n)$  be an integer sequence satisfying  $c_{n+1} = b_n c_n - D m_n$  for integer sequences  $(b_n)$  and  $(m_n)$ . For these general sequences, form  $W$  and  $F$  by the recursions in Section 8; an escaping window must have  $W \neq 0$ , and its modulus is  $|W|$ . In the factorisation below the quotients  $d_n$  are required to be integers; writing  $c_n = D_{\text{sm}} d_n$  asserts divisibility, not just an identity in  $\mathbb{Q}$ .

**Proposition 11.1** (conditional denominator reduction). *If  $c_n = D_{\text{sm}} d_n$  for every  $n$ , with  $D_{\text{sm}} > 0$ , then the recurrence and window identity for  $(d_n)$  have multiplier  $B$  in place of  $D$ . Moreover, for every  $n$  and every real  $t$ ,*

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$$0 < c_n \leq Dt \iff 0 < d_n \leq Bt.$$

*Proof.* Substitute  $c_n = D_{\text{sm}} d_n$  and  $D = D_{\text{sm}} B$  in the recurrence and divide by  $D_{\text{sm}}$ . Dividing  $0 < c_n \leq D_{\text{sm}} Bt$  by this positive factor gives the stated bound equivalence. Dividing the window identity  $c_{\ell+h} = W_{\ell,h} c_\ell - D F_{\ell,h}$  likewise gives  $d_{\ell+h} = W_{\ell,h} d_\ell - B F_{\ell,h}$ .  $\square$

For a general bound  $c_n \leq G(D, n)$ , division and integrality give only

$$d_n \leq \left\lfloor \frac{G(D, n)}{D_{\text{sm}}} \right\rfloor,$$

not automatically  $d_n \leq G(B, n)$ . The bounds used for the actual tails are linear in the multiplier before rounding. For any real  $t$ ,

$$\left\lfloor \frac{\lfloor Dt \rfloor}{D_{\text{sm}}} \right\rfloor = \lfloor Bt \rfloor.$$

Indeed, an integer  $z$  satisfies  $D_{\text{sm}}z \leq \lfloor Dt \rfloor$  exactly when  $D_{\text{sm}}z \leq Dt$ , or  $z \leq Bt$ . Thus rounding does not change the reduced bound in this case.

Growth of the radix product is not enough:  $b_n = 3, m_n = 1, D = 2$  and  $c_n = 1$  satisfy the recurrence, but 2 never divides  $c_n$ . The precise condition follows by reducing the recurrence modulo  $D_{\text{sm}}$ . For any starting index  $A$  and every  $n \geq A$ ,

$$c_n \equiv \left( \prod_{j=A}^{n-1} b_j \right) c_A \pmod{D_{\text{sm}}},$$

$$D_{\text{sm}} \mid c_n \iff \frac{D_{\text{sm}}}{\gcd(D_{\text{sm}}, c_A)} \mid \prod_{j=A}^{n-1} b_j.$$

The congruence is an induction, since  $D_{\text{sm}} \mid D$ . For the equivalence, cancel the positive greatest common divisor; the remaining factor of  $c_A$  is coprime to the remaining modulus. Thus it is growth of the required prime valuations, not growth of the product as a real number, that ensures divisibility.

For the actual radices the product is  $P_n/P_A$ . Its valuations at 2, 3, 5 all tend to infinity, so every fixed  $D_{\text{sm}}$  eventually divides  $c_n$ , for any integral initial carry  $c_A$ . Under rationality  $S = N/D$  in lowest terms, one may start at  $A = 1$ :  $c_1 = DX_1 = N - D$  is integral and coprime to  $D_{\text{sm}}$ . Since  $P_1 = 2$ , the criterion becomes  $D_{\text{sm}} \mid P_n/2 = h_n$ , exactly the onset already obtained from the finite-prefix identity in Proposition 7.4. This is an alternative proof of the divisibility step, not an additional arithmetic hypothesis left to verify for  $S$ . The finite-prefix proof in Theorem 7.3 remains valid.

**Proposition 11.2** (escaping windows exclude a positive bounded integer solution). *Let  $(b_n)$  and  $(m_n)$  be sequences of nonnegative integers, let  $G : \mathbb{N}_{>0} \times \mathbb{N} \rightarrow \mathbb{N}$ , and assume the residue condition (16) for these sequences and  $G$ , using  $|W_{\ell,h}| > 0$  as the modulus. Fix  $B > 0$  coprime to 30. There is no integral sequence  $(d_n)$  satisfying simultaneously  $d_{n+1} = b_n d_n - B m_n$ ,  $d_n > 0$  and  $|d_n| \leq G(B, n)$  for every  $n \geq 0$ .*

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*Proof.* Choose one escaping window  $(\ell, h)$ . The window identity gives  $d_{\ell+h} \equiv -BF_{\ell,h}$  modulo  $|W_{\ell,h}|$ . The endpoint state is positive and at most  $G(B, \ell + h)$ , whereas the least positive residue of the right-hand side exceeds that bound, so Proposition 8.1 applies.  $\square$

The escape hypothesis is the substantial arithmetic assumption. It requires a residue above the bound, not just a rapidly growing product of bases. For example, the constant data  $b_n = 2, m_n = 1$  have the positive solution  $d_n = B$ . For any bound  $G(B, n) \geq B$ , every window has  $W = 2^h, F = 2^h - 1$  and least positive residue at most  $B$ , so the hypothesis fails. For the actual three-prime coefficients and  $G = K$ , Theorem 8.2 makes the escape hypothesis equivalent to irrationality of  $S$ . That equivalence does not hold for an arbitrary choice of  $G$ .

Coprimality with 30 selects the denominators covered by the hypothesis; once a window is fixed, the finite contradiction does not use it. The conventions cover the edge cases: a zero window base is excluded, a zero residue is represented by the full modulus, and positivity prevents the endpoint carry from vanishing. The version stated with

a nonzero smooth factor assumes the exact factorisation  $c_n = D_{\text{sm}} d_n$ , the corresponding recurrence for  $(c_n)$  and the positive upper bound for  $(d_n)$ , and gives the same contradiction.

Even retaining the actual radix sequence and quadratic growth does not force irrationality if the numerators are changed. For the same  $b_a$ , prescribe the positive integer carries  $(a+3)^2$  and set

$$\widehat{m}_a = b_a(a+3)^2 - (a+4)^2.$$

The shift by three ensures positivity even at  $a = 0$ : since  $b_a \geq 2$ ,

$$\widehat{m}_a \geq 2(a+3)^2 - (a+4)^2 = a^2 + 4a + 2 > 0.$$

Both these integer coefficients and the prescribed carries have quadratic order, because  $2 \leq b_a \leq 30$ . Nevertheless, finite telescoping gives

$$\sum_{a=0}^{N-1} \frac{\widehat{m}_a}{P_{a+1}} = 9 - \frac{(N+3)^2}{P_N} \rightarrow 9.$$

This example keeps the exact radices, not the actual shell multiplicities or their numerical upper bound  $Q$ . It shows why the arithmetic of the specific  $m_a$ , rather than the radix alphabet and growth orders alone, must enter a proof for  $S$ .

### 11.3 Worked finite examples

The values of the running least common multiple at the first ten integer cutoffs are tabulated in Section 2. They illustrate both parts of Proposition 2.2: the value is constant on  $\{5, 6, 7\}$  and on  $\{9, 10\}$ , and each change multiplies by a single prime, by 2 at  $x = 2, 4, 8$ , by 3 at  $x = 3, 9$  and by 5 at  $x = 5$ . Also  $L(10) = 8 \cdot 9 \cdot 5 = 360$  is the least common multiple of the smooth numbers 1, 2, 3, 4, 5, 6, 8, 9, 10.

For Proposition 2.4 take  $(p, q, r) = (2, 3, 5)$  and the box  $\mathcal{B}(1, 1, 1)$ , whose eight points carry the smooth values 1, 2, 3, 5, 6, 10, 15, 30 and the heights

| $p^i q^j r^k$ | 1 | 2 | 3 | 5  | 6  | 10  | 15  | 30    |
|---------------|---|---|---|----|----|-----|-----|-------|
| H             | 1 | 2 | 6 | 60 | 60 | 360 | 360 | 10800 |

Six heights occur, two of them twice: the points 5 and 6 share the height 60, and 10 and 15 share the height 360. The identity reads

$$1 + \frac{1}{2} + \frac{1}{6} + \frac{1}{60} + \frac{1}{60} + \frac{1}{360} + \frac{1}{360} + \frac{1}{10800} = 1 + \frac{1}{2} + \frac{1}{6} + \frac{2}{60} + \frac{2}{360} + \frac{1}{10800} = \frac{18421}{10800},$$

and the two coefficients 2 carry the whole content of the regrouping on this box. Where the heights are pairwise distinct the identity is a relabelling.

Multiplying block radices along a run of blocks gives the product of the prime multipliers at the jumps in that run: for instance  $b_1 b_2 = 60$  is the product of the four multipliers at 3, 4, 5, 8. Block 4 is the only one among the first six containing an internal power of each odd prime, whereas block 5 contains neither, so its radix falls back to the terminal factor alone.

## 11.4 The finite-scan histogram

The larger scan described in Section 9 covers  $B \leq 5000$  coprime to 30 and  $100 \leq \ell \leq 3000$ , with search depth 24. Its saved run ([scan-5000-100-3000.json](#)) gives 3,869,934 pairs, an escape in every case, and the following first-success histogram:

| $h$ | 4 | 5   | 6   | 7    | 8     | 9      | 10     | 11      | 12      | 13     | 14    | 15   | 16  | 17 | 18 |
|-----|---|-----|-----|------|-------|--------|--------|---------|---------|--------|-------|------|-----|----|----|
| #   | 1 | 104 | 812 | 5437 | 51409 | 237423 | 735450 | 1431226 | 1132756 | 236752 | 34910 | 3076 | 521 | 49 | 8  |

The first case attaining the maximal observed length 18 has  $B = 917$  and  $\ell = 2980$ . The smaller run ([scan-1000-100-500.json](#)) over  $B \leq 1000$  coprime to 30 and  $100 \leq \ell \leq 500$  gives 106,666 pairs and maximal first successful length 14, first attained at  $B = 359$  and  $\ell = 291$ . The two cases have the following data.

|                        | Larger scan             | Smaller scan      |
|------------------------|-------------------------|-------------------|
| Denominator $B$        | 917                     | 359               |
| Start $\ell$           | 2980                    | 291               |
| Length $h$             | 18                      | 14                |
| Endpoint jump index    | 6179                    | 627               |
| Window base            | 18139852800000000       | 5038848000000     |
| Accumulated numerator  | 13196471407660025821045 | 25864575212865807 |
| Least positive residue | 76322101735             | 213175287         |
| Upper bound            | 3896420420              | 15932659          |

These histograms concern a bounded region. The window-growth law gives a necessary lower bound on a possible escape length, not a distribution law for residues or first successful lengths. No generic-residue model is used as evidence for cofinal escape.

## 11.5 Source inventory

Each entry pairs a mathematical statement with the Lean source that states it, at a fixed earlier revision of the sources. The declaration names are for lookup only.

Two conventions in the Lean sources require care. The Lean version of the half-height  $h_a$  is a natural number computed by integer division: its value at 0 is 0, whereas  $h_0 = 1/2$ . It agrees with  $h_a$  for  $a \geq 1$ , as the half-height identity assumes. The Lean version of the tail  $X_a$  divides after casting to  $\mathbb{R}$ , so agrees with  $X_a = h_a T_a$  at every index.

The denominator-clearing and reduced-carry lemmas assume  $T_1$  rational. The first shell consists of 1 alone, so  $T_1 = S - 1$ ; thus  $S = N/D$  supplies

$$T_1 = \frac{N - D}{D}, \quad \gcd(N - D, D) = \gcd(N, D).$$

Lean proves this translation, which preserves the denominator, its smooth factor and the clearing onset.

| informal statement                        | linked Lean source                                        |
|-------------------------------------------|-----------------------------------------------------------|
| integer represented by an exponent triple | <a href="#">smooth3 val</a>                               |
| pure-power height                         | <a href="#">three prime height</a>                        |
| lattice kernel                            | <a href="#">three prime kernel q</a>                      |
| exponent triples below the cut-off        | <a href="#">smooth prefix exponents</a>                   |
| running least common multiple             | <a href="#">smooth prefix lcm</a>                         |
| prefix value divides the height           | <a href="#">smooth3 val dvd three prime height of mem</a> |
| divisibility into the height              | <a href="#">smooth prefix lcm dvd three prime height</a>  |
| first pure-power membership               | <a href="#">pure first mem smooth prefix exponents</a>    |
| second pure-power membership              | <a href="#">pure second mem smooth prefix exponents</a>   |
| third pure-power membership               | <a href="#">pure third mem smooth prefix exponents</a>    |
| running-lcm identity                      | <a href="#">smooth prefix lcm eq three prime height</a>   |
| cell relation                             | <a href="#">same three prime log cell</a>                 |
| cell constancy of the height              | <a href="#">three prime height eq of same log cell</a>    |
| cell constancy of the running value       | <a href="#">smooth prefix lcm eq of same log cell</a>     |
| cell constancy of the kernel              | <a href="#">three prime kernel q eq of same log cell</a>  |
| positive power sets                       | <a href="#">positive prime powers</a>                     |
| counting positive prime powers            | <a href="#">positive prime powers card</a>                |
| exclusion of the origin                   | <a href="#">one not mem positive prime powers</a>         |
| disjoint positive prime powers            | <a href="#">positive prime powers disjoint</a>            |
| union of positive prime powers            | <a href="#">three prime positive jump set</a>             |
| positive jump count                       | <a href="#">three prime positive jump set card</a>        |
| jump set with the origin                  | <a href="#">three prime jump set with origin</a>          |
| jump count with the origin                | <a href="#">three prime jump set with origin card</a>     |
| first height step                         | <a href="#">three prime height first log step</a>         |
| second height step                        | <a href="#">three prime height second log step</a>        |
| third height step                         | <a href="#">three prime height third log step</a>         |
| first coordinate step                     | <a href="#">smooth prefix lcm first log step</a>          |

|                                                |                                                            |
|------------------------------------------------|------------------------------------------------------------|
| informal statement                             | linked Lean source                                         |
| second coordinate step                         | <a href="#">smooth prefix lcm second log step</a>          |
| third coordinate step                          | <a href="#">smooth prefix lcm third log step</a>           |
| exponent box                                   | <a href="#">smooth exponent box</a>                        |
| point height                                   | <a href="#">smooth point height</a>                        |
| height fibre                                   | <a href="#">smooth height fiber</a>                        |
| fibre sum                                      | <a href="#">smooth height fiber kernel sum</a>             |
| grouping equal denominators<br>in a finite sum | <a href="#">finite smooth kernel sum grouped by height</a> |
| upper bound by the cube of<br>the cutoff       | <a href="#">three prime height le cube</a>                 |
| origin value                                   | <a href="#">kernel 235 origin</a>                          |
| value at two                                   | <a href="#">kernel 235 two</a>                             |
| value at three                                 | <a href="#">kernel 235 three</a>                           |
| value at six                                   | <a href="#">kernel 235 six</a>                             |
| example excluding a product<br>of two factors  | <a href="#">kernel 235 not rank one</a>                    |
| variable-base tail step                        | <a href="#">tail state step</a>                            |
| expanded tail step                             | <a href="#">tail state step eq</a>                         |
| smooth exponent shell                          | <a href="#">smooth exponent shell</a>                      |
| short-interval uniqueness                      | <a href="#">exponent unique in short interval</a>          |
| first-coordinate projection                    | <a href="#">smooth exponent shell card le drop first</a>   |
| third-coordinate projection                    | <a href="#">smooth exponent shell card le drop third</a>   |
| sorted quadratic estimate                      | <a href="#">sorted pair quadratic</a>                      |
| quadratic shell bound                          | <a href="#">smooth exponent shell card quadratic</a>       |
| dyadic internal power                          | <a href="#">dyadic internal power</a>                      |
| internal-power uniqueness                      | <a href="#">dyadic internal power exponent unique</a>      |
| dyadic block base                              | <a href="#">dyadic block base235</a>                       |
| exact dyadic radix alphabet                    | <a href="#">dyadic block base235 cases</a>                 |
| bounded-radix consequence                      | <a href="#">dyadic block base235 mem interval</a>          |
| a nonzero $2 \times 2$ determinant             | <a href="#">kernel 235 minor eq neg one fifteen</a>        |
| no integer rotation orbit                      | <a href="#">no integer orbit logb of prime</a>             |
| height factorisation                           | <a href="#">three prime height factorisation</a>           |
| kernel factorisation                           | <a href="#">three prime kernel q factorisation</a>         |

| informal statement                            | linked Lean source                                                           |
|-----------------------------------------------|------------------------------------------------------------------------------|
| two-prime outer product                       | <a href="#">two prime kernel q eq outer product</a>                          |
| two-prime vanishing minors                    | <a href="#">two prime kernel q minor two eq zero</a>                         |
| uniform minors, general form                  | <a href="#">exists uniform nonsingular three prime kernel minor</a>          |
| uniform minors for primes                     | <a href="#">exists uniform nonsingular three prime kernel minor of prime</a> |
| no finite separation                          | <a href="#">not finite separable three prime kernel</a>                      |
| second-order minor                            | <a href="#">kernel 235 minor2 eq neg one fifteen</a>                         |
| third-order minor                             | <a href="#">kernel 235 minor3 eq one over 81000</a>                          |
| misleading proportional row                   | <a href="#">kernel 235 row three eq smul row zero</a>                        |
| failure of that proportionality               | <a href="#">kernel 235 row three ne smul row zero at four</a>                |
| integer numerator for one dyadic shell        | <a href="#">dyadic ordered block digit235</a>                                |
| normalised state step                         | <a href="#">dyadic normalized tail state r235 succ</a>                       |
| shell summability                             | <a href="#">summable dyadic shell mass r235</a>                              |
| infinite tail recurrence                      | <a href="#">dyadic normalized shell tsum tail r235 succ</a>                  |
| integer tails or separation from the integers | <a href="#">dyadic shell tsum tail integer or cofinal far</a>                |
| boundary divisibility                         | <a href="#">smooth height mul prime dvd boundary height</a>                  |
| half-height identity for $a \geq 1$           | <a href="#">two mul height normalizer235</a>                                 |
| window clearing identity                      | <a href="#">height normalizer235 mul window mass eq int</a>                  |
| clearing a rational $T_1$ at $a \geq 1$       | <a href="#">qsmul normalized tail state eq int of value eq rat</a>           |
| normalised-state collision                    | <a href="#">exists normalized tail state collision of value eq rat</a>       |
| least positive residue                        | <a href="#">least positive residue</a>                                       |
| positive representative range                 | <a href="#">least positive residue pos le</a>                                |
| representative congruence                     | <a href="#">least positive residue mod eq</a>                                |
| escape condition                              | <a href="#">residue escapes window</a>                                       |
| excluding a bounded positive natural number   | <a href="#">no bounded positive state of residue escape</a>                  |
| contrapositive form                           | <a href="#">residue le bound of bounded positive state</a>                   |
| excluding a bounded positive integer          | <a href="#">no bounded positive int state of least positive residue</a>      |
| identifying the least positive representative | <a href="#">least positive residue eq nat abs of pos le mod eq</a>           |

| informal statement                                 | linked Lean source                                                        |
|----------------------------------------------------|---------------------------------------------------------------------------|
| window base                                        | <a href="#">window base</a>                                               |
| accumulated numerator in a window                  | <a href="#">window forcing</a>                                            |
| affine window identity                             | <a href="#">affine recurrence window</a>                                  |
| scaling the accumulated numerator                  | <a href="#">window forcing const mul</a>                                  |
| integral carry window                              | <a href="#">integral carry window</a>                                     |
| endpoint residue identification                    | <a href="#">least positive residue window forcing eq carry</a>            |
| divisibility by the smooth part of the denominator | <a href="#">smooth3 val dvd three prime height of le</a>                  |
| common-factor cancellation                         | <a href="#">integral carry cancel common factor</a>                       |
| bound after dividing out the common factor         | <a href="#">reduced carry pos le of common factor</a>                     |
| window identity after division                     | <a href="#">reduced integral carry window</a>                             |
| cofinal window hypothesis                          | <a href="#">cofinal local window escape</a>                               |
| escape excludes a reduced positive integer carry   | <a href="#">no positive reduced carry of cofinal local window escape</a>  |
| the same contradiction with a common factor        | <a href="#">no positive absorbed carry of cofinal local window escape</a> |
| real window identity                               | <a href="#">true normalized state window</a>                              |
| escape from irrationality, general bound           | <a href="#">cofinal local window escape of irrational of beaten</a>       |
| escape from irrationality, quadratic family        | <a href="#">cofinal local window escape of irrational of quadratic</a>    |
| escape from irrationality, actual bound            | <a href="#">cofinal local window escape of irrational</a>                 |
| residue criterion equivalent to an irrational tail | <a href="#">actual cofinal local window escape iff</a>                    |
| residue criterion equivalent to an irrational sum  | <a href="#">actual cofinal local window escape iff irrational value</a>   |
| rational $T_1$ gives eventual integer carries      | <a href="#">exists reduced carry of value eq rat</a>                      |
| residue condition for the actual series            | <a href="#">actual cofinal local window escape</a>                        |

|                                                                                 |                                                                                 |
|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------|
| informal statement                                                              | <a href="#">linked Lean source</a>                                              |
| irrationality from the residue condition                                        | <a href="#">irrational value of cofinal local window escape</a>                 |
| bounded-radix alternative                                                       | <a href="#">bounded radix zero or cofinal far</a>                               |
| rational value from an integral scaled tail                                     | <a href="#">rational of scaled tail integer</a>                                 |
| zero block sums as differences of potentials                                    | <a href="#">channel block null iff channel potential</a>                        |
| a nonzero error exceeds the bound at one index                                  | <a href="#">no carry lift of error bound below two pow</a>                      |
| a nonzero error cannot stay uniformly bounded                                   | <a href="#">no bounded carry lift of block null two anchors</a>                 |
| first-block sum                                                                 | <a href="#">binary lift two anchors force first block sum one</a>               |
| four states cannot meet all three conditions                                    | <a href="#">no unit accurate lift with two anchors and first two block null</a> |
| carry as a residue plus a potential difference                                  | <a href="#">carry eq residue digit add coboundary</a>                           |
| carry interval                                                                  | <a href="#">carry residue mem interval</a>                                      |
| digit interval                                                                  | <a href="#">residue digit mem interval</a>                                      |
| map from formal expressions to functions                                        | <a href="#">cartier realisation</a>                                             |
| finite-dimensional span of the resulting functions                              | <a href="#">finite realised span of factorisation</a>                           |
| interval containing a Cantor-series tail                                        | <a href="#">state mem ioo</a>                                                   |
| state gap bound                                                                 | <a href="#">state gap lower bound</a>                                           |
| small-form criterion                                                            | <a href="#">irrational of small nonzero forms</a>                               |
| irrationality from clearing and small nonzero gaps                              | <a href="#">irrational of clearing and small gaps</a>                           |
| indices giving the threshold staircase                                          | <a href="#">staircase indices</a>                                               |
| finite threshold-column rank, including the extreme-column defect               | <a href="#">finite cut rank</a>                                                 |
| uniform admissible modular minors, with indices chosen before modulus and layer | <a href="#">uniform modular minors</a>                                          |

## 12 What the results use and what they do not prove

The main arguments have different inputs and different conclusions. In particular, the kernel result does not imply irrationality, and the residue criterion still requires an arithmetic proof of escape.

| Result                          | Dependency and limit                                                                                                                                                                                                                                                                                                               |
|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Height and cells                | Prime-power divisibility and unique factorisation. Their geometry alone gives no irrationality statement.                                                                                                                                                                                                                          |
| Two-prime values                | Earlier factorisation plus the cited external Hecke–Mahler value theorem. The reduction to that theorem is checked in Lean; the theorem itself is a cited input (Section 12).                                                                                                                                                      |
| Distinct-height sums, every $P$ | Integrality of the scaled tails, rearranged chains at a return of the logarithmic flow, Kronecker’s theorem and an Euler characteristic count on a two-dimensional subtorus. Ordinary proof checked by a second agent; no Lean proof. It stops for the repeated sums because their scaled tails are unbounded.                     |
| Single-prime sub-sums           | The same two-tail comparison with the carry $f_p$ and a chain chosen near the origin of the torus; no two-wall lemma. Ordinary proof checked by a second agent; no Lean proof. Says nothing about the whole repeated sum.                                                                                                          |
| Distinct-height sums, dyadic    | Dyadic blocks, finitely many affine maps with disjoint images, integrality of the scaled tails and the irrationality of $\log p / \log q$ . The theorem for $\{2, 3, 5\}$ is checked in Lean; the general criterion has an ordinary proof, and Corollary 4.7 adds exact finite computations. No statement about the repeated sums. |
| Arbitrary-order rank            | Diagonal rescaling, separate one-dimensional densities and a staircase determinant; checked in Lean.                                                                                                                                                                                                                               |
| Tail recurrence                 | Counting the original multiplicities, normalising by half the boundary height, and convergence; an abstract recurrence would not identify the sum.                                                                                                                                                                                 |
| Denominator reduction           | A finite prefix with cleared denominators and an upper bound for positive integer tails. The first valid index is determined exactly (Proposition 7.4); checked in Lean.                                                                                                                                                           |
| Residue criterion               | The actual recurrence, a dominating bound and window products comparable to $8^h$ ; escape at arbitrarily late starts remains unproved.                                                                                                                                                                                            |
| Weighted differences            | Exact reindexing and the contributions of the new boundary strips; integer enumeration checks the displayed counterexample to third-difference cancellation.                                                                                                                                                                       |
| Finite computations             | Direct checks settle only their individual windows. The two large exclusions have published witnesses and exact checkers, and the larger scan a published saved run; none is a theorem input.                                                                                                                                      |

None of these results proves irrationality of the repeated sum  $\mathcal{R}_p$  for three or more primes.

### What the Lean proof of the two-prime theorem assumes

Every theorem, lemma, proposition and corollary of this record has a Lean proof, except Theorem 1.1, Lemmas 4.1–4.4, Theorem 4.9, Proposition 4.8, Lemma 4.5, Theorem 4.6 and Corollary 4.7, whose proofs are ordinary proofs and, for the corollary, exact computations. Only the Lean proof of Theorem 1.3 assumes an input. It verifies the reduction of that theorem to the transcendence of the Hecke–Mahler boundary sum at an irrational  $\theta \in (0, 1)$  and nonzero algebraic  $\beta, \alpha$  with  $|\beta| < 1$  and  $|\beta||\alpha|^\theta < 1$ , which is Theorem 1.1 of Bugeaud and Laurent [14] in the case  $\rho = 0$ , due to Loxton and van der Poorten [15, Theorem 8, p. 40]. Lean does not prove that input; the Lean theorem assumes it. Everything the proof adds is checked: the running least-common-multiple identity, the selection of the single power of  $q$  between consecutive powers of  $p$ , the factorisation of the repeated sum, the identification of the distinct running values, the telescoping identity, the boundary identity (3) expressing  $A$  through the Hecke–Mahler sum, the irrationality of  $\log p / \log q$ , and each side condition verified before the value theorem is applied. The two identities expressing  $\mathcal{D}_{\{p,q\}}$  and  $\mathcal{R}_{\{p,q\}}$  as polynomials in  $A$  are unconditional and carry no additional input.

A theorem whose own statement is conditional is formalised exactly as stated. Theorem 1.3 is asserted unconditionally in this record, and its Lean counterpart carries the additional named input described here.

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